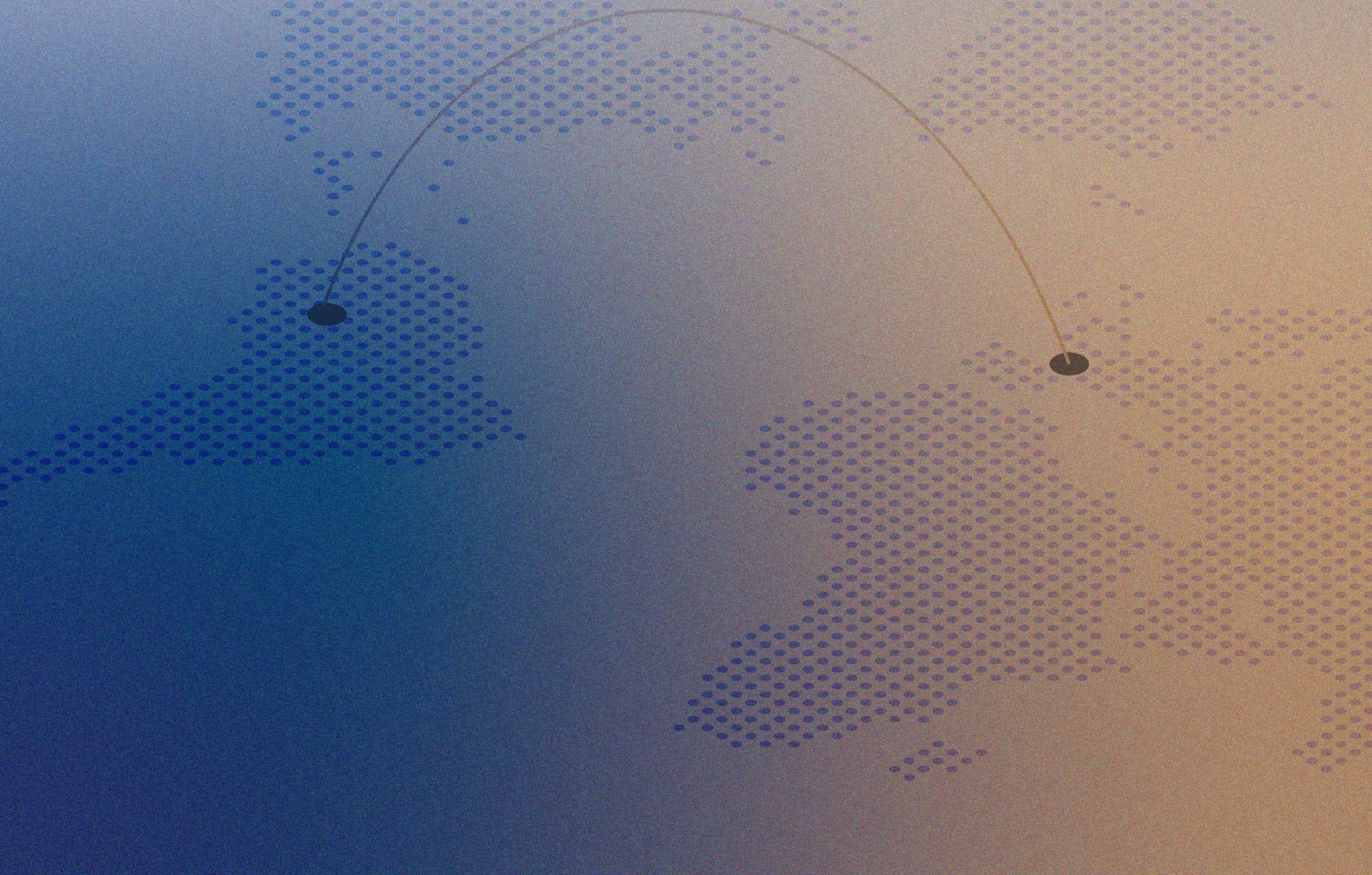
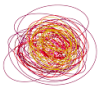
EUI SCHOOL OF TRANSNATIONAL GOVERNANCE

# REDemocracIA

## Handbook for a Democratic Approach to Artificial Intelligence in Ibero-America




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The Chair in Artificial Intelligence and Democracy is part of the Florence School of Transnational Governance at the European University Institute and is funded by the Spanish Secretary of State for Digitization and Artificial Intelligence and the European Union's NextGenerationEU fund. Since 2021, the Chair's team conducts research, transfer of knowledge, dissemination, teaching and innovation activities on matters related to the impact of artificial intelligence on democratic politics. Its mission is to contribute to the analysis of the relationship between artificial intelligence, society, and the individual, and to actively participate in and promote the international debate about the practical and conceptual challenges that algorithmic systems have introduced for democratic decision-making.

This document was edited by Marta Cantero Gamito and Lucía Bosoer.

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# FOREWORD

## Daniel Innerarity

Digital transformation is one of the most powerful forces shaping the future of our societies. For some, it represents a promise of efficiency and progress; for others, a threat that destabilises traditional structures. Beyond these opposing views, it is fundamental to understand that digitalisation is not just a technological phenomenon but is also a profoundly social and political process. Its impact cannot be measured solely in terms of innovation or competitiveness; it is necessary to analyse how emerging technologies influence societies, and at the same time, how they are shaped by them.

The debate on artificial intelligence (AI) and digitalisation is usually reduced to extreme and polarised positions. While some see in technology a power that is capable of substituting weakened and inefficient institutions, others blame it for eroding democratic governance and the capacity for political action. Both perspectives are based on the same assumption: that technology can replace politics. However, this vision simplifies a far more complex reality. Technology is not an autonomous entity, nor does it lead to an inevitable destiny. Rather, it is the result of multiple interactions between diverse actors, negotiated processes and political decisions that determine its impact and reach.

This guide starts from one key premise: no technology operates in a vacuum. It is a social construction that interacts with institutions, values and practices. Innovations like AI or quantum computing do not move in a single direction, nor do they necessarily lead to the same type of society. Their development and implementation depend on more or less deliberative processes, regulatory frameworks and socioeconomic structures that determine how they are integrated into everyday life.

If we seek to ensure that technological progress serves social development and welfare, then we need to promote spaces for broad reflection, democratic deliberation and inclusive governance. The main obstacle for this is the belief that we are facing an overwhelming and irreversible force. Often, the urgency 'to not fall behind' has led to a tendency to prioritise

speed over reflection and reaction over planning. Historically, many of the transformations that failed did so because they were implemented without sufficient consultation, which in turn generated resistance and amplified inequality.

We are facing an ongoing process whose destiny is not yet fully defined but in which we undoubtedly have a key role to play. Despite the fact that technology has become a new field of contestation for global leadership and that the digital space tends to fragment, the horizon must continue to be the construction of a shared environment based on common standards for digital technologies. However, these rules cannot be homogenous nor imposed from a single vision: they must reflect the diversity of perspectives, interests and needs of different communities, with special attention to those who have historically been neglected in technological development.

The following chapters seek to enrichen the debate on the governance of AI in Ibero-America from a plural and inclusive perspective. They examine how the development and use of AI in the region must draw on diverse perspectives, and they consider its impact on different social sectors, urban and rural contexts, as well as on communities with different levels of access to technological resources. Digitalisation cannot be a unilateral process that is designed and decided on by the few, while everyone else simply adapts. For the digital transformation to be truly equitable and sustainable, it must be built in a deliberative way that incorporates diverse voices and ensures that technology serves society as a whole, not just the few.


## Daniel Innerarity

*Director of the Institute of Democratic Governance and Chair in Artificial Intelligence and Democracy at the Florence School of Transnational Governance (European University Institute).*

# INTRODUCTION

**Marta Cantero Gamito and Lucía Bosoer**

This handbook was created within the framework of the **Ibero-American Network for Democracy and Artificial Intelligence** (REDemocracIA), promoted by the Chair in Artificial Intelligence and Democracy at the Florence School of Transnational Governance of the European University Institute.The exponential development of algorithmic systems and their growing role in all areas of our lives calls for a rethinking of the state-citizen relationship and a necessary reflection on what democracy means and how it manifests itself in the age of artificial intelligence (AI). In this regard, AI is not just a technical issue, but rather a political one: its regulation, design and implementation can strengthen or weaken democratic structures. REDemocracIA aims to promote the analysis of-and action-on the political and social effects of technology on democracy, emphasising the importance of a regional approach coordinated in Ibero-America, with particular focus on Latin America. We believe that the European and Latin American countries that make up the Ibero-American space share particularities and common interests that allow for mutual learning when it comes to developing policies that are beneficial for our societies, but that the conversation must start from the recognition of the specificities of each context.

The opportunities and challenges posed by AI systems have led to the proliferation of different strategies and draft legislations that seek to guide their development, maximise their benefits, and mitigate their risks. Many of these challenges – such as the growing concentration of power in the hands of a few technology companies – transcend borders and are common to different geographical contexts. Therefore, the exchange of knowledge and experience is essential to build collective responses and avoid the creation of isolated regulatory frameworks. At the same time, however, the governance of AI in Latin America faces particular challenges that require strategies that are adapted to the regional context. For example, the limited availability of financial resources for the development of technologies at a local level, or the growing influence of foreign regulatory models, like the AI Act of the European Union (EU), generate tensions that must be addressed with regional coordinated policies. While the regulatory framework of the EU can provide valuable references and its formulation process offers important lessons, it was designed on the basis of the EU's experience in digital regulation, its institutional maturity and its regulatory capacity, which are elements that do not always correspond to the realities of Latin American countries.

Latin America must develop its own regulatory frameworks and approaches, drawing from international experiences when it is pertinent, but ensuring that these respond to the particularities of its institutional, economic, social and technological panorama. In this regard, **intra and inter-regional dialogue** and the **design of regulatory frameworks that respond to local needs** are fundamental to ensuring that AI benefits the whole population, and not just the few actors with greater access to capital and technological infrastructure.

## Notes on methodology

This handbook aims to provide a **critical and multidimensional** approach to the governance of AI in the region, providing tools for the formulation of public policies that guide technological innovation towards the common good and that guarantee the protection of fundamental rights. The following chapters are the result of a collaborative process developed in the framework of REDemocracIA, through a series of virtual meetings involving specialists in technology, law, and public policy, along with actors from the private sector and civil society. During these discussions, participants exchanged knowledge and experiences to help shape a comprehensive vision for AI governance in Ibero-America.

Based on discussions with local actors, a survey of interests and concerns was carried out, and ten virtual meetings were organised. These addressed some of the most important issues related to the impact of AI on democratic life. As a result of these discussions, the main challenges and opportunities around their regulation and responsible use were identified, with the objective of offering concrete recommendations adaptable to local contexts. This document, therefore, is not merely a synthesis of ideas, but a reflection of **an ongoing inter-regional dialogue**.

The first part of the handbook presents the main structural challenges identified, while the second part centres on the impact of AI on three strategic areas: **education, justice** and the **labour market**. Each chapter analyses a specific problematic and attempts to provide an analytical framework for the formulation of adequate policy responses, following a normative approach centred on human rights and sustainable and inclusive development. For each theme, the handbook offers a summary of the discussions that took place in the meetings, followed by specific recommendations. Likewise, at the end of each virtual meeting, we invited participants to share their reflections on and contributions to each of the topics discussed. These valuable contributions have been incorporated into the third part of this document. Some of them go into more detail on aspects addressed in previous chapters, while others open up new perspectives and dimensions to consider each of the issues.

*Multidisciplinary and transnational approach*

The regulation and governance of AI in Ibero-America cannot be approached from a single viewpoint. The incorporation of technology in society is a complex phenomenon that cuts across diverse disciplines, sectors and legal frameworks. For this reason, this handbook adopts a multidisciplinary approach that allows an analysis of its implications from different perspectives: legal, economic, ethical, technological and social, among others.

Moreover, this approach recognises that AI regulation cannot be limited to the creation of isolated norms, but that it must contemplate the interactions between different areas and actors. To ensure a democratic balance between innovation and the protection of fundamental rights, collaboration between governments, the private sector, academia and civil society organisations is crucial. Likewise, the implementation of regulatory frameworks must respond to the realities of the region, avoiding the simple adoption of models from other latitudes without a process of contextual adaptation. For this, REDemocracIA seeks the integration of experiences of governance of emerging technologies, the learning and exchange of good practices between different countries, and the generation of solid institutional capacities to face up to the current and future challenges of AI in the region.

## Where are we headed?

This handbook does not aim to offer definitive answers, but rather to continue opening up **spaces of dialogue and reflection** on how to address the challenges and opportunities posed by AI in Ibero-America. Its content is aimed at public policy makers, regulators, academics, legal professionals, technological innovators, members of civil society and citizens interested in understanding and tackling the impact of AI in the region.

Given that technology will continue to evolve and expand into new areas, it is fundamental that these discussions continue. The development of normative and ethical frameworks for AI in Ibero-America cannot be reduced to the passive adoption of governance models developed in other contexts. In a global scenario marked by intense geopolitical struggle, it is imperative to promote a **technological governance grounded in local perspectives**, and commited to equity, inclusion, and sustainable development.

**Latin America has generated specific responses to the challenges of AI** that deserve to be valued and disseminated. The experiences that have worked—and those that have not—offer key lessons for both the region and for other parts of the world. Moreover, beyond any technological horizon, the regulation of AI must be aligned with international principles on human rights, guaranteeing that its application does not deepen inequalities or violate fundamental guarantees. **Technology must be at the service of society, not the other way around**. It is not a matter of adapting to the impositions of technological development, but rather of democratically guiding such development. This handbook aims to be a step in that direction.

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This handbook is the result of a collective effort developed within the framework of the seminars organised by REDemocracIA, in which scholars, legislators, experts in technology and representatives of civil society exchanged their knowledge and experience to build a comprehensive vision of AI governance in Ibero-America. The result, embodied in this document, would not have been possible without the collaboration and commitment of multiple actors who contributed their vision and enriched this project with their experience, knowledge and time. For this reason, we would like to express our most sincere thanks to the participants of the REDemocracIA meetings, whose reflections and debates have been fundamental to the realisation of this project.

We are especially grateful to the institutions that facilitated spaces for discussion and promoted the generation of knowledge about AI governance in Latin America and, in particular, to our strategic partners who participated in and disseminated our activities: Fundación CiGob, Fundación Kamanau, the Organisation of Ibero-American States (OEI), the Ibero-American General Secretariat, and the Mercosur Permanent Review Tribunal. Lastly, we wish to thank María Laura Minoli for her creativity and talent in the graphic design of this manual and the graphic pieces of REDemocracIA.

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# Structural and Regulatory Challenges of AI

PART ONE

# DEBUNKING BIASES IN ALGORITHMS: A CROSS-SECTORAL DIALOGUE

The biases in artificial intelligence (AI) systems and the consequent **algorithmic discrimination** have occupied a central place in discussions on the risks and challenges of AI in recent years. Algorithmic biases are defined as an anomaly in the results produced by AI, the origin of which may be related to the data used in the training phase, or to the biases present during a system's development phase.

There are many cases in which the algorithmic systems used in the public and private sectors have reproduced biases that already exist in society, many times amplifying them and creating new instances of discrimination. As a response to this phenomenon, diverse initiatives have emerged seeking to raise awareness among society and decision-makers about the biases that can be present in technology. These initiatives offer tools to contribute to a fairer and more equitable AI.

### How do biases in AI arise?

Biases can emerge in distinct phases of the development of an AI system, from the creation of the databases and the design of the algorithm, to the interpretation and execution of the results. The implementation of AI systems is developed over various phases, each of which is carried out by different working groups, often involving different people.

Frequently, algorithmic biases are related to the **databases** chosen to train an algorithm. For this reason, it is fundamental to understand who determines which data are collected and which population is included in such data. The data collected may not be representative of the entire population, and even if they are, they tend to reflect a social reality in which biases and discrimination abound.

For example, in 2018 it became known that years earlier the technological company Amazon had implemented a new AI system to recruit new staff. The algorithmic system used had been trained with the data of the individuals who had sent their curriculums to

the company over a period of ten years. As the majority of individuals who had sent their curriculums were men, the system understood that the most suitable candidates for the job were men and automatically rejected women's profiles. In Latin America, the problem of biased databases is particularly important given that many of the AI systems used in the region have been developed in other parts of the world with databases that are not representative of the population nor of the local context.

Biases can also originate in the **development phase of the algorithm** and the choice of the variables that are included in this. In 2017, the government of Salta, in Argentina, signed a deal with Microsoft to develop an AI system that could determine the probability of the occurrence of teenage pregnancies. The variables chosen to inform the algorithm included, among others, peoples' addresses, socioeconomic situations, and even migrant backgrounds. It was a system that, owing to the design of the algorithm and the variables considered to determine the possibility of adolescent pregnancies, inevitably threw up biased results, and thus contributed to the perpetuation of a reality of social discrimination.

AI biases are in large part a reflection of the society that we live in. Algorithms automatise biases that are present in human beings and convert them into a facilitating tool that accelerates discrimination that already exists. Each developer, client, and any other person involved in the development of an algorithm reproduces and introduces their own biases and their own subjectivity in said process. Assuming that algorithms are neither neutral nor objective, the main question then becomes how to control or minimise the consequences of biases.

### What to do in this scenario?

Faced with this reality, it would be necessary for all the actors involved in the process of development, execution and evaluation of an AI system to take on a role in the management of biases and algorithmic discrimination. The creation of **diverse datasets and training data** is a first step towards fairer AI. In the field of health, for example, if an AI model used for the diagnosis of illnesses is principally trained on the data of patients from one single region or demographic group, it might not be as precise when it comes to analysing cases of under-represented populations. In this regard, broadening the diversity of training data would help the AI system to make more precise diagnoses for all patients. Nevertheless, it is advisable to avoid framing the discussion around just one phase of algorithm development in isolation. Rather, we should think about the interaction between the different phases and how to systematically influence them as a whole. In this way, for example, a problem that might arise in the creation of a database could be solved during the development phase of the model.

The **transparency** of algorithms and **citizen participation** to guarantee legitimacy and accountability are key factors in controlling or minimising the impact of AI biases. There is a need for greater openness of the algorithms behind each system, the design of which is often highly opaque. This opacity often causes AI to be perceived as abstract and unseen, like a closed **black box**. In many cases, civil servants using AI to deliver basic services do not understand what is actually being used or offered to citizens. This is highly problematic and limits autonomy in decision-making. While it is crucial to encourage a greater awareness of the function and consequences of AI systems and to create new capacities, given the growing complexity of these systems, it is also important to avoid placing all responsibily on the user.

### Black boxes

The so-called "black boxes" are AI models whose reasoning process are inexplicable for the user and even the developer of the model. As AI techniques evolve and become more complex, black boxes are becoming more and more frequent, which poses important challenges in terms of transparency and accountability.

Public institutions should consider setting up **cross-sector dialogue tables** to assess the usefulness of AI systems pre- and post-implementation. These roundtables should include technicians and developers, regulatory bodies, small and medium-sized businesses for local economic and resource development, and citizens and civil society organisations. A holistic approach is required: the private sector must assume responsibility and commit to the development of algorithms that respect ethical and transparency standards; the state must ensure its presence as the guarantor of the citizens' rights; and civil society organisations must continue to bring affected groups and, in particular, marginalised communities to the negotiating table. To prevent the implementation of AI systems from contributing to reproducing and perpetuating existing inequalities, conscious efforts are needed to effectively engage a broad segment of society and ensure that diverse voices and perspectives are included in the conversation.

Likewise, it is necessary to count on regulatory frameworks that take into consideration local cultures and values. In Latin America interesting exercises to regulate AI from a local perspective are taking place. Initiatives like those in Brazil and Mexico stand out. These have incorporated considerations on gender and cultural diversity into their AI strategies, showing the path towards intersectionality and inclusion through the removal or mitigation of biases.

A call for shared responsibility, beyond programmers and technicians, towards public authorities, businesses and citizens, is crucial to ensure that AI systems are developed and used in ways that benefit society as a whole. The conversation must not only be technical, but also, and principally, political, and it must consider what kind of society should be built with the help of AI. In this sense, AI challenges society as a whole to reflect on issues that go beyond the technology itself.

### EDIA

EDIA[1] (Stereotypes and Discrimination in Artificial Intelligence) is a project developed by Fundación Via Libre[2] of Argentina with the aim of involving more people in the evaluation and auditing of AI systems. The tool permits anyone interested, without the need for technical knowledge, to explore the biases and discriminatory stereotypes present in language models like ChatGPT. The project aims to stimulate critical literacy, collaboratively create datasets to better audit AI systems, and promote broader citizen engagement around AI, bias and discrimination.

1 https://ia.vialibre.org.ar/

2 https://www.vialibre.org.ar/

## Recommendations to mitigate the impact of biases in AI

- One of the key pillars in the mitigation of algorithmic biases is effective regulation. Governments must develop clear normative frameworks that require the evaluation and reduction of biases in AI systems. These frameworks must be dynamic so that they can adapt to rapid technological development, and they must consider mandatory and periodic audits of algorithms used in the public sector. In addition, transparency and accountability must be guaranteed so that the disclosure of the results of these audits, with the participation of external auditors and affected groups, can strengthen both equity in automated decision making and public trust in that decision making.

- Education and training are another fundamental pillar. Digital literacy and knowledge about AI must not only be present in traditional educational settings, but also in the formation of civil servants, professionals in the technology sector and citizens in general. Only with a broad understanding of how these systems operate can a critical and responsible view of their use be fostered.

- Technology companies also play a central role in this process. The implementation of external and periodic audits to identify biases in their models must be required as standard practice. Moreover, diversity in development teams is crucial for reducing the reproduction of sociocultural biases: counting on experts from different disciplines and socioeconomic backgrounds will improve the ability to design more inclusive and equitable technologies. Transparency is also a guiding principle in this area. Companies must provide clear information about the data used to train their models, explaining how it is obtained, processed, and how it impacts algorithmic decisions.

- In turn, civil society plays a key role in monitoring and demanding good practices in the development and use of AI. Promoting awareness campaigns about the risks of algorithmic biases, fostering the creation of citizen observatories, and demanding that companies and governments provide details about their mitigation strategies, are some of the key actions needed to strengthen accountability.

- Lastly, the development of AI must be based on inclusive and intersectoral dialogue. Collaboration between governments, businesses, academia, civil society, and the general public will allow the creation of participatory mechanisms that integrate the concerns of the communities most affected by algorithmic biases.

# SURVEILLANCE APPLICATIONS AND THEIR IMPACT ON DEMOCRATIC GOVERNANCE

Even though surveillance applications are not an entirely new phenomenon, recent technological developments are perfecting their reach and effectiveness, generating important consequences for modern societies. According to the philosopher Byung-Chul Han, we are living in a "digital panopticon", an era characterised by hyperconnectivity, digitalisation and the mass collection of personal data.[3] Every aspect of our life is integrated in this digital network, from the daily use of online platforms to the voice assistants present in our homes, places of work and means of transport, all connected to the Internet of Things (IoT). Moreover, advances in neurotechnology are promoting the development of mechanisms that are capable of monitoring cerebral waves, thus coming to explore (and exploit) previously inaccessible levels of intimacy. Technologies like remote biometric identification systems, which often operate in a way that is imperceptible for citizens, exemplify how digital control can be exercised without the conscious knowledge of people. In this way, a surveillance infrastructure that remains largely invisible has been consolidated.

In Latin America, a region affected to a greater or lesser extent by organised crime, citizen insecurity and institutional weakness, surveillance applications acquire particular relevance, as both potential tools for improving security and as threats to fundamental rights. As a result, the relation between mass surveillance, citizen security, individual freedoms and democratic governance is becoming increasingly complex.

### What are *surveillance technologies?*

The notion of surveillance applications or technologies is normally used as a broad concept that encompasses a range of emerging technologies used to monitor human beings. These include diverse tools, devices and systems designed to monitor, collect, store and analyse information about people, behaviours, movements or environments. These technologies can be used by governments,

3 Han, B. C. (2015). The transparency society. Stanford University Press

companies or individuals with diverse motives, such as security (to prevent crime and manage emergencies), social control (to supervise and regulate collective behaviours), or commercial ends (to personalise the user experience and analyse data for marketing strategies.)

Currently, one of the most common examples of surveillance technologies is remote biometric identification, and in particular, facial recognition systems. These systems are commonly used to identify people at distance through the comparison of biometric data of a person with biometric data contained in reference databases. Although this technology has been developing over various decades, the emergence of new image acquisition and processing technologies, advances in the field of big data and improvements in deep learning techniques for the training of algorithms have led to the growth of facial recognition systems in different areas.

The accuracy of these systems depends on various factors: access to a database that allows users to identify images with previously catalogued persons; the quality of the images; the way in which the algorithms have been trained; and the design of the software, which indicates the precise parameters according to which the comparisons are made. A problem in each of these instances can lead to what is known as "false positives", and "false negatives".

- **A false positive** occurs when a system wrongly identifies one person as another. That is, the algorithm establishes a match between two faces that in reality belong to different individuals.
- On the contrary, a **false negative** occurs when the system does not recognise a person that it should have been able to identify. In this case, the algorithm does not detect a match between two images of the same individual.

In recent years, there has been a proliferation of research demonstrating that facial recognition systems have a very high false positive rate, which has led to serious situations of discrimination[4] and the violation of fundamental rights.[5] The limited accuracy of these systems is made worse by the fact that foreign companies tend to train their algorithms on databases that are not representative of the populations of countries or cities that acquire their products.[6] An algorithm can be fed with data that is representative of a given population, but if that same algorithm is used in a different context, the same data could be biased. This is particularly relevant in the Latin American context, where governments are increasingly acquiring biometric recognition systems developed abroad.

4 https://www.intercept.com.br/2019/11/21/presos-monitoramento-facial-brasil-negros/

5 https://www.pagina12.com.ar/209910-seis-dias-arrestado-por-un-error-del-sistema-de-recono-cimien

6 Valle Escolano, R. (2022). "Transparencia en la inteligencia artificial y en el uso de algoritmos: una visión de género", Cotino Hueso L., & Castellanos Claramunt. J. (coord.). p. 99. https://www.uv.es/cotino/publicaciones/libroabiertotp22.pdf

## Latin America: fertile ground for surveillance technologies?

The exponential development of digital technologies on a global level has impacted Latin America in multiple ways. One of the main tendencies, as mentioned before, has to do with the growing adoption by governments of surveillance technologies developed by foreign companies. According to the report Surveillance Technology in Latin America: Made Abroad, Used at Home[7], compiled by AccessNow in 2021, many of the systems used in the region come from international companies, particularly from Asia (China, Israel and Japan), Europe (the United Kingdom and France) and the United States. These technologies tend to be acquired directly or through local intermediaries who facilitate their integration. Many of these companies have troubling human rights records and operate without transparency about the functioning of their systems. Moreover, some of their systems have evidenced algorithmic biases and serious vulnerabilities.

This is coupled with the existence of a regional context characterised by diverse levels of political instability, institutional weakness and high insecurity rates, all of which provided fertile ground for the implementation of surveillance technologies, especially in the field of public security and the fight against organised crime and drug trafficking. Indeed, facial recognition systems are seen by many local administrations as a seemingly fast and effective solution to tackle growing crime rates.

Against this backdrop, one of the main challenges has to do with the lack of solid legal frameworks and effective mechanisms to guarantee transparency and the respect of human rights in the deployment of these technologies. In countries with important structural injustices, with records of corruption, discrimination and police abuse, the implementation of surveillance technologies could intensify existing problems, reinforce inequalities and perpetuate antidemocratic practices. Governments could find in surveillance technologies a tool to consolidate their power and justify intrusive practices under the premise of guaranteeing security. In this context, the lack of regulation and effective supervision not only puts individual freedoms at risk but also compromises democratic governance in already vulnerable societies.

### Facial Recognition Cameras in Buenos Aires

In 2019, the government of the Autonomous City of Buenos Aires, Argentina, announced the acquisition of facial recognition software for security cameras and monitoring centres in order to identify fugitives of justice. The deployment of facial recognition cameras across the entire city was implemented quickly following the announcement and was carried out via a direct contract, without any public tendering process or public participation. In view of the opacity of this process, diverse civil society organisations, like the Observatorio de Derecho Informático de la Argentina (ODIA), the Centro de Estudios Legales y Sociales (CELS), and the Fundación Vía Libre began to demand more information and started legal action to stop the deployment of the programme.

Throughout the judicial process, the government claimed that the system had been procured as a preconfigured and

7 https://www.accessnow.org/tecnologia-de-vigilancia-en-america-latina/

finished system, without knowing about or having access to the data with which it had been trained or the details of its operation. Likewise, it was demonstrated that system did not meet basic security requirements and that its records contained data on individuals who were not included in the data base of persons with arrest warrants. In other words, the government had accessed data on individuals whom it was not authorised to access under current law. As a result, the judicial authorities ordered the suspension of the facial recognition system for the identification of fugitives.[8]

### Pegasus software

Pegasus is the surveillance software that has gained most notoriety in the last decade. It is a spyware designed by the Israeli company NSO Group and has been used by governments across the world for mass surveillance purposes. A spyware is a type of software that is installed onto a computer or mobile phone without the user's knowledge and allows third parties to access emails, text messages, photos, and any other type of content. It is also able to monitor what the user is doing in real time, to access the camera, and to listen through the microphone.

While the spy programme has contributed to the fight against organised crime, facilitating, for example, the dismantling of child abuse networks or the arrest of notorious drug traffickers, it has also been used by various governments to spy on human rights defenders, journalists, and other opposition figures. Since 2021, organisations such as Amnesty International, the Forbidden Stories consortium of journalists and the University of Toronto's CitizenLab, among others, have denounced the spying on hundreds of journalists and activists around the world through Pegasus. In Latin America, the spy programme has been used to monitor journalists, activists, and members of the opposition in Mexico[9] and El Salvador.[10]

### Surveillance routes

As mentioned above, the deployment of surveillance technologies in Latin America has been characterised in most cases by a lack of transparency. The procurement of systems has generally taken place in an opaque manner, with little available information about suppliers and conditions of use. These characteristics are exacerbated by the inherently closed nature of this industry, which has historically operated within a paradigm of lax accountability.

8 For more information about this case, we recommend accessing the Reconoceme project by O.D.I.A and Democracia en Red, available at https://reconoceme.org/#home

9 https://www.nytimes.com/es/2023/04/18/espanol/pegasus-mexico-gobierno-ejercito.html

10 https://www.accessnow.org/press-release/pegasus-el-salvador-software-espia-arremete-periodistas-declaracion/

China has established itself as one of the largest exporters of these technologies to countries in Latin America and other regions of the Global South. Its advantage lies in offering solutions known as "turnkey" projects, which not only include advanced hardware and software, but also installation, maintenance and training for users. These offers, coupled with more affordable prices than those of Western competitors, are particularly attractive to countries without a sufficiently developed local technological ecosystem. In some cases, companies have even offered their technologies free of charge. This allows them to test their systems in real-world contexts, in a sort of experimental laboratory, whilst also securing future contracts.

Surveillance technologies in general, and facial recognition cameras in particular, have extended across the whole world, not just in Latin America. Even if in recent years many countries and cities have moved towards the prohibition or strict regulation of the use of facial recognition cameras in public spaces,[11] l11 these continue to be used, although not without controversy. For example, in Europe, Article 5 of the AI Act prohibits the use of biometric identification systems in real time in public spaces for security purposes, except in certain circumstances, such as the search for specific victims, the prevention of a specific, substantial and imminent threat, or the identification of a suspect. Despite this, during the 2024 Olympic Games, France implemented AI-based video surveillance systems to guarantee security,[12] which generated significant criticism from experts and civil society organisations.

11 In the United States, facial recognition in public spaces has been prohibited or placed on indefinite moratorium in Berkely, Boston, Cambridge, King County, Madison, Minneapolis, New Orleans, Northampton, Oakland, Portland and San Francisco, among other cities.

12 https://es.wired.com/articulos/inteligencia-artificial-nueva-policia-de-juegos-olimpicos-de-paris-2024

## Recommendations for the deployment of surveillance technologies within a framework of legality, transparency and respect for human rights

The deployment of surveillance technologies poses important democratic challenges. While many governments argue that these tools represent an opportunity for public security, their unrestricted and opaque use may violate fundamental rights such as privacy, freedom of expression and non-discrimination. To guarantee that its implementation is framed within democratic and ethical principles and the respect for human rights, governments, legislators, companies, civil society and academia must all actively collaborate in the regulation and control of these systems. Below, we offer a series of recommendations in this regard:

- **Regulation, transparency and accountability.** Any solid normative framework should delimit, among other things, who can access these technologies and with what ends, ensuring that their acquisition and use occurs transparently. In addition, governments must establish mechanisms of independent supervision, limit the use of state secrecy in procurement and conduct human rights impact assessments before deploying any surveillance system. In turn, accountability must be guaranteed through public authorities and processes of control to prevent the misuse of these tools. No surveillance technology should operate in the shadows or be exempt from citizen oversight. Lastly, it is important to design clear and accessible processes to guarantee transparency and justice, and to establish compensation mechanisms for individuals affected by the misuse of surveillance technologies.

- **Protection of privacy and cybersecurity.** A further crucial aspect is the integrity and security of data collected through surveillance technologies. Strict access controls must be established, and losses, leaks or accidental alterations prevented.

- **Corporate responsibility and commitment to human rights.** The companies that develop and market these technologies must adopt exhaustive due diligence mechanisms to evaluate the impact of their products, ensure that their systems do not reproduce biases and perpetuate discrimination, and prevent their use in authoritarian contexts. Similarly, these companies must guarantee transparency in their operations and adopt a policy of active collaboration with regulatory agencies and other bodies responsible for protecting human rights.

- **Monitoring, social activism and research.** The role of civil society and academia is central to the construction of a democratic framework for the adequate use of these technologies. Civil society must continue developing spaces for independent monitoring, promoting public debate on the use of AI in surveillance contexts and demanding mechanisms for redress. Moreover, another fundamental contribution is the development of empirical studies that analyse the impact of surveillance technologies in local contexts, including their effectiveness in addressing tpublic safety issues and their influence on citizens' perceptions of insecurity. In this regard, it would be necessary to explore not only technical aspects, such as the accuracy and biases of these technologies, but also their social, ethical and psychological effects.

# DECOLONIAL RESPONSES TO ALGORITHMIC SYSTEMS[13]

In a world in which algorithms are increasingly rooted in all spheres of human life, it is fundamental to analyse their impact on marginalised communities and on global power structures. The adverse effects of digitalisation and technological development have had a greater impact on the countries of the so-called Global South, as well as on historically marginalised communities, exacerbating pre-existing inequalities and perpetuating asymmetrical power relations between centres and peripheries.

Against this backdrop, various initiatives have emerged that seek to reimagine AI from a decolonial perspective, anchored in local contexts, values and needs. Constructing decolonial AI is not just a technical question, but is also a profoundly political challenge. Any process towards decolonial AI will require us to question existing structures, imagine other forms of development and open up spaces for multiple voices to participate in the development of technologies that are not only advanced, but are also fair and responsible towards their environment

## Algorithms, inequalities, and the colonial legacy

AI, like any other technology, feeds on and contributes to reproducing the norms, values and assumptions that predominate in society. AI encapsulates and transfers specific worldviews, oftentimes disguised behind the apparent objectivity of technology. However, algorithmic systems are not neutral; they are constructions that reflect the decisions and priorities of those that design and implement them. The main problem lies in the fact that political and ontological conceptions of technology remain invisible for most people given the growing opacity and complexity of algorithms. Understanding the ecosystem of

13 The dialogue of REDemocracIA on this topic was inspired by ideas developed from decolonial thinking (Quijano, 2000; Mignolo, 2007; Ricaurte, 2019; Milán y Treré, 2019; among others). In this document, we decided not to delve into them, as addressing these theoretical foundations in the detail they deserve excedes the scope and objectives of this work.

algorithms—from natural resources and infrastructure to mathematical models and largely invisible labour input—requires time, financial resources and technical expertise, all of which makes "opening up" AI a particularly complex challenge.

Due to their very nature, algorithms tend to classify and pigeonhole individuals within predefined categories, such as male or female, conservative or liberal, vegetarian or omnivore. This logic of classification limits the fluidity and complexity inherent in human identity, ignoring the multiple dimensions that make-up people and creating tensions with the dynamic and diverse realities of the communities in which algorithms are embedded. As such, the potential risk is that technology fosters a uniform culture where only certain values are visible, such as efficiency, optimisation, rapidity, productivity or success. A culture that rests on a specific set of premises and themes, and excludes or limits the visibility of others.

One of the most problematic elements of algorithmic systems is the asymmetrical dependency relationship that is established between those who produce the data and those who use it to generate economic benefit. This model is founded on an extractive logic, where captured data is transformed into monetizable resources, often at the expense of the rights and dignity of the communities involved. In this way, AI, in addition to perpetuating dominant norms and visions, also fits into a broader framework of historically asymmetrical power relations. For example, the biases inherent in many algorithms tend to perpetuate and amplify existing social inequalities. Well-documented cases include facial recognition systems that discriminate against African Americans[14] or hiring algorithms that automatically exclude women and minorities[15], reinforcing patterns of systematic exclusion. Beyond biases, the development and use of AI is also linked to growing job insecurity. Those responsible for training algorithms often work in highly vulnerable conditions, and the digital platforms that operate based on algorithmic systems have transformed labour markets to the detriment of workers' stability and rights.

The environmental impact of technology cannot be ignored either. The development of AI is in large part linked to an instrumentalist vision of nature. This perspective views the planet as an unlimited resource at the service of humankind, thus permitting the exploitation of its resources to drive technological progress. AI and its related infrastructures depend on enormous amounts of energy and natural resources. The data centres used to train and execute these systems, for example, consume huge quantities of energy. Likewise, the hardware components central to AI development, like Graphic Processing Units (GPUs), depend on the extraction of rare-earth minerals, oftentimes linked to deforestation, pollution, and ecosystem degradation.

### The limits of ethical frameworks

In recent years, there has been a proliferation of statements and documents seeking to offer ethical frameworks for the regulation and governance of AI. Most of them refer to an AI that is transparent, secure, robust and reliable, among other basic principles. While efforts to provide guidelines to steer technological development towards the common good are important, most of these documents also suffer from significant limitations. On the one hand, there is a risk of what is known as 'ethics washing', i.e. the adoption of ethical principles by technology companies or governments as a superficial means of improving their legitimacy without any real commitment to policies and concrete changes in their practices.

14 https://www.intercept.com.br/2019/11/21/presos-monitoramento-facial-brasil-negros/

15 https://r3d.mx/2018/10/10/amazon-retira-sistema-de-inteligencia-artificial-para-reclutamien-to-porque-discriminaba-perfiles-de-mujeres/

On the other hand, from a decolonial perspective, many ethical frameworks are universalist in nature and based on utilitarian and individualistic conceptions that do not necessarily reflect the diversity of values and local realities. Furthermore, they are often designed from Eurocentric perspectives, which assume homogeneous standards applicable globally and ignore other historical and cultural legacies. For example, a principle such as privacy, or the idea of 'human-centred' AI, which are so prevalent in global ethical debates, may not resonate with indigenous communities that value the collective over the individual or that understand technology from a more holistic relationship with nature.

### From Reflection to Action

Thinking from a decolonial perspective is not without its difficulties, partly because technologies originating from global centres of power have permeated deeply into our lives. These tools influence social, economic and cultural dynamics, but they also act at a deeper level by impacting the very definition of subjectivities. Clarifying and disentangling these technological forms from everyday life becomes, in this sense, a particularly complex challenge, as it involves questioning deeply rooted habits and structures.

In this context, it is fundamental to go beyond analysis and begin to redesign devices and systems that allow us to question, problematise, and "decolonise" the way in which algorithms configure social relations and structures. It is essential to intervene in the materiality of technology: to design devices and tools that not only challenge hegemonic narratives, but that also activate pedagogical processes, stimulate critical thinking and promote alternative forms of participation. These practical actions are essential for building new ways of engaging with technology, based on principles of justice, diversity and commitment to local realities.

#### The Montevideo Declaration on Artifical Intelligence and its Impact on Latin America

The Declaration of Montevideo on AI and its impact on Latin America[16] was signed in June 2023 on the occasion of the Latin American Meeting in Artificial Intelligence (Khipu). It outlines a series of principles that should guide the development of AI from the design stage, such as the imperative to minimise its environmental impact and the need for its effect on labour productivity to be accompanied by an improvement in working conditions and the quality of employment. The signatories also make an urgent call for the integration of the particularities of Latin American cultures in the design and development of AI systems that serve the region.

16 https://zenodo.org/records/8208793

## Recommendations for Decolonial AI

The development of AI is not a neutral process, nor is it unrelated to global power structures. Most current technologies reflect dynamics of digital colonialism in which knowledge, data, and economic benefits are concentrated on certain actors and regions while other communities remain marginalised or are otherwise directly harmed. To counter these trends and move towards decolonial AI, we need to reflect on its foundations from an inclusive, diverse and more responsible perspective for the planet.

- One of the first steps in this process is to **promote a broader and more democratic reflection on technological development** that facilitates collective decision-making regarding the areas in which AI will be used and for what purposes. This conversation must consider the possibility that the communities involved may decide to halt or reject certain technological applications.

- **The voices and knowledge of communities that have historically been excluded must be actively incorporated** at all stages of AI development. This does not only mean greater human representation, giving space to indigenous, Afro-descendant, peasant and other marginalised perspectives, but also an openness to forms of knowledge that consider interdependence with ecosystems and other forms of life.

- Furthermore, we must promote **interdisciplinarity in the study and development of AI**, integrating disciplines such as philosophy, international relations, natural sciences, humanities and the arts, which provide a richer and more contextualised understanding of the impact of these technologies. To imagine alternative, fairer and more sustainable futures, a collective exercise in speculation is also required, where tools such as science fiction and the arts play a key role in the formulation of transformative technological scenarios.

- **Transparency** and technodiversity are key pillars for fairer AI. Opacity in AI systems reinforces asymmetrical power relations and reduces the capacity for communities to understand, question and take ownership of technologies. To reverse this trend, we must open up technological processes by promoting standards of transparency and accessibility in their design and implementation. Furthermore, it is urgent that we diversify technological infrastructures, designing systems that reflect local realities and integrate traditional knowledge, rather than imposing technological models that are alien to the contexts in which they are implemented. A key step in this direction is the development of local data infrastructures that avoid excessive dependence on systems that are designed and controlled exclusively by external actors and that do not reflect the specificities of the contexts in which they are used.

- Likewise, the **governance of AI must be inclusive and multisectoral**. For decision-making on these technologies to be truly democratic, we must build bridges between academia, the public sector, civil society and businesses and to foster spaces for dialogue and ongoing collaboration between these sectors. In this process, it is important to critically assess the ethical frameworks that are often applied in a universalist manner and consider whether they truly reflect the values and priorities of the communities involved. Rather than adopting global standards without adapting them to the local context, it is advisable to develop ethical principles and norms that emerge from local realities and to recognise the cultural and epistemic diversity of each context.

- Finally, to move towards decolonial AI, ways of **reconfiguring the relationship between technology and nature** should be considered. AI is neither immaterial nor neutral from an ecological point of view: its development depends on an infrastructure that exploits natural resources, consumes massive amounts of energy and reinforces dynamics of unequal accumulation. It is, therefore, crucial to rethink AI by taking greater responsibility for the planet and by designing technological systems that respect ecological limits and promote fairer relationships with natural resources. This means leaving behind extractivist logics of accumulation and excessive consumption and moving towards models of technological development that prioritise sustainability and global equity.

# NEURO-RIGHTS: A NEW DIMENSION IN THE PROTECTION OF HUMAN RIGHTS?

One of the most pressing issues regarding the impact of technology on human rights is the emergence of neurotechnologies that can measure and influence brain activity, for example, to treat diseases, but also to control technological devices with the mind. Neurotechnologies that integrate AI promise significant advances in the field of medicine, but at the same time they pose potential risks to human rights, such as by undermining our privacy, identity and autonomy.

Recent developments in this field raise ethical and legal concerns and are giving space to proposals to update existing legal frameworks and create human rights to protect mental processes and brain data. With Chile at the forefront, Latin America has positioned itself as the region where the most progress has been made in this area, influencing and driving the debate in organisations with global reach.

## Neurotechnologies and the emergence of Neuro-Rights

**Neurotechnologies** are defined as any technology capable of monitoring or manipulating brain activity. Among their main advances, these technologies promise to revolutionise the treatment of mental and neurological diseases such as Alzheimer's, schizophrenia, strokes, post-traumatic stress disorder, depression and addiction. In addition, their potential application transcends the medical field and offers tools to enhance human capabilities and redefine human interaction with the environment.

Within the field of neurotechnologies, **brain-computer interfaces** (BCIs)—devices that connect the human brain to computers or other devices—have generated particular interest. BCIs enable two-way communication, i.e. they capture brain data while also modifying brain activity. They are classified into two types: invasive, which require surgical implantation, and non-invasive, such as helmets or external sensors. While invasive interfaces are regulated within the medical field, non-invasive interfaces, which are

available to consumers, operate under regulations designed for conventional technologies that are therefore insufficient.

Initially conceived of for medical and research applications, neurotechnologies have begun expanding into the consumer market, generating deep ethical and social implications. Companies such as Meta, Apple, Snap, and Neuralink have patented or are developing wearable neurotechnologies that will soon enter the market.

### Neuralink and the promise of brain implants

Several companies are working on the development of brain-computer interfaces. The overall goal is to help people with neurological diseases communicate more easily, either by allowing them to move via the computer cursor or by translating their brain activity into speech or text. The company that has attracted the most attention in recent years is Neuralink, founded in 2016 by Elon Musk. In May 2023, Neuralink received approval from the US Food and Drug Administration (FDA) to begin human clinical trials. In January 2024, the first implant was carried out on a human patient, and the company has since continued to make progress with other patients. However, the media and experts have signalled a lack of transparency and scientific information about the company's research practices and results.

Neurotechnologies, in particular brain-computer interfaces, pose significant challenges for the protection of fundamental rights. For example, companies could exploit neural data obtained through non-invasive neurotechnological devices to influence consumer behaviour or for other marketing purposes. Moreover, unequal access to the opportunities offered by neurotechnologies could exacerbate existing social inequalities.

The concept of **neuro-rights** has emerged in response to the ethical, social and legal challenges posed by advances in neurotechnology. The pioneer in consolidating this term was Rafael Yuste, a Spanish neuroscientist and ideologist of the BRAIN Initiative[17] in the United States. In 2017, Yuste and a multidisciplinary group of experts and researchers formally proposed neuro-rights as a set of fundamental protections to safeguard mental integrity and individual autonomy in the face of advances in neurotechnology. This approach emphasises the need to anticipate the risks associated with technologies capable of capturing, manipulating or altering neural data.

### Five Neuro-Rights

- **The right to personal identity:** Guaranteeing control over physical and mental integrity, as well as preserving the sense of "self".
- **The right to free will:** Ensuring free will and the ability to decide one's own actions.
- **The right to mental privacy:** Protecting thoughts and brain data from unauthorised access.
- **The right to fair access to mental augmentation:** Guaranteeing that the benefits of neurotechnologies are distributed fairly among the population.

17 https://braininitiative.nih.gov/

- **The right to protection from algorithmic bias:** Preventing technologies from introducing or perpetuating discriminatory biases.

## Regulatory Challenges

Although in some areas the discussion on neuro-rights is often viewed with some suspicion as it is considered as a futuristic issue, the impact of neurotechnologies is already tangible: companies are marketing non-invasive devices, such as electroencephalography headsets, which do not require medical protocols and are often exempt from any type of regulation. These devices are promoted for various purposes. In the workplace, for example, there are proposals to use neurotechnologies to assess employees' levels of fatigue or anxiety.

Regulating neurotechnologies is a particularly complex task, given that it is an emerging and rapidly developing field. In this regard, any legislative initiative requires a comprehensive, multi-level approach that combines the updating of existing instruments with the development of new frameworks at the national, regional and international levels. In this context, soft law instruments—guidelines, international standards, and non-binding codes of conduct—offer a practical and flexible solution while more robust regulations are being developed.

## National, regional and international initiatives

### *Chile*

Chile has become an international reference for the regulation of neurotechnologies and the protection of neuro-rights, positioning itself as the first country in the world to legislate in this area. In October 2021, the Chilean Congress approved a constitutional reform that establishes the protection of mental integrity as a fundamental right, recognising the need to safeguard the brain and its processes from possible abuse or manipulation resulting from technological advances.

Similarly, Bill No. 13.828-19 seeks to regulate in detail the protection of people's mental integrity and the development of research related to advances in neurotechnologies. Among its proposals are the prohibition of any form of interference or intervention in neural connections without the free, express and informed consent of the person, as well as the classification of neural data as sensitive data, thus granting it special protection.

Following the Chilean initiative, in 2023, the state of Rio Grande do Sul in Brazil adopted similar legislation. In addition, several Latin American countries, such as Uruguay, Ecuador, Mexico, and Brazil, have promoted bills under discussion that seek to regulate the protection of neuro-rights at the federal level.

### *The Organization of American States (OAS)*

The OAS has recognised the need to establish regulatory frameworks that protect human rights in the context of advances in neuroscience and neurotechnology. In August 2021, the Inter-American Juridical Committee (CJI) approved the Inter-American Declaration of Principles on Neuroscience, Neurotechnology and Human Rights[18], , which establishes fundamental guidelines to ensure that scientific advances respect human dignity and

18 https://www.oas.org/es/sla/cji/docs/CJI-RES_281_CII-O-23_corr1_ESP.pdf

fundamental rights.

The Declaration addresses key issues such as the protection of identity and the autonomy and privacy of neural activity, considering neural data as sensitive personal data. Moreover, it emphasises the need to obtain express and informed consent for the use of such data, promoting equality, non-discrimination and equitable access to neurotechnologies. Likewise, it highlights the importance of applying therapies exclusively for the enhancement of cognitive abilities and neurocognitive integrity.

### *The United Nations*

The United Nations Educational, Scientific and Cultural Organisation (UNESCO) has made efforts to address the ethical aspects of neurotechnology. In July 2023, it organised the International Conference on the Ethics of Neurotechnology in Paris, under the theme 'Towards an Ethical Framework for the Protection and Promotion of Human Rights and Freedoms'.[19].

In addition, UNESCO has appointed an international group of experts to prepare a new global standard on the ethics of neurotechnology[20]. This group is tasked with developing a normative framework to address the ethical, legal and social aspects of neurotechnologies, so as to guarantee that their development and application respect human rights and fundamental freedoms.

Furthermore, in August 2024, the United Nations Human Rights Council Advisory Committee presented the report 'Impact, opportunities and challenges of neurotechnology with regard to the promotion and protection of all human rights' (A/HRC/57/61[21]). The document elaborates on the ethical and human rights implications associated with the advance of neurotechnologies, emphasising the need for adequate regulation to protect mental privacy and individual autonomy.

19 https://www.unesco.org/es/ethics-neurotech

20 https://www.unesco.org/es/articles/etica-de-la-neurotecnologia-la-unesco-nombra-un-grupo-internacional-de-expertos-para-preparar-una

21 https://documents.un.org/doc/undoc/gen/g24/133/31/pdf/g2413331.pdf

## Recommendations for the Fair and Responsible Development and Deployment of Neurotechnologies

The following recommendations aim to create a fair and equitable framework for the development and use of neurotechnologies, ensuring that fundamental human rights are protected and that technological advancements in this field are made in an ethical and responsible manner.

- First, **existing legal frameworks need to be reviewed and adapted**. Before creating regulations *exnovo*, legal frameworks need to be updated to address the specific challenges of neurotechnologies without creating entirely new rights. The reinterpretation and broadening of existing Human Rights protections should allow to regulate advancements in neurotechnology effectively, guaranteeing respect for mental privacy, individual autonomy and non-discrimination. Moreover, to protect neuro-rights it is essential to create clear and accessible mechanisms that allow people to exercise their rights in the case of violations, including control procedures, sanctions and access to information on the use of their neuro-data.

- Given that the impacts of these technologies transcend borders, it is also necessary to adopt a **multi-level focus** and to promote **harmonised regulation at the international level** through cooperation forums and the development of shared principles that guide the global governance of neurotechnologies.

- Companies must take on an active role in risk mitigation and in the **implementation of ethical standards** in all stages of the development and implementation of neurotechnologies. **Security, informed consent and user privacy** must be at the heart of innovation, thus avoiding applications that might compromise individual autonomy or exacerbate existing inequalities. Likewise, it is crucial to anticipate and prevent possible misuses of neurotechnologies, such as the manipulation of mental states or unauthorised access to neural data. Implementing t**echnical safeguards and robust internal procedures** will contribute to minimising these risks.

- Ensuring **equitable access** to these technologies is essential to prevent only certain privileged sectors from benefiting from their opportunities. Companies must align their economic incentives with the health and well-being needs of the population, promoting models that favour equity in access to and distribution of their benefits. Transparency and accountability are also key: collaboration with governments, scientists and civil society organisations in the creation of regulatory standards will strengthen the development and responsible oversight of the sector.

- Civil society plays a fundamental role in **democratising the debate** on neurotechnologies, Public education and awareness about neuro-rights will enable citizens to understand the implications of these technologies and to demand appropriate regulation. Moreover, constant monitoring of the implementation of policies and laws will help ensure that regulations do not just exist on paper, but are effectively enforced. Fostering informed public debate is fundamental to ensure that decisions about the future of neurotechnologies are not left exclusively in the hands of governments and corporations, but rather reflect a pluralistic vision that respects human rights.

- Finally, we need a comprehensive and **collaborative approach** to ensure that neurotechnologies are developed and deployed in an ethical, equitable and respectful manner towards human dignity. Only through a combination of effective regulation, corporate responsibility, citizen empowerment and international cooperation can we build a future where advances in neurotechnologies benefit society as a whole without compromising fundamental rights.

# INTELLIGENCE AND ALGORITHMS: THE FUTURE OF AI IN LIGHT OF INTELLECTUAL PROPERTY

The advent of AI has challenged established paradigms of intellectual property rights. Content generation by models like ChatGPT or DALL-E is raising fundamental questions: is it possible to grant copyright to creations generated by algorithms? Where is the line between inspiration and plagiarism? This chapter examines these questions and the impact of AI in the field of intellectual property (IP). From the legal implications of using data to train algorithms to new notions of creativity and authorship, it analyses the challenges and opportunities that AI presents for copyright. Through specific examples and recent cases from Latin America and around the world, it argues for the need to establish legal frameworks that balance technological development, creators' rights, and public interest.

## The relation between AI and intellectual property

Artificial intelligence and intellectual property interact in complex ways, ranging from the use of protected data to train models to the creation of artworks generated by algorithms. In this sense, the interaction between AI and IP can be divided into three main areas: training data, automated creative production, and the protection of algorithms themselves.

AI models require large volumes of data for training, and this data is often protected by copyright. This creates significant tensions around fair use and licensing agreements. For example, platforms such as Stable Diffusion and ChatGPT are trained using texts, images, and music that are part of massive databases, which raises questions about whether these practices constitute copyright infringement. In cases such as The New York Times v. OpenAI,[22] the debate has focused on how to balancethe rights of creators with technological advancement.

22 https://www.nytimes.com/2023/12/27/business/media/new-york-times-open-ai-microsoft-lawsuit.html

In terms of creative production, AI tools can generate high-quality content that mimics artistic and literary styles. This raises questions about what can be considered as an original creation. For example, AI can replicate works in the style of renowned artists,[23] which has led to a discussion about whether these creations should be considered as derivative works or whether they are themselves protected by copyright. The paradox here is that, although these tools democratise access to creativity, they also generate a potential displacement and precariousness for human creators.

Finally, the companies that develop AI models seek to protect their algorithms through patents, trade secrets, and copyrights. This approach, however, contrasts with the need for algorithmic transparency, especially in areas such as justice or education where opacity can hinder equity and public trust. The dilemma, therefore, between protection and access has become a constant point of friction.

### Ethical and legal consequences

The increasing relevance of AI in the field of intellectual property raises a number of legal and ethical questions. One of the most pressing is the **rights of original creators**. The discussion about the originality of AI-generated work is at the heart of the legal debate. Some countries, such as the US, have made it clear that human authorship is a necessary requirement for copyright protection.[24] However, this approach does not fully resolve the challenges posed by algorithmic creations, especially in contexts where the lines between inspiration and plagiarism are blurred. In many cases, **copyright-protected works are used without authorisation** to train AI models, creating an imbalance in the intellectual property system. This phenomenon can be particularly problematic in contexts where creative industries lack the legal and financial resources to engage in international litigation.

Another critical point regards **transparency in the use of data**. The opacity of AI training practices makes it difficult to evaluate whether such practices respect copyright and generates ethical concerns about compliance with personal data protection regulations. This lack of clarity also affects the ability of creators to protect their rights and to receive fair compensation. Increasingly, local artists are facing difficulties protecting their work in international environments, as well as a lack of clarity regarding how their content is used by AI systems. Big tech companies are able to monopolise both access to this data as well as the earnings coming from their technologies, while individual creators and cultural communities are marginalised.

In the debate on intellectual property and AI, the **problem of access to information** should also be addressed. On the one hand, as mentioned, AI depends on being able to access large volumes of data, including texts, images and music, which is often protected by copyright. On the other hand, limiting access to these resources could hinder technological progress and development opportunities.

One of the central features of this discussion is the danger of adopting a maximalist approach to copyright in the regulation of AI. This implies conceiving of copyright as the main solution to problems arising from the lack of compensation for creators and the lack of transparency in the use of data. However, this approach could further consolidate the power of big tech

23 El País, Cuando el robot pinta como Rembrandt. 3 febrero de 2020: https://elpais.com/economia/2020/01/31/actualidad/1580472914_468275.html

24 https://www.copyright.gov/rulings-filings/review-board/docs/a-recent-entrance-to-paradise.pdf

companies, like OpenAI or Google, which use copyright to protect their own algorithms and maintain their dominance. In contrast, open-source models such as Stable Diffusion represent a more equitable and accessible alternative. Moreover, **copyright has historically been used as a mechanism for censorship in the digital environment**. Examples like the Digital Millenium Copyright Act, or DMCA, in the United States illustrate how 'notice and takedown' orders have led to the mass removal of content without proportionate nor adequate analyses of exceptions to copyright. Platforms like YouTube have implemented automated systems, like Content ID, to identify and administer copyright protected content. Often these systems unfairly penalise content and limit creativity and freedom of expression.

Legislative progress in the region must also address the **issue of algorithmic transparency**. The lack of specific regulations, which in many countries and regions is currently being legislated, makes it difficult to understand how AI technologies use and process data. This affects not only creators, but also civil society, which depends on transparency to assess the social and economic impacts of these technologies.

Lastly, another significant challenge regards **authorship in AI generated creations**. The difficulty of tracing the origin of certain types of content, such as memes or collective works, raises questions about who could be considered as author and how compensation might be distributed. In Guatemala, for example, las tejadoras mayas (Mayan weavers) have promoted initiatives to recognise collective rights over cultural expressions, which could serve as a model for approaching similar problems in the context of AI.

### The global and regional perspective

In general, legislative initiatives on the subject attempt to balance the promotion of innovation with the protection of fundamental rights. More specifically, legislators face the challenge of finding the right balance between protecting creators' rights and ensuring that the necessary data for the development of AI are available in a fair and ethical way. This includes exploring legal exceptions, like fair use or obligatory licencing, which can facilitate both innovation and the preservation of fundamental rights of information access.

At a global level, the European Union has led regulatory efforts with initiatives like the AI Act, which seeks to establish a clear framework for transparency and accountability in the use of AI. Although the European regulation does not include specific provisions for intellectual property, it does establish risk categories and specific requirements for systems that impact fundamental rights, including those related to intellectual property.

In Latin America, legislative developments around artificial intelligence and its relationship with intellectual property present major challenges. Unlike other regions such as Europe and North America, where regulatory initiatives have progressed more rapidly, Latin American countries face a lack of harmonisation, as well as the resources to implement robust legal frameworks. However, some countries are taking promising steps.

In Colombia, legislative discussions centre on ensuring transparency and the inclusion of ethical principles in the implementation of AI. Although there is no specific law, the government has promoted guidelines to regulate the use of AI tools in key sectors such as justice and education. These initiatives seek to protect copyright and promote the responsible use of technology. Additionally, some judges have begun using tools like ChatGPT to draft parts of court rulings. This has sparked a debate about the reliability of the generated responses and the need to ensure that the data used to train such tools

respects copyright.
In Brazil, the National Congress is debating a comprehensive AI bill that addresses transparency and accountability, as well as the protection of creators' rights. This framework also includes provisions on the use of protected data for algorithmic training. In addition, Brazil has implemented sectoral strategies that seek to balance the promotion of innovation with the protection of fundamental rights.

In Argentina, reforms to the Personal Data Protection Law are being discussed that could include regulations on the use of data in AI models. Progress is also being made on initiatives to promote transparency in algorithmic process and ensure fairness in compensation for creators.

Other countries, such as Mexico, are in the early stages of regulation. Although the Federal Copyright Law provides a basic framework, it does not yet address the specific challenges of AI. However, legislative proposals have been put forward to address the use of protected data and transparency in algorithmic processes.

These initiatives reflect a growing interest in addressing the impacts of AI on intellectual property. Nevertheless, the lack of harmonisation between countries, and in some cases, the limited resources to implement and supervise these regulations are significant barriers that must be overcome.

## Recommendations for the Sustainable Development of AI in relation to Intellectual Property

- The sustainable development of artificial intelligence regarding its interaction with intellectual property requires an approach that considers both the promotion of technological advancements and the protection of creators' rights. For this, it is crucial to **prioritise algorithmic transparency** as a guiding principle of public policy in this area. For example, requiring technology companies to share the databases they use to train their models would not only build public trust, but also permit creators to check whether their work has been used, and thus obtain fair compensation. This also includes establishing clear requirements for technology companies to disclose information about the data sources used in training AI models. This would include requiring companies to undergo independent audits to guarantee their compliance with intellectual property rules.

- The **promotion of open-source technologies** is another key pillar. Tools like Stable Diffusion have demonstrated that technological openness does not just reduce the concentration of power in the hands of a few companies, but that it also permits different communities to access and personalise these technologies to their needs. This model, in contrast to patent-protected and inaccessible solutions, democratises creative capacity and fosters diversity of perspectives.

- Furthermore, it is essential to **develop innovative models of collective compensation.** An illustrative case is the possibility of creating funds financed by big tech companies to support artistic and cultural projects. This approach could help ensuring a fairer distribution of the economic benefits deriving from AI, and promoting greater cultural inclusion and the preservation of creative local expression.

- At the same time, copyright laws must be adapted to the challenges of the 21st century and include exceptions and limitations that allow for the fair use of protected data in the training of AI models. Reforms that introduce **exceptions for educational, research, and non-commercial uses** are essential to prevent outdated regulations from restricting the advancement of AI. For example, facilitating access to public domain databases or establishing flexible licences would help balance the interests of technology creators and users. Legal frameworks should also include exceptions and limitations, such as fair use or compulsory licensing, to ensure that AI tools can be developed without disregarding ethical and legal principles.

- Finally, **education and awareness-raising** must also play a role in policy-making in this area. Legislators, creators and users alike need to understand the impact of AI on intellectual property. This can be achieved through the implementation of training programmes for legislators, creators and other key players on the challenges and opportunities presented by AI in this field. This will help to foster greater understanding and preparedness to address the rapid changes in technology. Thus, information campaigns, workshops and educational programmes can foster a cultural environment that allows for the integration of technology without ignoring ethical and legal concerns.

# REGULATORY SANDBOXES FOR AI

In recent years, the regulation of artificial intelligence (AI) has become a major issue in public policy at a global level. The need to balance digital transformation and technological innovation with the protection of fundamental rights has given place to experimental regulatory approaches. Among these, regulatory sandboxes stand out as a tool that allows regulators and private actors alike to experiment with emerging technologies in controlled environments. This chapter examines the usefulness of regulatory sandboxes in the context of AI, drawing on the dominant discussions surrounding existing policy and legislative initiatives on this regulatory mechanism.

## What is a regulatory sandbox?

> A regulatory sandbox is defined as a controlled and supervised space where companies can test new technologies, products, services or marketing models under the supervision of a regulator.

The concept of regulatory sandboxes emerged in the context of technological innovations that require testing before large-scale deployment. Originally used in sectors such as finance, this approach has been extended to other areas, such as AI development, due to its ability to manage the traditional incompatibility between innovation and regulation. Regulatory sandboxes allow companies and regulators to collaborate, which in turn facilitates the identification of the inherent risks of new technologies before their implementation at a large scale. Therefore, sandboxes allow developers to operate within a flexible regulatory framework, facilitating experimentation and the development of solutions without the immediate application of stricter regulation. This approach supports both the advancement of the technology and its effective integration with existing regulation. In addition, it allows legislators to identify problematic features in the marketing or use of a good, service or system before they can cause real and even irreversible damage.

A number of countries are experimenting with regulatory sandboxes to support AI innovation. The United Kingdom, for example, has created isolated spaces for AI in healthcare, like the Care Quality Commission AI Sandbox,[25] which is centred on AI applications for medical diagnostic services. Similarly, initiatives in countries like Colombia, Chile, Brazil and Spain have proposed the implementation of sandboxes focused on the responsible development of AI, especially for SMEs and start-ups, thus ensuring that companies can implement AI systems in accordance with the legislation that will soon be applicable in their respective countries.

### What are the objectives of a regulatory sandbox for AI?

The main objective of a regulatory sandbox is to **evaluate the risks and benefits** of technologies before their large-scale deployment in the market. In the AI context, the main aim of this approach is to analyse the ethical, social and technological impacts of AI within a regulatory framework. At the same time, it seeks to prevent technological innovations from being limited by regulations that could restrict innovation, and even ensure that AI development is aligned with regulatory requirements from the design phase. Regulatory sandboxes for AI, therefore, offer a space where technologies can be evaluated from various perspectives, such as ethics, data security, privacy and human rights. These spaces allow regulators to interact with technologies and gain a practical understanding of how they work, enabling them to develop regulations and make more informed decisions about the impacts of the technology.

### Key objectives and benefits

1

***Innovation and competition without compromising public safety***

Regulatory sandboxes help to stimulate innovation by providing a space where new AI technologies can be tested and adjusted without being immediately constrained by the full regulatory burden. This can contribute to a country's competitiveness by fostering the development of advanced technologies. For example, Spain's initiative[26] aims to ensure that AI systems developed within the sandbox comply with the regulatory requirements of the AI Act, while promoting the development of innovative AI applications.

2

***Promoting trustworthy AI***

One of the main objectives of AI sandboxes is to ensure that AI systems are developed responsibly, thereby minimising risks related to security, health, privacy and other fundamental rights. This is particularly important given AI's potential to affect fundamental rights. Through experiences gained in the sandbox, legislators and companies will be able to identify practices that guarantee that AI systems are trustworthy and adjust to human-centred approaches.

25 https://www.cqc.org.uk/sites/default/files/20200324%20CQC%20sandbox%20report_machine%20learning%20in%20diagnostic%20services.pdf

26 https://digital-strategy.ec.europa.eu/es/news/first-regulatory-sandbox-artificial-intelligence-presented.

3

***Regulatory clarity and the reduction of legal uncertainty***

By testing AI systems in isolated spaces, regulators can better understand how new technologies interact with existing legislation. This allows them to refine regulations and ensure they are fit for purpose. In addition, the sandbox environment offers regulators and developers a unique opportunity to collaborate and co-create solutions that are effective and flexible for emerging technologies.

4

***Capacity building***

Sandboxes also play a decisive role in developing the regulatory and technical capacity of authorities. For example, the AI sandbox in Spain will support the creation of a National Supervisory Authority and will contribute to the development of standards and tools for AI regulation. Such knowledge transfer is key for regulators to effectively oversee the complex and rapidly evolving AI sector.

5

***Global cooperation***

Given that AI regulation has turned into a global priority, international cooperation has become an essential component of the regulatory process. For example, Spain's AI sandbox is being developed in close collaboration with the European Commission, and the lessons learned will serve as a basis for other initiatives in other regions. This type of collaboration could lead to a certain level of regulatory harmonisation and thus avoid the problems arising from regulatory fragmentation.

## How do regulatory sandboxes support AI implementation readiness?

Regulatory sandboxes are essential tools that help public sectors and governments **prepare for the adoption and regulation of AI**. They function as collaborative spaces where key actors—politicians, regulators, technology companies and, in some cases, end users—come together around specific AI projects. This creates meeting spaces that allow different disciplines and interests to converge in one place, thus promoting a **comprehensive understanding of how the technology works and its regulatory implications,** and ensuring the necessary flexibility for its implementation within a safe and effective regulatory framework.

Within a sandbox, policymakers and regulators have the opportunity to learn directly from technology developers about the practical and technical aspects of AI. In turn, technology companies benefit by coming to understand the concerns and expectations of the public sector, which facilitates a dialogue that would otherwise be difficult to achieve. This type of interaction allows access to technology, encourages interaction with it and helps to demystify many of the social concerns associated with AI. To what extent will AI replace us? How does it work? What is needed? How can the legitimate use of personal data be guaranteed? The sandbox, therefore, becomes a space where these issues can be addressed jointly, ensuring a more informed and socially responsible adoption of technology.

In addition, the inclusive approach of regulatory sandboxes marks one of their greatest contributions to AI regulation. Used as a tool, sandboxes create **space for collaboration and mutual learning**. Sandboxes allow for the incorporation of a variety of actors beyond regulators and technical developers who do not normally participate in policymaking,

including philosophers, ethics specialists, and other specialised technical profiles. This not only responds to companies' demand for clear regulatory frameworks that allow them to operate safely, but also promotes comprehensive governance that combines legal norms with ethical codes and technical standards.

Ultimately, regulatory sandboxes can promote the development of policies that support innovation while **helping to ensure that AI implementation is aligned with human rights**. Thus, sandboxes represent an essential tool for building a regulatory environment tailored to the challenges and opportunities of AI, as they respond to both the needs of industry and the demands of society.

### Are there existing regulatory sandbox initiatives for AI?

#### *Colombia: From regulatory innovation in the financial sector to AI*

In Colombia, the implementation of regulatory sandboxes has been a key strategy for promoting technological innovation in various sectors. Initially, these testing spaces focused on the financial sector, with the Financial Superintendency of Colombia (Superintendencia Financiera de Colombia, or SFC) leading initiatives such as "laArenera"[27], a sandbox designed to evaluate innovative technological developments in financial, stock market and insurance services. This environment allowed companies to test new technologies under supervision, ensuring regulatory compliance and consumer protection.

Recognising the need to expand these spaces beyond finance, the Colombian government, through policies such as the CONPES 3975 document of 2020, proposed the creation of sandboxes in areas such as data protection, competition and transport. Colombia has attempted to move forward in this area by issuing a decree (Decree 1732 of 2021)[28] that allows regulatory sandboxes to operate in different sectors and test various standards. In the field of AI, the initiative took shape with the Superintendency of Industry and Commerce (Superintendencia de Industria y Comercio, or SIC), which launched a sandbox focused on privacy by design and by default in AI projects.[29] This space seeks to enable companies to develop AI solutions that comply with data protection regulations from the outset, promoting responsible and ethical practices in the handling of personal information. In addition, in October 2023 the Ministry of Information and Communications Technology (Ministerio de Tecnologías de la Información y las Comunicaciones, or MinTIC) announced the structuring of ten regulatory sandboxes aimed at accelerating innovation ecosystems in Colombia. These spaces will permit entrepreneurs to test innovative products, services and business models in a regulatory environment with greater flexibility, and have the aim of contributing to the country's economic growth and digital transformation.

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27 https://www.superfinanciera.gov.co/publicaciones/10114253/innovasfclaarenerasandbox-del-supervisor-10114253/

28 Decree 1732 of 2021. Regulating article 5 of Law 2069 of 2020, in relation to exploratory regulatory mechanisms for innovative business models in regulated industries and special surveillance and control environments or regulatory sandboxes, and adding Chapter 19 to Title 1, Part 2 of Book 2 of Decree 1074 of 2015.

29 For more information, see: https://www.sic.gov.co/sites/default/files/files/2021/150421%20Sandbox%20sobre%20privacidad%20desde%20el%20dise%C3%B1o%20y%20por%20defecto.pdf.

### *Chile: The path towards regulation aligned with UNESCO Recommendations*

Chilean legislation has made progress in the regulation of AI through a proposed bill that seeks to establish a regulatory framework for the development and use of AI systems in the country.[30] One interesting feature in this bill is the creation of sandboxes to facilitate the experimentation and validation of emerging technologies in a supervised environment. The bill establishes that state administrative bodies will be able to provide these sandboxes for the development of AI systems, assigning clear responsibilities to providers and establishing penalties for non-compliance with the established rules. In addition, it is envisaged that the Ministries of Science and Economy will implement measures to give smaller companies priority access to these spaces, thus promoting awareness and skills development in the use of AI systems.

However, during the public consultation process, various opinions emerged regarding the effectiveness and adequacy of the proposed regulation.[31] Some experts consider that the proposal is adequate for the functioning of sandboxes, while others express concerns regarding data management by public and private entities, as well as the role of the state in the technical implementation of these spaces. It is also suggested that the regulation applicable to companies that develop or operate AI systems in sensitive areas such as health, labour or justice be reviewed, especially those that are not based in Chile or whose servers are located abroad. Opinions are divided on the prioritisation of smaller companies in accessing sandboxes. Some consider the proposal insufficient and suggest greater contribution and participation by the state, for example, through the provision of curated data sets or technological equipment. Others consider the measure irrelevant or unnecessary, arguing that it could unfairly favour certain actors and limit the competitiveness and efficiency of public and private organisations.

### *Brazil: A sandbox that guarantees Ethical AI and Personal Data Protection*

Although there are currently no specific laws that directly regulate AI in the country, Bill No. 2,338/2023, approved in December 2024, aims to provide a dedicated regulatory framework for AI.[32] In parallel with this legislative effort, Brazil's National Data Protection Authority (ANPD) in cooperation with the CAF, has launched a regulatory sandbox pilot programme focused on artificial intelligence and personal data protection. This sandbox aims to provide a controlled environment for testing AI technologies within a regulatory framework, while ensuring compliance with data protection standards, particularly under Brazil's Data Protection Law (LGPD).

The ANPD's sandbox initiative enables collaborative experimentation and involves a variety of stakeholders, including technology and innovation companies, academics, and civil society organisations. The main objectives of the programme include promoting algorithmic transparency, fostering best practices in data management, and protecting fundamental rights, all while supporting Brazil's National AI Strategy (EBIA). The programme has been

30 Draft Bill of 7 May, available at: https://www.camara.cl/verDoc.aspx?prmID=17048&prmTIPO=INICIATIVA.

31 For further information, see R Roberts, expert consultation: Draft Bill on artificial intelligence systems (Bulletin 16821-19), July 2024, available at: https://obtienearchivo.bcn.cl/obtienearchivo?id=repositorio/10221/36220/2/Resultado_Consulta_experta_IA___BCN.pdf.

32 https://legis.senado.leg.br/sdleg-getter/documento?dm=9347622&ts=1720545987618&disposition=inline.

informed by successful examples from other data protection authorities around the world, such as the UK's ICO, Singapore's PDPC, Colombia's SIC, Norway's Datatilsynet and France's CNIL. The programme is expected to last between 18 and 24 months, beginning with a call for proposals and concluding with the publication of final reports and results.
Recently, the ANPD ended the call for contributions phase, receiving 71 submissions, mainly from private entities, civil society, academia, and individual citizens, and thus reflecting strong interest and commitment at both the national and international levels. With this sandbox, Brazil aims to improve its regulatory approach to AI and build a framework that fosters responsible innovation while respecting ethical and data privacy standards.

### *Spain: The pilot project for European AI regulation*

The sandbox offers a framework to evaluate AI systems in critical sectors such as biometric identification, education, employment, public services, migration management, and justice administration. European legislation considers the use and development of AI systems in these sectors as high risk because they have a high social impact and may pose significant risks to health, safety, or fundamental rights. In addition to high-risk systems, the Royal Decree[33] covers the participation of general-purpose AI models and foundational models (e.g., text or image generation systems) in the sandbox, which have broad applicability in the market. The project has a limited duration and will close 36 months after its launch, or when the European AI Act comes into full force. This deadline means that the sandbox can be used as a temporary preparatory measure, rather than a permanent regulatory system.

Participation in the sandbox is voluntary and open to both suppliers and users established in Spain who meet criteria for innovation, social impact, and technical maturity. Participants are required to conduct a self-assessment of compliance and submit the results to the State for a more thorough evaluation. Once admitted to the sandbox, participants must follow the preliminary guidelines and technical specifications provided by the Secretary of State for Digitalisation and Artificial Intelligence. These guidelines are aligned with the standards set out in European regulations and may be updated during the trial period. Participants must demonstrate that their systems comply with the requirements, including with regards to transparency, ethics and fundamental rights. At the end of the evaluation process, participants must carry out a self-assessment of compliance, which is only valid within the controlled environment of the sandbox. This self-assessment does not replace any mandatory compliance assessment that may be required at a later stage under specific sectoral regulations.

Once tests are completed, it is expected that a report will be compiled based on experiences gained from the project. This report will contain conclusions, best practices, and technical recommendations, and will serve as a guide for other participants and as a reference for the implementation of the AI Act across the Europea Union.

33 Royal Decree 817/2023, 8 November, establishes a controlled testing environment for testing in compliance with the proposal for regulation made by the European Parliament and Council which lays down harmonised standards in the field of artificial intelligence. BOE no. 268, 9 November 2023, pp. 149138 - 149168. BOE-A-2023-22767. https://www.boe.es/eli/es/rd/2023/11/08/817.

### What are the major challenges in the development of AI regulatory sandboxes?

1. **Legal certainty and regulatory capture**. The application of the concept of experimentation in law, and flexibility in the application of rules, naturally raises some difficulties. Nonetheless, a key question arises: what exactly is granted in a sandbox? There is no single model; it may be a licence, a waiver or a special permit. The aim of a sandbox is to allow compliance with standards in a different way, as authorised by the regulation itself. On the other hand, one of the recurring problems in any regulatory process is the possibility that actors with greater resources, especially large companies, may capture the regulatory process in their favour.

2. **Design and institutional coordination.** The institutional organisation and design around a sandbox is a constant challenge. Who leads the sandbox? How are different entities involved in it coordinated? Experience shows that this is not usually the responsibility of a single authority. Several entities are involved: technology authorities, public innovation policy makers and regulatory authorities. Establishing an effective leadership and coordination structure is therefore essential for the success of a sandbox. However, although coordination between institutions should be a smooth process, this is not always the case. Conflicts, tensions and different perceptions of the same issue may arise between the various entities involved. In Colombia, for example, the Presidency coordinated various entities and ministries to work together on the development of a sandbox through an advisory body for digital transformation. This coordination facilitated and accelerated implementation, even attracting other actors, such as the Development Bank of Latin America (CAF), which contributed specialist knowledge. However, in other countries the lack of such coordination becomes an obstacle, and even if ministries promote initiatives, collaboration with the authorities can be difficult. The involvement of an entity that acts as coordinator and facilitator can be key, although it is also necessary to respect the autonomy of each authority.

3. **Advanced innovation and resource constraints.** For sandboxes to be truly effective, they must test technologies that present significant regulatory challenges. This ensures that the efforts of public entities and other actors are focused on technologies that really justify the process. Sandboxes, therefore, must have sufficient resources to ensure that advanced technologies can be tested. For example, in countries that are not innovation hubs or that have limited access to tech companies, attracting this type of technology becomes a challenge. As a result, developing countries often face difficulties in attracting cutting-edge technology companies or acquiring the necessary expertise to manage these testing environments effectively.

4. **Human Rights.** The necessary considerations for implementing new developments in AI must be precisely defined so that public bodies can exercise appropriate oversight and control. There is a solid foundation for these processes to function properly, but additional clarity is needed in certain areas to optimise oversight processes and guarantee effective protection of human rights in the context of regulatory sandboxes.

## Recommendations for the Development and Implementation of Regulatory Sandboxes for AI

- **Establishing clear criteria for legal certainty**
  This involves clearly specifying the types of permits, exemptions and regulatory exceptions that may apply, and ensuring that these procedures do not compromise fundamental rights. Transparency in legal criteria will reduce the risk of regulatory capture and ensure a fair and equitable environment for experimentation. Furthermore, given that sandboxes are a temporary tool for experimentation, we recommend establishing specific time frames for their operation, with periodic reviews. This will allow objectives to be adapted and will ensure that sandboxes fulfil their purpose.

- **Developing an institutional coordination strategy**
  Sandboxes should be backed by a well-defined governance structure that facilitates cooperation between involved entities. The designation of a coordinating entity to oversee the operation of sandboxes and ensure smooth collaboration between technology, regulatory, and public policy authorities is recommended. This may include the creation of inter-agency committees and the collaboration with international organisations and development banks to provide resources and expertise.

- **Encouraging high-impact innovation and allocating adequate resources**
  Sandboxes should prioritise experimentation with advanced AI technologies that pose significant regulatory challenges. This will require the creation of specific incentives to attract innovative technology companies, as well as funds to finance the infrastructure needed to manage these testing environments. For countries with limited resources, it is advisable to seek support from international organisations and partnerships with the private sector to increase the viability of sandboxes.

- **Incorporating the protection of human rights into the design of sandboxes**
  Sandboxes should include specific monitoring and evaluation protocols to protect human rights and personal data. This implies establishing privacy, transparency, and non-discrimination guidelines that are integrated into the design and testing phases of the technology, as well as the formation of ethics committees to monitor respect for fundamental rights during the experimentation process.

- **Creating spaces for learning and training in AI regulation**
  Regulatory sandboxes should include a training component for both regulators and developers. Regulators should receive training on the technical aspects of AI, while technology companies should come to understand the regulatory requirements and the rights they must protect. This will foster mutual understanding and better prepare stakeholders for future regulatory implementations.

- **Promoting international collaboration**
  The regulation of AI is a global priority. Therefore, international collaboration in the implementation of regulatory sandboxes is crucial. Sandboxes should include a component for cooperation with other jurisdictios and participate in international AI regulatory forums. This could foster the harmonisation of standards, the reduction of fragmentation and the sharing of best practices

- **Implementing mechanisms for the evaluation and publication of results**
  Sandboxes must conclude with outcome reports that include best practices, key learnings, and recommendations for future policy. These reports must be accessible to the public and stakeholders so that they can also serve as reference resources for other jurisdictions wishing to implement regulatory sandboxes for AI.

# AI DATA PROTECTION, AND PROCESSES OF REGIONAL INTEGRATION

The protection of personal data has become a central concern in the context of the digital revolution. Since the emergence of the Internet and the proliferation of digital platforms, both citizens and regulators have identified the risks associated with the misuse of personal information in the virtual space. With the rise of artificial intelligence (AI), the protection of personal data has become even more important. Given that AI systems rely on large volumes of data to function effectively, the challenges associated with privacy and the protection of personal information have intensified.

In this context, regulatory harmonisation stands as a key objective for the protection of citizens' rights at a global level. A prime example is the European Union's General Data Protection Regulation (GDPR), which establishes clear and strict standards for the handling of personal information belonging to European citizens or those located within EU territory. In Latin America, although there is no equivalent regional regulatory framework, some countries have made significant progress in updating their laws or designing new regulatory frameworks, while others still face challenges establishing robust standards for personal data protection.

## What is personal data?

Personal data is defined as any information that can be used to directly or indirectly identify a person. This definition covers a wide range of data, from basic information such as name, address or telephone number to more sensitive data such as sexual orientation, religious beliefs or biometric data. The key is that any data that can be linked to a natural person must be protected under privacy regulations.

## AI and personal data

AI poses new and important challenges for the protection of personal data. AI systems rely on large volumes of data, both personal and non-personal, to train their algorithmic models. The larger and more diverse the

databases that feed an algorithm, the better its results will be. Massive processing involves not only basic data such as names or addresses, but also special categories of personal data, such as biometric data, health data or behaviour patterns, which increases exposure to risks of misuse or unauthorised access.

AI also introduces specific risks that go beyond traditional privacy concerns. Algorithmic biases, for example, can perpetuate inequalities by relying on incomplete or unrepresentative data sets. Furthermore, the lack of transparency inherent in some AI systems, known as 'black boxes,' makes it difficult to understand what data has fed into automated decisions and how these decisions affect individuals.

An emerging concept that seeks to address the challenges associated with the use of data in digital environments and technologies such as AI is that of dynamic-informed consent. This approach recognises that the conditions under which data is collected and used may change over time, necessitating a more flexible model tailored to user needs. Data collected for one purpose today may be reused in the future for completely different purposes, often without the knowledge or explicit approval of users. This change requires a regulatory framework that allows individuals to continuously update and manage their preferences regarding how their data is used.

An example of the new challenges for personal data protection posed by AI was provided by tech giant Meta, when in June 2024 it announced a new privacy policy that generated significant controversy. Under this policy, the company planned to use public content from users on its platforms to train generative AI models. Although the company claimed that it would only process public information, there was widespread criticism of the lack of transparency and the complexity of the informed consent process. In Europe, Meta was forced to suspend the implementation of its new privacy policy due to its incompatibility with European law. In Brazil, the National Data Protection Authority (ANDP) initially also demanded that Meta suspend its new privacy policy but later lifted the measure after reaching an agreement that introduced clearer mechanisms to ensure informed consent from users.[34]

### EU General Data Protection Regulation

The European Union's General Data Protection Regulation (GDPR) came into force in 2016 and establishes a set of fundamental principles for the processing of personal data. Among other requirements, it highlights the need for data to be processed fairly, lawfully and transparently. This means that companies and organisations must inform users about how their data is collected, stored and used, ensuring that the process is clear and accessible.

One of the key principles of European regulations is that of data minimisation, which states that only data that is strictly necessary for the specific purpose for which it is being used should be collected. In addition, data must be accurate, up to date and kept for no longer than is necessary for the purposes for which it was collected.

Data protection by design and by default are key concepts in the GDPR, implying that data

34 https://www.eleconomista.com.mx/tecnologia/Meta-informara-a-los-brasilenos-sobre-como-utiliza-sus-datos-personales-para-entrenar-la-IA-20240903-0132.html

protection must be a priority from the initial stages of in the development of products and services.

- Data protection by design: companies must consider the protection of privacy and data security when designing their processes, systems and products. From the outset, they must implement security measures and ensure that data is processed securely and in accordance with the law.
- Data protection by default: companies must adopt default settings that maximise the protection of the privacy of the data subject. For example, in the case of cookies, platforms must configure their systems so that users must give their explicit consent before their data is collected or processed. Default settings should not permit the use of personal data without the user's consent.

Although the GDPR is the most advanced legislation on personal data protection and has become a global standard, the regulation has not been without criticism. Civil society organisations and supervisory agencies have pointed out its lack of effectiveness against the power of big tech companies and the difficulties faced in its application, among other limitations, which has resulted in uneven implementation across Europe.[35]

**The limits of informed consent**

A key aspect of personal data protection is informed consent. Informed consent means that the individual is aware of and understands the data processing activities and their purpose when giving consent to the processing of their personal data.

Currently, the business model of many major digital platforms is based on the collection, storage and mass processing of personal data for commercial purposes. One of the main criticisms levelled at this model is the lack of genuine consent from users when they provide their data.

In many cases, companies adopt an opt-out approach so that users can refuse to receive personalised advertising, rather than an opt-in approach that lets users actively decide whether they want their data to be used in this way. In fact, the way consent is presented on many platforms is designed to make the user accept the terms and conditions without fully understanding how their data will be used.

Furthermore, the concept of legitimate interest is often invoked – and abused – by companies as a legal basis for legitimately obtaining and processing personal data without the need to obtain explicit consent. Yet, in many cases, the legitimate interest invoked by companies is not aligned with the interests and rights of users.

## Personal data protection in Latin America

Unlike the European model, in Latin America personal data protection lacks a unified regional regulatory framework. The regulation of this matter falls under the responsibility of the States, which has led to a variety of approaches and different levels of progress at the regional level. Brazil has set an important precedent with the enactment of its General

35 Gentile, G., & Lynskey, O. (2022). Deficient by design? The transnational enforcement of the GDPR. International & Comparative Law Quarterly, 71(4), 799-830.

**Data Protection Law** (LGPD) in 2018. Countries such as Argentina, Chile, Colombia and Costa Rica have personal data protection laws which, although establishing essential principles such as the right to privacy and transparency in data processing, need to be updated to effectively address the challenges posed by emerging technologies. In Bolivia and Paraguay, the processes to enact initial legislation on the matter have fluctuated back and forth, and in countries such as Honduras and El Salvador, the debate is not yet on the legislative agenda.[36]

The GDPR has had an enormous influence in Latin America. Since it came into force, most Latin American countries have sought to enact new laws or adapt existing ones to comply with European personal data protection requirements in order to facilitate international trade and cooperation. International data transfers have become one of the key issues in this area, given the cross-border nature of the digital ecosystem and growing global interconnection. Currently, the EU recognises Argentina and Uruguay as the only two countries in the region with adequate legislation for international transfers of personal data.

While compliance with European regulations is extremely important, when considering data protection at the local level, Latin American countries should bear in mind that adopting rules inspired by foreign models may prove ineffective if they are not adapted to the specific characteristics of the region, such as varying levels of institutional weakness and persistent levels of inequality. Therefore, it is essential to design regulatory frameworks that respect local forms of data governance and do not end up reproducing structures that reinforce existing gaps.

Rather than focusing exclusively on enacting new laws, countries should aim to guarantee the effective implementation of existing rules by adopting proactive measures that promote trust in the handling of personal data. In many cases, existing laws with appropriate adjustments, may already provide answers to current challenges. For example, transparency and the duty to inform are recognised principles in several Latin American laws, but in practice their implementation is often insufficient or even non-existent.

### The advantages and challenges of harmonisation

A key priority for Latin America must be moving towards regulatory harmonisation within the region. The lack of a unified framework creates obstacles for cooperation and hinders the implementation of shared standards that respond to common challenges in personal data protection. Having a unified regulation within the region would not only strengthen trust among Latin American countries, but also allow for greater integration in joint initiatives, especially in contexts that require technological interoperability and coordination in the handling of cross-border data.
The harmonisation of personal data protection in Latin America has made some significant progress over the last decade. The Ibero-American Data Protection Network (RIPD) has promoted the Ibero-American Standards[37], a series of common principles and rights for personal data protection, with the aim of moving towards the harmonisation of regulations at the regional level. Similarly, the 2021 MERCOSUR Electronic Commerce Agreement[38] mentions the need to establish common measures for the protection of personal data and to take into consideration relevant international instruments. In addition, Argentina,

36 https://www.accessnow.org/press-release/meta-extraer-datos-personales-en-america-latina/
37 https://www.redipd.org/documentos/estandares-iberoamericanos-2017
38 https://normas.mercosur.int/public/normativas/4018

Uruguay and Mexico have acceded to Council of Europe Convention[39], which reinforces their commitment to international standards. However, despite these advances, disparities between national regulatory frameworks are hindering progress towards real integration.
Regional integration in data protection offers significant strategic advantages, especially in the context of the digital revolution and the advance of emerging technologies. From an international law perspective, one of the core values of integration is legal harmonisation. In the case of personal data protection and AI, legal harmonisation can translate into a strengthened internal market that fosters both technological innovation and the effective protection of citizens' rights.

Indeed, cooperation and greater integration in this area would enable Latin American countries to develop a regulatory environment more conducive to the digital economy. It would also strengthen their negotiating position vis-à-vis extra-regional actors, such as large tech companies, and in other international markets, promoting standards that balance economic growth with ethics and responsibility. Regulatory fragmentation weakens countries' abilities to demand fair conditions for the processing of personal data or the provision of digital services on an individual basis. In contrast, a region with consistent and robust legislation is better placed to impose standards that reflect its own priorities and to protect its interests in global negotiations.

39 https://rm.coe.int/16806c1abd

## Recommendations for Data Protection in the Context of Regional Integration

- **Fundamental principles for data protection**. Data minimisation must be a central standard in public policy and in companies' internal policies, which means limiting the collection of personal information to what is strictly necessary and establishing clear time limits for its retention. **Transparency** is another key principle: users must be fully aware of how their data is used, the purposes of processing and the consequences of its use in AI models, even when the opaque nature of certain algorithms makes this difficult. Furthermore, **privacy by design and by default** must become standard practice, ensuring that emerging technologies incorporate protective measures from the outset and offer default settings that prioritise user safety.

- Beyond general principles, risk **management and proactive responsibility** must be at the heart of any regulatory strategy. To ensure fair and reliable systems, it is necessary to establish rigorous controls over the data used in AI training models, constantly review the validity of their inferences, and detect potential biases.

- **Internal, external and independent audits** must be a recurring practice, as they allow for the continuous monitoring of the impact of AI systems on people's rights. Furthermore, given that cybersecurity remains a critical vulnerability in many countries in Latin America, regulatory frameworks must urgently address the protection of their digital infrastructure against potential breaches and threats.

- At the regional level, regulatory harmonisation remains a major challenge in the face of increasingly cross-border data flows. While it is important to draw inspiration from international standards, regulations must be adapted to the social, economic and cultural specificities of Latin America. The diversity and inequalities within the region demand collaborative approaches that permit the development of common frameworks without losing sight of the needs and capacities of each country. Moving towards agreements that strengthen cooperation on AI and data protection will enable the region to position itself strategically in the global digital ecosystem.

- Innovation and local technological developme nt must also be promoted through regulations that balance the protection of rights with the opportunities offered by technological advances. In this regard, it is important to avoid overly rigid regulation and opt for flexible and adaptable frameworks that can evolve alongside technological advances.

- Finally, training and awareness are essential to guarantee the effective implementation of any regulatory framework. In order to make informed decisions, those involved in the formulation, oversight, and enforcement of these regulations must have a thorough understanding of the digital ecosystem and emerging technologies. Likewise, the development of awareness programmes on personal data protection can help reduce information gaps and foster a citizenry that is well-informed about its rights.

# Fields of Application

PART TWO

# THE USE OF AI IN THE JUDICIAL SYSTEM

The use of technology offers various advantages for judicial decision-making, such as improving the efficiency of legal proceedings by automating tasks which have been performed manually for centuries, like document reviews and legal research. Digital transformation and the more recent use of AI-enabled tools in judicial administration can reduce costs and facilitate access to justice by automating administrative tasks, analysing cases and identifying relevant legal precedents. Furthermore, the use of AI in justice has the potential to facilitate decision-making that is, in principle, more informed and consistent with existing legal interpretation and case law, and that is free from certain biases and human error. In this way, AI has the potential to transform the legal system and improve access to justice.

However, despite these potential advantages, there are also significant ethical, technical, and legal concerns. For example, when algorithms are opaque, difficult to understand, or poorly designed, they can perpetuate or even amplify existing biases and discrimination, which can undermine public trust in the institutions that use them. This has an impact on individuals' rights, particularly with regard to procedural safeguards. Thus, it is important to focus on how procedural principles and other fundamental rights can be translated into the development of algorithms.

**AI in the judicial context**

Applying AI to the judicial system involves the use of technologies that are capable of analysing large volumes of legal data, identifying patterns in previous decisions and assisting in administrative and procedural tasks. These technologies, which include machine learning algorithms, large language models (LLM) and predictive analytics, can be designed to respond to the demands of judicial systems that are often overburdened and lack sufficient resources.

A prime example is the use of large language models, such as GPT-4, which can process and synthesise legal information to support the drafting of judgments or rulings. These

tools make it possible, for example, to identify relevant case law in seconds, something that could take hours or days for a human judicial operator. In addition, the automation of administrative processes optimises the time of judges, prosecutors and judicial staff, freeing them from repetitive and mechanical tasks so that they can concentrate on the most complex and strategic issues in each case.

AI is also proving useful in predictive tasks, helping to forecast possible case outcomes based on historical patterns. However, the use of AI for judicial decision-making is not without controversy, especially when it comes to decisions that affect fundamental rights. In the United States, for example, there have been reports of lawyers submitting legal documents that included references to non-existent rulings generated by AI-powered chatbots, which highlights the need for rigorous human oversight in the use of these technologies.[1]

## What are the policy benefits associated with AI?

Public policy objectives regarding AI in the judicial system focus on three main areas: increasing efficiency, democratising access to justice, and ensuring transparency and accountability. However, these objectives are deeply interconnected with a series of critical challenges.

### Increased efficiency

AI has the potential to become a key tool for addressing saturated judicial systems, especially overloaded courts and tribunals that suffer from structural delays in the resolution of cases. For example, in Colombia, in 2021, there were two million cases pending, with an average of 12 years for decisions. The use of AI can radically change this situation.

### The democratisation of access to justice

AI also has the potential to reduce economic and administrative barriers, allowing citizens to interact with the judicial system in a more direct and efficient way. In Brazil, for example, digital tools allow citizens to file lawsuits and seek legal advice without the need for intermediaries. In regions where physical distances make access to courts difficult, these digital tools represent a significant advance. However, democratisation is not without risks. Algorithms could exclude certain groups if they are not developed with cultural and social diversity in mind. UNESCO has signalled that the inclusive design of these tools is key to preventing the perpetuation of existing inequalities.

### Guaranteeing transparency and accountability

AI systems must be designed so that their decisions are understandable and auditable. The lack of clarity regarding the internal processes of AI systems, often referred to as 'black boxes,' is a major barrier to their acceptance by citizens, which explains the need to prioritise the development of explainable algorithms. Systems must be able to justify how and why a specific decision was reached. This is particularly critical in cases where fundamental rights are affected. For example, in the United States, the use of the COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) system has generated controversy due to its alleged racial bias. This algorithm, used to assess defendants' risk of recidivism, has been criticised for assigning higher risk scores to African Americans than to white defendants, even when their backgrounds were similar. These differences not only reflect biases in the historical

1 UNESCO Guidelines for the Use of AI Systems in Courts and Tribunals. UNESCO [71854]. CI/DIT/2024/GL/01 CI/DIT/2024/GL/01. Available at: https://unesdoc.unesco.org/ark:/48223/pf0000390781_spa.

data used to train the system, but also perpetuate structural inequalities that affect impartiality in judicial decision-making. For this reason, both UNESCO and the European Union have emphasised the need to develop technologies that are explainable and meet high standards of oversight.

## Experiences in Latin America

### Prometea in Argentina

Prometea is an AI tool developed in the Autonomous City of Buenos Aires that seeks to transform task automation in the judicial system. This initiative, initially implemented in the Public Prosecutor's Office of the Autonomous City of Buenos Aires, aims to increase efficiency, reduce procedural times, and improve the quality of judicial rulings. Prometea was designed based on multidisciplinary principles and with a focus on data governance, taking into account the reality of judicial systems in Latin America and the inherent limitations of the region, such as the lack of technological standardisation and inequalities in available resources.[2]

Prometea uses natural language models and machine learning algorithms to automate repetitive tasks within the judicial system, such as the drafting of rulings. The system ingests large volumes of structured and unstructured legal data, learns patterns from this information, and generates draft documents, which are subsequently reviewed and validated by judicial professionals. This approach seeks to allow judges and prosecutors to spend more time on complex decisions and less time on administrative tasks.

A key feature of Prometea is its ability to make predictions based on previous cases. This includes identifying relevant case law and suggesting decisions based on historical patterns. However, this process is complemented by rigorous human review, thereby ensuring that final decisions respect fundamental principles of justice and fairness.

One of Prometea's core pillars is data governance, understood here as a comprehensive data management strategy that spans the collection, organization, and optimized use of information in judicial decision-making. In a region like Latin America, where judicial systems often lack an adequate technological infrastructure, Prometea was designed to operate in environments with specific limitations. These range from the lack of the digitalisation of court files to the absence of specific regulations governing the use of AI in the judicial domain.

### AI in the Uruguayan judicial system

In Uruguay, experiences vary considerably between jurisdictions. Some judicial operators, especially those with manageable workloads, are more cautious about adopting AI, while those facing a collapse in case management see these technologies as a critical solution. This polarisation reflects a common reality in Latin America: the disparity in resources and capacities between jurisdictions. Likewise, it highlights a recurring theme, which is the need to adapt technology to local contexts. For instance, small courts dealing with

2 Estevez, E. C., Linares, S., & Fillottrani, P. (2020). Prometea: Transformando la administración de justicia con herramientas de inteligencia artificial. Available at: https://publications.iadb.org/es/publications/spanish/viewer/PROMETEA-Transformando-la-administracion-de-justicia-con-herramientas-de-inteligencia-artificial.pdf.

specific issues may benefit from simpler, more tailored solutions, whereas national courts face broader challenges such as ensuring interoperability and managing large-scale data. These challenges also include the need for standardized systems and the integration of diverse data sources to support consistent and fair decision-making.

In this context, discussions about 'black boxes' in AI are particularly relevant. Advanced algorithms, such as generative models, pose challenges not only in terms of explainability, but also in terms of the cost and feasibility of auditing their processes. As an expert in AI implementation in Uruguay pointed out, the real challenge is not just understanding how the algorithm works, but also recognising how much it costs to ensure that it will work in an ethical and effective way.

### The use of ChatGPT in Colombia

The Colombian judicial system faces critical overload, with more than two million pending cases and an average case resolution time of 12 years. The lack of personnel, budget, and technological resources exacerbates this situation. This scenario has led to the exploration of technological solutions, including AI, to improve efficiency and reduce resolution times. Since 2017, pilot projects and training programmes for judges have been developed, reaching more than 3,000 judicial operators. The approach to digitalise the Colombian justice system has been incremental and has prioritised transparency and legality in the use of these technologies, especially AI. Its practical application, however, has been limited due to high implementation costs, challenges in data structuring, and the need for specialised knowledge.

Still, in 2023, the Colombian case drew significant attention when two Colombian judges publicly documented the use of ChatGPT to draft parts of judicial decisions. In one case, a judge used the tool to verify the consistency of his decision in a legal ruling, while in the other, a magistrate used ChatGPT to structure a cybersecurity protocol for a hearing in the metaverse. Although these uses were limited to complementary functions, the cases generated criticism on social media and exposed the challenges of relying on generative models that can produce 'hallucinations'.

These cases highlighted how ChatGPT, as a multipurpose tool, is not specifically optimised for the legal context. Indeed, its use in the legal context carries significant risks, such as generating non-existent legal references, which can increase judges' workloads rather than reduce them. One specific example involved a judge who spent a day verifying an 'agreement' that turned out to be completely fictitious, causing frustration and mistrust in the tool.

## Main challenges

1. **The lack of transparency in algorithms.** Algorithmic opacity remains a critical barrier. AI systems, especially generative models, are complex and difficult to audit. In the judicial sphere, where the explainability of decisions is key to ensuring fairness, this lack of clarity can undermine trust in decisions. In various parts of the world, the use of algorithmic risk assessment tools has been questioned for their inability to justify the reoffending probabilities assigned to suspects, as in the famous COMPAS case, which in turn has generated controversy over possible racial biases that can lead to

algorithmic discrimination. As a result, AI pilot projects are integrating specific audits to evaluate the internal processes of algorithms. These examples highlight the need to implement clear standards for explainability and accountability in judicial systems that use AI.

2

**The persistence of algorithmic biases and their impact on public trust.** AI models trained with historical data often perpetuate pre-existing inequalities. In contexts where data reflects past discrimination, AI can reinforce those patterns, disproportionately affecting vulnerable groups. In 2023, Fundación Vía Libre of Argentina conducted an exercise with secondary school students in Montevideo, using AI to predict their future careers. The results showed that AI exhibited clear gender biases, suggesting that girls would become mothers rather than professionals, while boys were assigned more diverse professional roles. The perception that AI systems are 'black boxes' not only puts human rights at risk, but can also erode public trust. In the judicial context, errors or biases not only damage the credibility of the technology, but also that of the judicial system as a whole.

3

**Insufficient training for judicial operators.** Due to the rapid pace of tech development, many judges and judicial operators still lack the technical training necessary to supervise or use AI tools. This limits their ability to interact critically with these technologies, while potentially increasing dependence on automated systems. A practical example is the case of Colombia, where initial training on the use of AI in the judicial system reached more than 3,000 judges, but was limited to basic aspects, thus leaving significant gaps in technical knowledge. Similarly, the implementation of predictive algorithms in public safety showed how a lack of technical knowledge on the part of operators can lead to automated decisions with high levels of bias.

4

**Implementation and maintenance costs.** The development and integration of AI tools tailored to the judicial field requires considerable resources and investment in terms of technological infrastructure, specialised knowledge and data organisation. In countries with budget constraints this is a major obstacle to adoption, especially in overburdened judicial systems. Added to this is the cost of ensuring that systems operate ethically and effectively. For example, auditing complex algorithms such as advanced generative models requires specialists, specific tools and time, which significantly increases costs. In this regard, understanding how parameters are adjusted in these models or ensuring that decisions are not biased means substantial technical and financial effort. Furthermore, as mentioned above, the existence of errors in implemented systems can erode public trust. This adds an intangible yet critical cost: that of restoring trust, in addition to the expense associated with correcting errors.

5

**Automation vs. customisation**

Another emerging challenge in the implementation of AI in the judicial system is the difference between different local judicial systems in terms of needs and capabilities. In small or specialised jurisdictions, where cases often have unique or specific characteristics, the implementation of AI faces different barriers than in more centralised or large-scale systems. In local systems, technological tools must be highly

adaptable to specific problems, which increases the complexity of system design and deployment. For example, an AI system in a rural jurisdiction may need to be integrated with traditional procedures that are not digitised, while in a national court, the challenge lies in managing large volumes of data efficiently and transparently.

6

**Insufficient regulation.** The lack of specific regulatory frameworks creates uncertainty about the ethical and legal use of AI. Without specific regulations, AI systems could be implemented without adequate safeguards, exposing the judicial system to unnecessary risks.

## Recommendations for the Ethical Implementation of AI in the Judicial System

The adoption of AI in Latin American judicial systems presents a unique opportunity to modernise and optimise processes. However, its success depends on the adoption of a holistic approach that addresses technical and ethical challenges, as well as on collaboration between multiple actors. Transparency, continuous training, and adequate regulation are essential pillars if we are to guarantee that these technologies not only improve efficiency but also strengthen trust in the judicial system.

- The development of **ethical regulatory frameworks** that are specific to the local context is a key priority. This includes establishing regulations that address the classification of risks associated with AI systems, defining mandatory criteria to ensure transparency, accountability and ethical impact assessments. These regulations must also be dynamic and must be periodically reviewed in order to be adaptable to technological advances. They must also clearly define the responsibilities of developers, operators and regulators, and they must establish legal mechanisms to address potential failures or misuse of the technology. AI models must also be adapted to local contexts through techniques such as fine-tuning with specific legal data. This improves the accuracy of the tools and reduces the risks associated with multipurpose models. Latin American countries should opt for specific solutions that reflect their legal frameworks and judicial needs.
- Another crucial aspect is **data governance**. The data used in judicial systems must be representative, reliable and free from bias. This requires massive efforts to digitalise justice administration, as demonstrated in Brazil and Argentina, where the digitalisation of case files has enabled the creation of more robust databases for training predictive tools. In addition, strict data protection measures must be implemented to ensure that the privacy and confidentiality rights of individuals involved in judicial proceedings are respected and protected. In addition to personal data protection, the creation of representative and bias-free datasets is also necessary to ensure the success of AI systems in the judicial system.
- The **technical and ethical training of judicial operators** cannot be underestimated. Training programmes must range from a basic understanding of how algorithms work to their critical supervision. Judges, prosecutors and other professionals must be equipped to interact critically with AI tools. This involves not only understanding how algorithms work at a technical level but also having the capacity to assess their limitations and ethical risks. In this regard, comprehensive educational programmes are essential to ensure that these operators can effectively supervise and use these technologies. For example, practical workshops that integrate simulations with AI tools can demonstrate how they work in real life and help judges to understand their limitations. In addition, collaborations with academic institutions, such as those carried out in Colombia with local universities can be an effective strategy for integrating technology into judicial practices in an incremental manner.
- The **design and use of more efficient algorithms**, such as the One-Shot training strategy, which allows an AI model to learn from a single example or 'prompt', or algorithms based on synthetic data, can transform current paradigms, reducing both dependence on large volumes of data and the costs.

- The **implementation of pilot projects with a gradual approach** is another key strategy. AI systems must be subject to ongoing technical and ethical audits to identify and mitigate biases, ensure transparency, and verify their effectiveness. This includes simulations to analyse the impact of automated decisions in different scenarios through pilot phases of AI implementation. In Latin America, these audits could be combined with judicial observatories that continuously evaluate the performance of technologies. Starting with specific areas allows for the impact of technology to be evaluated in a controlled manner before it is extended to the entire judicial system. These projects must include regular audits and constant review to ensure that algorithms are accurate and aligned with local needs.

- Finally, **transparency and explainability in AI systems** are essential to maintaining public trust. In this regard, it is essential to develop specific rules for AI in justice, based on principles of transparency, fairness, and accountability. These regulations must include requirements for the certification and auditing of AI systems to ensure that developers detail how the algorithms were trained and what data they used. Algorithms must also be designed to justify their decisions in a way that is understandable to judges, lawyers and the parties involved. In addition, human oversight over all automated decisions that affect fundamental rights is essential, so that algorithmic analysis is complemented by critical human judgement. This hybrid approach would not only improve trust in technology, but could also ensure greater accuracy and legitimacy in outcomes.

# AI AND THE FUTURE (AND PRESENT) OF WORK

The impact of AI on jobs and the workplace is no longer science fiction or solely a concern for technologists. In factories, offices, hospitals and digital platforms, among other spaces, AI is reshaping how work is done, who does it and under what conditions. While some tasks are being automated and disappearing, new job opportunities are also emerging that require skills that many workers do not yet possess. In Latin America, this transformation is particularly challenging, as high levels of informality, the digital divide and the lack of specific regulation leave millions of people vulnerable to automation.

Beyond the fear of job displacement, the key question is how to ensure that AI will become an asset for improving working conditions and expanding access to quality jobs. What implementation models are working in the region? How can public policy, education, and trade unions influence a more equitable labour transition? This chapter explores how AI is reshaping the Latin American labour market, identifying both its risks and the strategies needed to prevent it from becoming another factor of precarious work.

**How is AI integrated into the labour market?**

The implementation of AI in the workplace is being driven by a combination of many rapid technological advances as well as by the growing need to adapt work processes to an increasingly competitive global context. As in other contexts, AI is being integrated into work through tools that automate repetitive tasks and optimise complex processes. Throughout history, technological development has enabled productivity improvements and cost reductions. In sectors such as industry and agriculture, there is a growing use of algorithms to analyse data, predict patterns and make more efficient operational decisions. However, the impact of this transformation is not equally distributed, and many communities face exclusion due to a lack of access to digital technologies and/or the skills required to interact with them.

At the same time, the labour market is not homogeneous, and AI, therefore, acts differently depending on the context in which it is deployed. In Latin America, for example, while AI has created opportunities for jobs related to data management and technological development in some emerging industries, its integration has highlighted and accentuated existing inequalities in sectors with a high incidence of unregulated employment. The lack of technological infrastructure and adequate training programmes has intensified this imbalance, making it difficult for large sectors of the population to participate in the opportunities brought about by automation.

Despite these inequalities, there have also been success stories in the region that show how AI can be used as a tool to improve working conditions. For example, applications have been developed that support small farmers when making decisions about crops, based on climate and soil data analysis. These initiatives not only increase productivity but also help mitigate the risks associated with climate instability. However, it should be noted that while these types of solutions highlight the inclusive and adaptive potential of AI, this is so only as long as it is deployed with sensitivity to the local context.

Similar to other areas, the advance of AI in the workplace poses not only technical challenges, but also ethical and regulatory issues. Growing automation has led many people to worry about job security and the economic sustainability of those who perform tasks that are susceptible to automation. The implementation of general-purpose AI, such as generative models, adds a layer of complexity to the situation, as these technologies are not always designed for the specific uses assigned to them in the workplace.

**Public policy objectives and case studies from Latin America**

In Latin America, public policy objectives related to AI in the workplace are beginning to take shape, with a focus on responding to the region's specific challenges, such as informal labour, the digital divide and socioeconomic inequality. Although the development of these policies is still in its infancy compared to other regions of the world, several countries have taken significant steps towards integrating AI and giving it a greater role in their national technological development and digital transformation strategies. These initiatives not only seek to drive technological modernisation, but also to ensure that its implementation promotes inclusive and sustainable economic development, with a focus on workforce training and the creation of opportunities in different productive sectors.

One of the main public policy objectives in the region is to promote connectivity and reduce the digital divide – an essential foundation for the effective integration of AI in the workplace. In Colombia, the National Development Plan 2022-2026 includes as a priority the expansion of connectivity in rural areas through initiatives such as 'Conexión Total' (Total Connection), which seeks to guarantee high-speed internet access in rural and hard-to-reach communities. This type of initiatives, like 'Internet para Todos' (Internet for All) in Peru, has made it possible to connect previously excluded communities. Such strategies not only create the necessary conditions for the use of AI-based tools, but also allow more workers, especially in informal sectors, to access training opportunities and jobs in the digital economy.

Another key objective is to strengthen job skills through training programmes in technological and digital skills. In Mexico, the 'Jóvenes Construyendo el Futuro' (Youth Building the Future) programme has integrated technology training modules that include AI-related skills. This

programme enables young people between the ages of 18 and 29 to receive training in sectors such as advanced manufacturing and data analysis—areas closely related to AI. In addition, this initiative promotes the inclusion of young people at risk of exclusion from an increasingly technologically advanced labour market.

In the area of the formalisation of employment, several governments have implemented policies to use AI as a tool to integrate informal workers into more structured systems. In Argentina, for example, the 'Argentina Programa 4.0' programme not only seeks to provide free training in technological skills, but also uses AI systems to analyse labour market demands and link participants with specific opportunities in sectors such as software development and the technology industry. This approach seeks to reduce informal labour and narrow the gap between market needs and the availability of skilled talent.

In Brazil, the Brazilian Strategy for Artificial Intelligence (EBAI) aims to strengthen digital infrastructure and promote research and development in AI. This plan includes the creation of innovation centres to work on the application of AI in sectors that are critical to the Brazilian economy, such as health and agriculture. For example, initiatives such as those that use AI to monitor and optimise agricultural performance have helped small producers to improve their productivity and have demonstrated how AI can have a direct positive impact.

## The main challenges in implementing AI in the labour market

1. **The digital divide and unequal access.** One of the most pressing challenges in the region is the unequal access to digital technologies and infrastructure. Rural areas and low-income communities lack access to high-quality internet, technological devices and/or basic digital skills, which limits their ability to benefit from the opportunities offered by AI. This problem excludes certain sectors of the population from the digital labour market and perpetuates inequality between urban and rural workers.

2. **Informal employment and the lack of social protection.** In Latin America, a large proportion of workers operate outside the legal and regulatory framework of the labour market. This hinders the effective implementation of AI-based solutions. Informal workers, who lack formal employment contracts and access to social benefits, are at a disadvantage when it comes to technological advances. The lack of specific regulatory frameworks to integrate AI into this sector exacerbates job insecurity and limits opportunities for labour formalisation through technology.

3. **Ethical risks and algorithmic biases.** The use of AI in the labour market poses serious ethical risks, particularly with regard to privacy and discrimination. Similar to other sectors, the use of AI in the labour market can reproduce and amplify pre-existing biases in data, perpetuating structural inequalities. For example, AI-based tools used in recruitment processes can discriminate against certain groups due to implicit biases in the algorithms. Also, algorithms for performance evaluation or shift allocation may contain biases that negatively affect certain groups of workers compared to others. For example, in companies such as Amazon, automated productivity assessment systems monitor the work pace of employees in distribution centres and can generate warnings or even automatic dismissals if certain performance levels are not met.

These algorithms can disproportionately affect older or disabled workers, as they do not consider factors such as physical limitations or necessary breaks for health reasons and thus penalise them compared to younger or physically fitter employees. Another case has been identified in delivery and transport platforms, where algorithms that assign shifts or orders may discriminate against certain workers based on previous activity patterns. This can occur in companies such as Uber Eats or Rappi, where AI systems are suspected of prioritising delivery drivers who have accepted more orders at difficult times or worked longer hours. This can disadvantage workers who need greater flexibility for family or health reasons. This lack of consideration for worker diversity in the allocation of job opportunities demonstrates the importance of monitoring and regulating these systems to prevent AI from reinforcing pre-existing inequalities in the labour market. This problem is particularly worrying in contexts with significant cultural diversity and historical inequalities.

4

**Job displacement and insufficient reskilling.** Although AI has the potential to create new jobs, it also poses the risk of job displacement in sectors that rely on repetitive and automatable tasks. In Latin America, many workers lack access to adequate training programmes to acquire the skills needed in a technologically advanced labour market. Reskilling and upskilling remains a critical challenge, as without significant investment in education and training, existing labour inequalities could deepen.

5

**Implementation costs.** Implementing AI solutions in key labour sectors can be prohibitive given the high costs associated with developing advanced technologies, training workers and the creating the necessary digital infrastructure. This is particularly relevant in contexts where financial and technical resources are limited, such as in some Latin American economies.

6

**Technological dependence and the concentration of power.** Another important challenge is the growing dependence on technologies developed and controlled by large tech companies. This not only raises issues of technological sovereignty, but also creates a concentration of power that limits the capacity of countries in the region to adapt AI solutions to their local contexts. Furthermore, this dependence increases the risk that the implemented solutions will not be culturally relevant or ethically appropriate.

7

**Lack of multidisciplinary skills.** The implementation of AI requires collaboration between disciplines such as technology, sociology, economics, philosophy, and politics. However, in many countries in the region, labour and educational institutions are not prepared to foster this integration. This limits the ability to design and implement AI solutions tailored to the specific labour challenges of Latin America.

## Recommendations for the Ethical and Sustainable Automation of the Labour Market

The integration of AI into the labour market in Latin America simultaneously represents an opportunity and a challenge, and requires comprehensive solutions tailored to the regional context. While its capacity to transform and modernise processes is undeniable, its deployment must be carefully managed to ensure that the benefits of AI are not concentrated exclusively in certain sectors or countries.This balance requires a joint effort between governments, companies and communities to ensure that AI not only optimises productivity but also contributes to social and economic equity in the region.

- **Reducing the digital divide** should remain a central objective of public policy. Unequal access to technology and digital infrastructure remains a key barrier in Latin America, that stops people from taking advantage of AI, especially in rural areas and marginalised communities. It is necessary to continue strengthening programmes that expand access to the Internet and technological devices. The replication and expansion of projects to increase connectivity in rural areas and to ensure access to basic technologies as part of national AI strategies, together with subsidies targeted at small entrepreneurs and informal workers, could facilitate more inclusive integration into the digital labour market.

- Another key objective is **training and retraining**. As automation replaces certain tasks, it is essential to ensure that workers have the tools they need to adapt to the demands of a changing labour market. In this regard, Mexico has developed technology training programmes focused on sectors such as manufacturing and logistics that promote specialisation in skills related to programming, data analysis and automated systems maintenance. These initiatives not only seek to mitigate the impact of unemployment, but also to foster economic resilience in communities dependent on traditional jobs. In addition, public-private partnerships could focus on developing specialised courses on AI tool management, data analysis and programming, thus enabling workers to effectively transition to sectors with high technology demands.

- Dependence on foreign technology solutions limits the ability of countries to adapt AI to their local contexts. It is necessary to **promote the development of local technology through tax incentives, financing for start-ups, and collaborations with universities**. Through its National AI Plan, Brazil is already moving in this direction by prioritising innovation in key sectors such as agriculture and health, demonstrating the potential of policies aimed at technological sovereignty.

- AI can play a key role in **the transition of informal workers to more structured employment schemes**. In Colombia, AI-based digital platforms have begun to link independent workers with formal jobs. These tools should be expanded and integrated with financial and social security services that offer benefits such as insurance and micro loans, thereby incentivising the formalisation of labour and improving workers' quality of life. For example, in the agricultural sector, platforms such as Agtech use AI algorithms to analyse climate and soil data, optimising crop decisions. These tools are particularly valuable for small farmers who face uncertain weather conditions and limited resources. In countries such as Brazil and Argentina, these solutions have helped increase agricultural productivity and reduce resource waste.

- In practical terms, case studies of AI in Latin America demonstrate its potential to address local challenges and improve working conditions. But for this to happen, it is essential to implement protective measures for workers who lose their jobs due to automation. **Social safety nets that include access to free training, temporary subsidies, and transitional employment programmes** are essential to mitigate the impact of job displacement. In addition, conducting regular studies on the impact of AI in different industries will allow for proactive and effective policy adjustments. At the same time, it is crucial to create independent bodies to oversee the use of AI in work processes, audit algorithms and ensure the protection of workers' rights, especially in vulnerable sectors.

- **Reforming education programmes** to include technical, social and ethical approaches will allow the training of professionals prepared to address the complex challenges of the digital labour market. Chile's National AI Policy recognises this need and offers a model to follow by prioritising collaboration between disciplines in the development of technological solutions. Similarly, regional cooperation funds could facilitate the exchange of experiences and the adaptation of good practices to different contexts, which in turn would promote inclusive and sustainable development.

- Last but not least, **trade unions** have a key role to play in regulating the impact of AI on the region's labour market. Their role should focus on the negotiation of collective agreements that establish guarantees for workers in the face of automation. These agreements must include provisions on transparency in the use of algorithms for labour decision-making, protection against algorithmic discrimination, mandatory trainings and retrainings, and mechanisms to ensure that technology is used to improve working conditions, rather than to replace jobs without viable alternatives. They must also promote the **oversight of AI at work**, by demanding audits and mechanisms to challenge automated decisions that may affect labour rights. In addition to their traditional role of defending rights, **trade unions must modernise to adapt to the digital transformation**. This involves actively participating in the design of public policies on AI and employment, promoting the inclusion of labour protection measures in technology regulations, and adopting digital tools that strengthen their advocacy and communication with workers.

# AI AND EDUCATION: WHERE ARE WE HEADED??

Classrooms are no longer what they used to be. With the arrival of AI, education is undergoing a radical transformation that goes beyond access to technology. We are experiencing a redefinition of the way we learn, teach and assess knowledge. From systems that personalise teaching to tools that automate administrative processes, AI is becoming a key player in the education ecosystem. However, its implementation is not without challenges, especially in regions such as Latin America where inequalities in access to technology and a lack of teacher training can turn AI into a factor that widens existing gaps.

This chapter analyses how AI is reshaping the education sector by exploring its promises and risks. Can AI improve equity in education or will it end up benefiting only those who already have access to advanced technological resources? How can we prevent algorithmic tools from reinforcing biases in learning? How can education systems integrate AI without losing sight of the development of fundamental skills such as critical thinking and creativity?

This analysis seeks to shed light on the future of education in the age of AI, and on the strategies needed to harness its potential without compromising the **fundamental values of learning.**

## The Impact of AI on Educational Assessment and Knowledge Certification

*The automation and customisation of assessments*

The use of AI in educational assessment has optimised assessment and feedback processes by providing more detailed student performance analyses.

Advanced algorithms can evaluate multiple-choice answers and open-ended questions, identifying learning patterns and providing personalised recommendations in real time. Algorithmic assessment systems can adjust the difficulty level of questions based on student performance, which permits a more accurate

measurement of their skills and knowledge. This approach has been implemented in some online learning platforms in the region and provides more accurate diagnostics by avoiding standardised tests that may not adequately reflect students' abilities.

At the same time, one of the most promising uses of AI in education is personalised learning. Through data analysis, advanced algorithms can identify specific needs of students and can adapt content and teaching methods to their pace and learning style in real time. This is particularly useful in highly heterogeneous environments, such as the education systems in Latin America, where there are significant inequalities between schools, and where even classrooms are often composed of students with different levels of knowledge. For example, in Brazil, the AI-based platform *Geekie*[3], has proven to be an effective tool for preparing students for national exams. By providing personalised recommendations, it helps students to improve in areas where they have most difficulties. These solutions can also help reduce dropout rates through engaging students further in their learning process.

However, the use of AI tools to personalise the educational experience also poses risks. As in other sectors, in the world of education there is the risk that algorithms could reinforce pre-existing inequalities if they are not developed following an inclusive approach. Thus, if the training data is biased or does not reflect cultural and social diversity, recommendations could favour certain groups over others. In this context, it is crucial to design algorithms that take local particularities into account and to regularly audited those algorithms to ensure fairness. The impact of AI on cultural and intellectual diversity should also be considered. In the following section, we discuss the impact of AI on cognitive abilities and, in particular, on critical thinking.

*Automated Monitoring and Risks of Bias*

AI-based proctoring systems have become more relevant with the expansion of online education, especially since the COVID-19 pandemic. Protoring tools monitor exams remotely by analysing facial expressions, eye movements and even writing patterns to detect irregularities. Universities and other higher education centers have implemented these types of technologies to ensure academic integrity in off-class assessments. However, the application of these tools poses challenges in terms of privacy and discrimination. There have been reports of cases in which these algorithms have been biased and have disproportionately affected certain groups of students. Technical problems, such as errors in facial recognition or restrictions in the detection of behaviour patterns can lead to unjustified sanctions. In addition, contextual factors such as connectivity problems or environmental conditions can interfere with results, which mainly affects students in vulnerable contexts.

*The Certification of Knowledge and the Validation of Skills*

With the growth of online education, validating knowledge acquired outside traditional settings has become a priority. AI and technologies such as blockchain are facilitating the issuance of digital credentials, which allow skills to be certified in a decentralised and secure manner. These credentials can be used to verify the authenticity of academic degrees and professional certifications while reducing dependence on traditional accreditation mechanisms. For example, in some countries educational institutions and technology companies have begun using digital credentials that certify specific skills acquired through online courses or practical experiences. This approach recognises self-taught learning and

3 https://www.geekie.com.br/

work-based training, facilitating the integration of workers into technological and digital sectors. These initiatives can help reduce barriers to labour market access for people who have developed skills outside the formal education system, thereby promoting greater inclusion in technological and digital sectors.

### Critical Thinking and Analytical Skills in the Era of AI

AI is very efficient on routine and low-cognitive tasks, but it is important to ensure that its use is not scaled to a point preventing students form developing critical and analytical skills in a world dominated by technology. The ability to question, interpret and synthesise information will be essential in a future saturated with data.

The UNESCO report on AI and education educación,[4] highlights the importance of integrating critical thinking into school curricula. A successful example of this philosophy is the Ceibal programme 'Pensamiento Computacional e Inteligencia Artificial' (Computational Thinking and Artificial Intelligence)[5] in Uruguay, which combines the use of AI-based learning platforms with activities that encourage reflection and debate. With more than 90,000 students enrolled to date and implemented in over 1,300 schools (more than 400 of them rural), this programme is an example of how such initiatives help balance technology with human development.

However, there are also worrying examples. In some cases, the excessive use of technological tools has led to a decline in students' ability to solve complex problems autonomously. This highlights the need to use AI as a complement to rather than a replacement for traditional learning.

### Ethical, Social and Cultural Implications

Despite its transformative potential, AI also risks exacerbating educational inequalities, especially in regions where access to technology remains limited. For example, while some urban schools in Argentina and Chile have begun to adopt innovative tools, many institutions in rural areas lack the basic infrastructure to implement advanced solutions.

Efforts to bridge the digital divide are therefore a critical first step. A case in point is Colombia, where the government has promoted the use of digital platforms to improve access to education through the 'Colombia Aprende' programme.[6] In Peru, the 'Aprendo en Casa' initiative,[7] launched during the COVID-19 pandemic and also present in seven other Latin American countries, is another example of how technology can expand access to education.

A recent report by Uruguay's National Public Education Administration (ANEP) points at significant limitations in the implementation of AI, including lack of teachers' training and skills in the use of these tools, unequal access to devices and connectivity, and the risks of excessive technological dependence. These barriers are particularly visible in rural areas and among disadvantaged communities, which further reinforces the need for public policies that guarantee equity.

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4 Holmes, W., Hui, Z., Miao, F., & Ronghuai, H. (2021). *Inteligencia artificial y educación: Guía para las personas a cargo de formular políticas*. UNESCO Publishing.

5 https://pensamientocomputacional.ceibal.edu.uy/inteligencia-artificial/.

6 https://www.colombiaaprende.edu.co/.

7 https://aprendoencasa.org/.

More broadly, the use of AI in education also raises ethical and legal dilemmas. Who controls the data generated by students? How is their privacy guaranteed? These questions are particularly relevant in countries where data protection regulations tend to be limited or non-existent. In this regard, experts have pointed out that digital tools often collect sensitive data without a clear framework for protection, which could violate students' rights. The creation of regulations governing the use of AI in education is an essential step to avoid these risks.

Culturally, it is also important that AI tools reflect and respect the diversity of local contexts. For example, in Bolivia, educational initiatives that incorporate AI have begun to include content in indigenous languages such as Quechua and Aymara, thus promoting the incorporation of technology that promotes a more inclusive education.

## Recommendations for the future of education in the age of AI

Looking ahead, the role of AI in education will depend largely on the decisions made today. In Latin America, this means prioritising equity, inclusion, and adaptability in the design and implementation of these tools. To ensure that AI has a positive impact on education, it is essential that we adopt a collaborative approach that involves governments, educational institutions, communities, and the private sector. Here are some key recommendations:

- The development of AI in education must focus on **accessibility** and must guarantee that no community is left behind. To this end, it is crucial that to strengthen technological infrastructure in remote and disadvantaged regions in order to ensure internet connectivity and access to electronic devices in rural schools and vulnerable communities.

- **Teacher training** must be a fundamental pillar in the implementation of AI. Teachers must understand how to apply these technologies to complement learning without relying on them exclusively. In addition, the design of ongoing training programmes that integrate the use of AI tools into teaching methods will not only enable educators to improve their teaching practice but will also foster the development of critical and analytical thinking among students. For example, specific workshops could address how to balance the use of AI with traditional teaching approaches.

- As in other sectors, it is necessary to establish **clear regulatory frameworks governing the use of personal data in education**. Transparency in the collection and handling of student data is crucial to protect privacy and to build trust in AI-based tools. For example, implementing independent audits to ensure compliance with data protection regulations could be a key step.

- The creation of **inclusive and contextualised educational content** must be a priority. This includes the development of platforms that reflect the cultural and linguistic diversity of each country. Inclusivity must also be reflected in the incorporation of content in indigenous languages and with the use of materials that are adapted to local contexts.

- **Inequalities between educational modalities must be minimised**. AI tools should not reinforce the divide between students with access to in-person education enriched by technology and those limited to basic digital education. To this end, programmes combining technology with meaningful human interactions can ensure an equitable educational experience for all.

- Promoting **public-private partnerships** between governments, technology companies and non-governmental organisations to finance AI-based educational projects would ensure their sustainable and effective implementation. These collaborations could also stimulate the creation of technological tools adapted to local needs and reduce dependence on imported solutions that do not always reflect local contexts.

- Finally, continuous evaluation and monitoring are essential to ensure the success of AI in education. Supervision and impact assessment systems are crucial to identify areas for improvement in the integration of AI in education. In addition, the creation of educational observatories to analyse the impact of these technologies would allow strategies to be adjusted according to the identified needs. This includes conducting pilot tests in specific communities before extending their use nationwide.

PART THREE

# Expert Views

# Gender Biases in Artificial Intelligence: Why Should it Be a Concern in the Public Sector?

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Several years ago, authorities from the Ministry for Early Childhood in the province of Salta, Argentina, considered that an algorithmic system would be a valuable tool for preventing teenage pregnancy and school dropout rates. After working in collaboration with Microsoft, the Technological Platform for Social Intervention ("Plataforma Tecnológica de Intervención Social") was launched in 2018, bringing with it an official promise: with just a name and address, it would be possible to predict, five or six years in advance, whether a girl or teenager was at high risk of becoming pregnant as a teenager.[1] This would enable preventive measures to be taken that were targeted at those who really needed them.

But how could this level of accuracy be possible? The answer was through an algorithmic system that would analyse data from 12,000 girls and teenargers aged 10 to 19, including their neighbourhood of residence, ethnicity, country of origin, family educational level, the number of people in the household, the presence of any disability, and the availability of certain public services.[2] According to some media reports, visits were made to the homes of these girls and young women to gather information on the ground, take photos and geolocate their residences, mainly in areas affected by poverty and with a high presence of indigenous people and migrants.[3]

This is an example of how the use of algorithmic systems can have negative consequences for certain population groups if the potential risks are not carefully studied before their implementation. In this specific case, the data and constructs behind the system contribute to perpetuating stereotypes about women, especially those living in poor areas and in indigenous and migrant communities: it is assumed that girls and young women are solely responsible for teenage pregnancies and, furthermore, those who find themselves in a situation of structural inequality are stigmatised.[4] Although the idea that algorithmic systems bring significant benefits to society—such as efficiency, effectiveness and accuracy—is becoming increasingly widespread, it is important to view these developments with a critical eye, asking questions such as: who is losing out in the race for the advancement of disruptive technologies? And, what is the role of the public sector in addressing these

1 Sternik, I. (2018). La inteligencia que no piensa. *Página 12*. https://www.pagina12.com.ar/109080-la-inteligencia-que-no-piensa

2 Peña, P. y Varón, J. (2020). *Decolonising AI: A transfeminist approach to data and social justice*. https://medium.com/codingrights/decolonising-ai-a-transfeminist-approach-to-data-and-social-justice-a5e52a-c72a96

3 Jemio, D., Hagerty, A., Aranda, F. (2022). El Caso del Inquietante Algoritmo Que 'Predice' el Embarazo Adolescente. *Wired*. https://www.wired.com/story/argentina-algoritmo-embarazada-prediccion/

4 Peña, P. y Varón, J. (2020). *Decolonising AI: A transfeminist approach to data and social justice*

important challenges?

## Gender Biases in AI

The analysis of gender in algorithmic and AI systems is part of a broader debate involving socio-technical approaches and feminist theories of technology.[5] Broadly speaking, these perspectives suggest that technology and society are mutually constructed: it is impossible to separate social and political issues from technological advances. This creates a vicious circle: values are incorporated into technological development and in turn, technology reflects, reinforces, perpetuates or transforms those values.[6] In this sense technology is not neutral, and regarding our study, nor is it gender neutral.

Considering this starting point, gender biases related to AI can be identified in two areas: data and design.[7] In the case of data, information may be incomplete. For example, when an AI system is fed data available on the internet it is likely that this data is not representative of the entire population, due, among other things, to the digital divide. In terms of design, some studies have highlighted that certain AI systems have characteristics that correspond to roles traditionally associated with women. For example, virtual assistants such as Siri, Alexa and Cortana have female names, and their responses and interactions help reinforce the stereotype that links women to supporting roles[8]. The same is true of chatbots[9] and some types of robots.[10] Some image generators, such as DALL-E 2 and Midjourney, also offer results that reflect existing gender inequalities: when asked for an image of a CEO or an influential person, they mainly show white men.[11]

## The Impact of AI on Women

What consequences could the implementation of AI systems have on the real lives of women and people who do not fit into the model of traditional masculinity if gender bias is not taken into account? Although analyses have been carried out on the implications of AI in certain areas, such as employment and women's working lives[12], here we will specifically focus on the possible impacts from the point of view of public policy and public services.

First, certain groups may be excluded from accessing public services. Here, we are not only referring to data and the digital divide but also to specific systems such as facial

5 Lagesen, V. (2015). Gender and Technology: From exclusion to inclusion? In J. Wright (Ed.), *International Encyclopedia of the Social & Behavioral Sciences*. Elsevier, pp. 723-728.

6 Balmaceda, T., Pedace, K., Lawler, D., Pérez, D., y Zeller, M. (2021). *Pensar la tecnología digital con perspectiva de género*. https://proyectoguia.lat/wp-content/uploads/2022/06/perspectiva-generoV6.pdf

7 Ortiz de Zárate, L. y Guevara Gómez, A. (2021). *Inteligencia artificial e igualdad de género. Un análisis comparado entre la UE, Suecia y España*. https://fundacionalternativas.org/publicaciones/inteligencia-artificial-e-igualdad-de-genero-un-analisis-comparado-entre-la-ue-suecia-y-espana/

8 West, M., Kraut, R., y Ei Chew, H. (2019). I'd blush if I could. Closing gender divides in digital skills through education. https://en.unesco.org/Id-blush-if-I-could

9 Feine, J., Gnewuch, U., Morana, S., & Maedche, A. (2020). Gender Bias in Chatbot Design. In Følstad, A., Araujo, T., Papadopoulos, S., Lai-Chong Law, E., Granmo, O., Luger, E., Bae Brandtzaeg, P. (Eds.) *Chatbot Research and Design* (11970:79-93). Lecture Notes in Computer Science. Cham: Springer International Publishing, doi: 10.1007/978-3-030-39540-7_6.

10 Manasi, A., Panchanadeswaran, S., Sours, E., y Lee, S. J. (2022). Mirroring the bias: gender and artificial intelligence. *Gender, Technology and Development*, *26*(3), 295–305. https://doi.org/10.1080/09718524.2022.2128254

11 Maslej, N., Fattorini, L., Brynjolfsson, E., Etchemendy, J., Ligett, K., Lyons, T., Manyika, J., Ngo, H. Niebles, J.C., Parli, V., Shoham, Y., Wald, R., Clark, J., y Perrault, R. (2023). *The AI Index 2023 Annual Report*. https://aiindex.stanford.edu/wp-content/uploads/2023/04/HAI_AI-Index-Report_2023.pdf

12 Collett, C., Neff, G., & Gouvea Gomes, L. (2022). *Los efectos de la IA en la vida laboral de las mujeres*. https://publications.iadb.org/es/los-efectos-de-la-ia-en-la-vida-laboral-de-las-mujeres

recognition software. In a well-known study, two researchers[13] analysed three gender classification systems and found that the results were more accurate with white male faces than with black female faces. It has also been highlighted how these systems may be even less suitable for recognising transgender and non-binary people.[14] In presence of a high margin of error in these types of systems, the rights of women and other groups may be affected, especially when they are used for identity verification for accessing public services, or for monitoring at airports or on roads, as is the case in Brazil.[15]

The case of the province of Salta is another example of how the use of stereotypical constructs in AI systems can lead to stigmatisation and biased decision-making by public administrations. If the algorithmic system is based on the premise that the area of residence and conditions of poverty or migration are considered variables of risk for teenage pregnancy, the results will reflect a biased worldview. In turn, public policies should consider this non-discriminatory perspectives as well as the opportunity to design holistic solutions that address the problem with genuine respect for people's rights.

Something similar has occurred in the case of VioGén, an algorithmic system developed by the Spanish Ministry of Interior that assesses the risk of gender-based violence facing women who have already suffered it. An external audit found that the model on which the risk analysis is based gives higher scores to women with children and/or foreign women, which means that they may receive greater protection from the authorities.[16] While the inclusion of these variables is based on studies on the subject, the fact is that women without children, for example, have in many cases been left without adequate protection. In this regard, it is essential to thoroughly evaluate the effects that the use of such tools can have when addressing such sensitive situations.

As with public policies in general, the solutions proposed by AI systems are framed within certain representations of the problems.[17] However, in these cases, it is even more important to look critically at the models on which technological development is based in order to avoid a deeper and more rapid perpetuation of stereotypes. Governments and public administrations have, in this sense, a dual responsibility: to regulate AI systems developed and implemented in the private sector, and to ensure that disruptive technologies deployed within their own structures respect the principles and values of democracies, including equality and non-discrimination.

### AI with a Feminist Perspective

To date, not only have technical measures been proposed to mitigate gender and other types of bias, but ideas to incorporate the gender perspective throughout the entire AI life cycle have also been put forward.[18] For example, it has been proposed that audits and tests be carried out during the

13 Buolamwini, J., y Gebru, T. (2018). Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification. *Proceedings of Machine Learning Research*, 81, 1-15.

14 Keyes, O. (2018). The Misgendering Machines: Trans/HCI Implications of Automatic Gender Recognition. *Proceedings of the ACM on Human-Computer Interaction*, *2*(CSCW), 1–22. https://doi.org/10.1145/3274357

15 Silva, M. R., y Varón, J. (2021). *Reconhecimento facial no setor público e identidades trans. Tecnopolíticas de controle e ameaça à diversidade de gênero em suas interseccionalidades de raça, classe e território.* https://codingrights.org/docs/rec-facial-id-trans.pdf

16 Eticas. (2022). *The External Audit of the VioGén System*. https://eticasfoundation.org/wp-content/uploads/2022/03/ETICAS-FND-The-External-Audit-of-the-VioGen-System.pdf

17 Bacchi, C. (2009). *Analysing Policy: What's the problem represented to be?* Pearson.

18 UNESCO. (2020). *Artificial Intelligence and Gender Equality. Key findings of UNESCO's Global Dialogue*. https://unesdoc.unesco.org/ark:/48223/pf0000374174

design, implementation and evaluation of these systems to identify potential biases and risks of exclusion.[19] The need to promote the presence of women and other historically underrepresented groups, as well as gender specialists, during all phases of the development, adoption and implementation of these systems has also been raised.

From a critical point of view, some authors[20] abogan por incorporar los principios advocate the incorporation of feminist principles in a cross-sectional way throughout the design and deployment process of disruptive technologies. This means thinking about the power relations behind the development of AI, looking more closely at who wins and who loses with certain AI systems, sharpening our critical gaze to locate possible gender exclusions in unexpected places, considering whether an AI system is really the most appropriate solution in certain contexts, and moving towards more collaborative approaches. To move forward with many of these measures, it is also essential to promote the transparency and explainability of AI systems so that their functioning—even if only in broad terms—and the consequences of their implementation can be understood.[21]

Considering these dynamics in the framework of the public sector, it is advisable to analyse each specific case, taking into account the impact of AI on the public policy cycle itself[22] and the gender dimension in each of these phases. In this regard, it is important to make use of gender mainstreaming strategies, which specifically promote the 'integration of a gender perspective into the preparation, design, implementation, monitoring and evaluation of policies, regulatory measures and spending programmes, with a view to promoting equality between women and men, and combating discrimination'.[23] This is a long-term commitment that not only requires a proactive attitude towards incorporating this dimension, but also, in some cases, addressing individual or institutional resistance that may arise throughout the process.[24]

## Final Reflections

With the increasingly widespread use of algorithmic and AI systems in the private and public sectors, it is essential that we approach their design and implementation with a critical eye. While these disruptive technologies offer a number of advantages for decision making, there are also challenges for the exercise of rights and for access to public services on equal terms. In this specific case, gender biases, which may be present in both the data and the design of these systems, may contribute to the perpetuation of stereotypes that harm women and certain historically discriminated population groups.

In this regard, it is indispensable that there is a shared effort between companies, the public sector, social organisations and citizens to address the ethical and political challenges of AI development and implementation in

19 UNESCO. (2020). *Artificial Intelligence and Gender Equality*.

20 Peña, P. y Varón, J. (2021). *Inteligencia Artificial opresiva: categorías feministas para entender sus efectos políticos*. https://notmy.ai/es/sem-categoria-es/inteligencia-artificial-opresiva-categorias-feministas-para-entender-sus-efectos-politicos/

21 Collett, C., Neff, G., & Gouvea Gomes, L. (2022). *Los efectos de la IA en la vida laboral de las mujeres*.

22 Valle-Cruz, D., Criado, J. I., Sandoval-Almazán, R., y Ruvalcaba-Gomez, E. A. (2020). Assessing the public policy-cycle framework in the age of artificial intelligence: From agenda-setting to policy evaluation. *Government Information Quarterly*, *37*(4), 101509. https://doi.org/10.1016/j.giq.2020.101509

23 Instituto Europeo para la Igualdad de Género (s.f.). *What is gender mainstreaming*. https://eige.europa.eu/gender-mainstreaming/what-is-gender-mainstreaming

24 Lombardo, E., y Mergaert, L. (2013). Gender Mainstreaming and Resistance to Gender Training: A Framework for Studying Implementation. *NORA - Nordic Journal of Feminist and Gender Research*, *21*(4), 296–311. https://doi.org/10.1080/08038740.2013.851115

contemporary societies. While it is true that each actor must take steps in this direction, the public sector has an critical role to play. Governments and public administrations have a responsibility to demand minimum standards from AI developers—something that is already beginning to happen in some national AI strategies and regulations, and to set precedents for the implementation of algorithmic and AI systems that respect human rights and democratic values.

# Surveillance Technologies in Ibero-America

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## Introduction

During the first two decades of the 21st century, the development, convergence and integration of the Internet of Things (IoT), big data, robotics, artificial intelligence and even neurotechnology have led, among other things, to an signficant surge of mass social surveillance technologies. Since November 2022, this situation has been greatly accelerated by the emergence and exponential development of generative AI ('GenAI'). This has led to an unprecedented revolution in surveillance technologies, the impacts and consequences of which are being felt strongly in modern societies.

It is important to note that, in recent times, there was a proliferation of mass social surveillance systems used to monitor compliance with social distancing measures during the COVID-19 to mitigate the spread of the virus. In addition, there have been numerous attempts to use AI in traffic management and public transport in cities to optimise mobility and road safety.

However, the extensive use of these tools has also generated ethical concerns, particularly regarding the privacy of citizens. Despite their effectiveness, it has been argued that digital surveillance measures require a balance between innovation and responsible behaviour to avoid violations of personal privacy and ensure the protection of civil rights.

According to Feldstein, more than 40% of countries worldwide use AI-related technologies for mass social surveillance based on image classification, facial recognition, video analysis or voice identification, creating biometric systems for human facial analysis.[1] It should be emphasised that within this group of countries, there are more democratic governments than authoritarian regimes.[2] The creation of smart or safe cities with thousands of sensors and cameras that transfer data and images in real time is another example of the use of AI-related technologies. Similarly, the application of smart policies in areas such as justice, crime prosecution and prediction, and the monitoring of social platforms to analyse and control online communications are additional examples of its use.

In the current global ecosystem of social surveillance technologies, it is crucial to recognise that China, through its major technology corporations, including ZTE, Huawei, Hikvision and Dahua, is the leading global supplier of tools that enable other governments and public institutions to carry out surveillance practices at all levels. At the

1 Feldstein, S. (2019). *The Global Expansion of AI Surveillance*. Carnegie Endowment for International Peace. https://carnegieendowment.org/files/WP-Feldstein-AISurveillance_final1.pdf

2 51% of advanced democracies, 41% of competitive autocratic states, 41% of illiberal democracies and 37% of closed autocratic states employ mass social surveillance systems (Feldstein, 2019).

same time, it should not be overlooked that there is a powerful group of large US digital corporations dedicated to this field, such as Amazon, IBM, Cisco, Palantir, etc., as well as prominent companies from third countries such as NEC (Japan) and NSO Group (Israel).

**Types of Surveillance and Analysis of Ibero-American Cases**

To understand the diversity of surveillance tools and practices, we can group them into three broad categories:

**Specific government hacking or espionage.** This includes all types of software or devices that allow remote or direct access to phones, computers and other communication systems. These tools can be used to spy on specific targets, from political actors to journalists or social leaders, without their knowledge or consent. The best-known example is the spyware developed by the Israeli company NSO Group and commercially distributed under the name Pegasus.

**Mass surveillance and biometric data collection.** These practices entail the systematic identification and continuous tracking of large groups of people. They includes traffic cameras, image analysis systems in stadiums and shopping centres, facial recognition software and other technologies that continuously record sensitive information. As these technologies are increasingly integrated into out everyday lives, they raise serious concerns because of their potential impact on fundamental rights such as privacy andfreedom of expression. The best-known examples here are monitoring and surveillance systems currently in place in many cities.

**Integration of official databases.** This involves aggregating data and various government records, such as medical records, criminal records and social security records, to create detailed profiles of the population. In some cases, this integration extends to cross-border collaboration between security forces, thus increasing the scope and complexity of surveillance systems. The most notorious example is the Chinese social credit system, which Venezuela has attempted to replicate with its 'Carnet de la Patria'.

All three types of technological surveillance pose serious challenges to democratic governance and share concerning issues regarding transparency, accountability and the protection of fundamental rights. However, it is the second category, centred on mass surveillance and biometric tracking, where particularly rapid growth has been observed, driven by widespread adoption in public settings and the deployment of large-scale state and corporate infrastructure.

The installation of cameras, sensors and control centres for mass social surveillance infrastructure has become an attractive solution for governments with a variety of objectives. The cases of Ecuador, Colombia and Spain are presented below.

**The Case of Ecuador. The ECU-911 Project**. One of the most representative examples of the adoption of Chinese surveillance technology in Latin America is the ECU-911 system in Ecuador. In 2011, during the government of Rafael Correa, a contract was signed with the Chinese state-owned company CEIEC (China National Electronics Import & Export Corporation) to deploy an emergency monitoring and management centre. Thousands of video surveillance cameras were interconnected with police stations, fire stations and health services so as to enable more rapid responses to crimes or natural disasters. Although the project was initially presented as an advance in public safety, allegations emerged about the political use of the cameras and facial recognition, which were allegedly being used to monitor protests and dissidents. After Correa left office in 2017, suspicions intensified that the sophisticated network of cameras and video analysis could be used not only against crime but also for social and political control. These accusations were

further supported by reports in international media outlets such as The New York Times, which warned of the risks of surveillance infrastructures connected to large databases and biometric recognition.[3]

**The Case of Colombia. Video Surveillance and Social Protests.** Since 2015, several Colombian cities (Bogotá, Medellín, Cali, and Barranquilla, among others) have installed Huawei and Hikvision cameras with the aim of reducing urban crime rates. Subsequently, starting in 2019, facial recognition capabilities were introduced to identify individuals with criminal records or who were wanted by the authorities. Without a clear legal framework regulating the collection and use of biometric data, organisations such as the Karisma Foundation have questioned the effectiveness and proportionality of these measures, as well as demanding greater transparency and data protection guarantees for those investigated and/or prosercuted as a consequence of suirveillance.[4]

During the 2021 National Strike protests in the country, video surveillance infrastructure acquired largely from Chinese suppliers was used to monitor protesters in key urban areas. While the government justified these actions as part of a strategy to prevent vandalism and protect citizens, human rights organisations denounced the identification of social leaders and the indiscriminate collection of personal data from protesters, evidencing a use of surveillance technology that goes beyond public security and stretches into social control.

**The Case of Spain. Pegasus and the "procés".** Although much of the focus on mass surveillance is on non-democratic or developing countries, the controversy surrounding Pegasus spyware in Spain shows that these tools also pose serious challenges in established institutional contexts. According to research by Citizen Lab, Pegasus was allegedly used to spy on leaders and activists linked to the Catalan independence movement, as well as other civil society figures.[5] The so-called 'procés' refers to the series of political and legal events that unfolded in Catalonia, particularly since 2017, when the Catalan government pushed for an independence referendum that was declared illegal by the Spanish authorities. This situation generated political tension, legal battles and intense debates about the legitimacy of state action in the persecution of leaders of the independence movement. In this context, Pegasus was allegedly used to tap phones of politicians, lawyers and journalists, accessing conversations, emails and private files without the knowledge of those affected.

These examples of technological surveillance in three Latin American countries illustrate the complex interaction between states, technology corporations and civil society. With infrastructures that enable mass surveillance and large-scale data processing, states face a dilemma: on the one hand, the promise of greater security and efficiency in emergency management; on the other, the latent risk of abuse of power, the criminalisation of social protest and the invasion of the privacy and intimacy of citizens.

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3 Álvarez-Velasco, C., & Vila-Seoane, M. (2023). La cooperación entre Ecuador y China en tecnologías de seguridad: El caso del ECU 911. *URVIO. Revista Latinoamericana de Estudios de Seguridad, 36.* https://doi.org/10.17141/urvio.36.2023.5847

4 Parra, J. P., & Botero, C. (2021). *Pistolas contra celulares.* Fundación Karisma, Bogotá, Colombia. https://web.karisma.org.co/pistolas-contra-celulares/

5 Scott-Railton, J., Campo, E., Marczak, B., Razzak, B. A., Anstis, S., Böcü, G., Solimano, S., & Deibert, R. (2022). CatalanGate: Extensive Mercenary Spyware Operation against Catalans Using Pegasus and Candiru. Citizen Lab, University of Toronto. https://citizenlab.ca/2022/04/catalangate-extensive-mercenary-spyware-operation-against-catalans-using-pegasus-candiru/

## Conclusion

Surveillance is not inherently malicious. In fact, as Alexis de Tocqueville suggested in the nineteenth century that the mutual and continuous scrutiny between the different spheres involved in and affected by modern democracies (state, market and civil society) is a prerequisite for their good health and development.

A clear example of this is that the integration of AI-based surveillance technologies and machine learning into modern societies has improved public safety in some respects by reducing for example traffic violations and crime. However, the indiscriminate deployment of surveillance technologies has also raised serious concerns related to democratic governance.

To preserve the latter, governments, legislators, civil society, technology companies and researchers must implement a set of measures and actions with the objective of balancing public safety and civil rights, to ensure that surveillance technologies are used in an ethical, responsible and transparent manner in the current context of technological acceleration. Among the main measures that should be carried out are the implementation of ethical guidelines and oversight mechanisms by governments; the development of specific legislation in the field of mass social surveillance; the creation, within civil society, of platforms and collectives dedicated to monitoring surveillance infrastructures; the implementation of codes of good practice and internal ethical guidelines for supplier companies; and the development of field studies and recommendations for use by researchers and experts.

The current situation of mass social surveillance is leading to the technological, algorithmic and centralised control of citizens, and is reducing the space for democratic participation and oversight.[6] Concerns about the use of mass social surveillance technologies the potential drift towards more authoritarian measures of control call into question the integrity of democratic institutions and threaten individual freedoms. To address these challenges, surveillance must be tightly regulated to serve clearly defined public safety goal while respecting fundamental rights and democratic accountability.

6 Calvo, P., & Saura García, C. (2024). Democracia de la vigilancia: datos, activismo y contrapoder. Revista Internacional De Pensamiento Político, 19, 395–417. https://doi.org/10.46661/ripp(ijpt).9612

# Reflections on the Preliminary OAS Process Toward an Inter-American Declaration on Neurosciences, Neurotechnologies, and Human Rights

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## Short background

The rapid development of new AI technologies, is giving rise to multiple concerns, particularly regarding the ethical and legal criteria that should guide these scientific innovations.

In the specific field of neurotechnologies, certain applications enable a two-way connection between an individual's central nervous system and an electronic device. This makes it possible to access and explore the data generated by the brain, record it on external devices, and even delete or modify it.[1]

Against this backdrop, various initiatives are being promoted to protect human rights. These immersive technologies not only transform the lives of societies, but also pose risks to autonomy, freedom of thought and physical integrity and, in doing so, they open up a new and largely unexplored frontier for the legal field.

## Progress at the Inter-American Level

At the regional level, significant progress has been made thanks to the development of international standards drawn up by the Inter-American Juridical Committee (IAJC) of the Organization of American States (OAS).

On 11 August 2021, the IACJ adopted the **Declaration on Neuroscience, Neurotechnologies and Human Rights: New Legal Challenges for the Americas**, becoming the first of its kind in the world.[2] In one of its sections, the Declaration mentions the lack of legal frameworks on the subject and underlines the need to call on key actors to guarantee the protection of human rights in the face of rapid technological development:

'The absence of specific regulations on neurotechnologies, as well as their scope and impact, generates a risk of illegitimate manipulation of emotions, feelings, and decisions by those who produce such technologies and/or control the large AI systems that decode neural information. Likewise, the use of such neurotechnologies can break the last natural frontier of a person, i.e. their mental privacy, and thus affect the identity and dignity of each human being.'

In this regard, the document warns that advances in neuroscience and the development of neurotechnologies require serious reflection on the part of all involved sectors (states, the private sector, academia and the scientific community), and calls for the adoption of specific measures to enable such innovations to contribute to the common good.

1 Yuste Rafael, Ganser Jared y Harrmann Sephanie. (2021). "It's time for neuro-rights". *Revista Horizon, N° 8*, p. 154 -156

2 CJI/DEC. 01 (XCIX-O/21) http://www.oas.org/es/sla/cji/docs/CJI-DEC_01_XCIX-O-21.pdf

The Declaration also proposed including the issue on the agenda of the OAS Inter-American Judicial Committee and developing a document on Inter-American Principles on Neuroscience, Neurotechnology, and Human Rights. Its objective was to establish international standards to guide and harmonise national regulations in this area. Since then, and with the collaboration of the Committee of Experts[3], that participated in the drafting of the Declaration, preparatory work has been carried out to advance the creation of more precise standards.

In addition, on 21 June 2022, also within the framework of the OAS, a public hearing was held before the Inter-American Commission on Human Rights (IACHR). The hearing was requested by a group of experts and went under the title "Neurotechnologies and Human Rights: Challenges, Impacts, and Risks to Privacy and Health". Its aim was to identify possible human rights violations arising from the unregulated use of neurotechnologies by states and private entities, especially in the field of health, and to propose recommendations to prevent risks.[4]

Subsequently, regional consultations were held with academics, independent experts, and civil society. Following this, a preliminary document was drafted and sent to the OAS member states for review, whose contributions were included in the final version of the text. Finally, on 9 March 2023, thanks to the efforts of Commissioner Ramiro Gastón Orias and the technical support of the Kamanau Foundation, the **Inter-American Declaration of Principles on Neuroscience, Neurotechnology and Human Rights**[5] was approved.

The document is a pioneering international initiative that seeks to harmonise national regulations to promote responsible innovation in neuroscience, neurotechnology and human rights in the region. Its development was based on an analysis of current international norms and standards, which were adapted to the specific characteristics of neurotechnologies with the aim of preventing and combating possible human rights violations. The document is structured over ten sections that reinforce existing guarantees, with a special focus on preserving individual identity and cognitive integrity in the face of neurotechnological interventions.

In addition, the document's principles include specific standards related to the protection of personal data, which emphasise the need to protect the rights of individuals from the early design stage of neurotechnologies all the way through to their implementation, evaluation, commercialisation and use. In this context, neural data is recognised as sensitive data, requiring express and informed consent for any manipulation of neurodata.

Regarding the right to equality, the document highlights the importance of guaranteeing equitable access to neurotechnologies and preventing discrimination against historically marginalised categories, such as race,

3 Committee of Experts is made up by: Eduardo Bertoni, Ciro Colombara, Francesca Fanucci, Verónica Hinestroza, Amelie Kim Cheang, Tomás Quadra Salcedo, Moisés Sánchez, Silvia Serrano Guzmán and Rafael Yuste. Moreover, in the period leading up to the Declaration, work was carried out on the results of a consultation conducted by the Pro Bono Foundation with the various member states to obtain information on the situation in each state.

4 The request for a hearing was made by the Neuro Rights Initiative at Columbia University, the Kamanau Foundation, the Pro Bono Network of the Americas, and Fundación Ronda, in conjunction with a group of experts of different nationalities and professions linked to the world of science and international human rights law, all of whom participated in the preparation of a document that outlined the challenges and impacts of neurotechnology on human rights and, in particular, on neuro rights. 184th session of the IACHR: https://www.youtube.com/watch?v=-JdUHdlXgdE

5 Inter-American Declaration of Principles on Neuroscience, Neurotechnology, and Human Rights: https://www.oas.org/es/sla/cji/docs/CJI-RES_281_CII-O-23_corr1_ESP.pdf

gender, nationality, religion and social status. On this same issue, the document raises the need to establish clear limits and exercise enhanced control over the enhancement of cognitive abilities. Finally, it promotes strategies for the efficient governance of neurotechnologies, including the creation of supervisory and oversight bodies, as well as the guarantee of access to effective legal protection.

### State Reforms, Projects and Initiatives

This section outlines the various projects and initiatives that have been presented on neurodiverse rights at the state level.

In **Chile**, in October 2019, on the initiative of the Chilean Senate's Future Commission and with the commitment of the president of the Republic, two bills seeking to regulate the protection of neurodiverse rights were submitted for legislative debate. On the one hand, this included a push for constitutional reform to consider the need for preventing interferences with the brain activity (Bulletin 13827-19), and on the other hand, a more detailed bill (Bulletin 13828-19). The constitutional reform was approved and published in the official gazette on 25 October 2021 (Law 21.383). It adds a new paragraph to the constitution's Article 19 No. 1, which states: 'Scientific and technological development will be at the service of the people and will be carried out with respect for life and physical and psychic integrity. The law will regulate the requirements, conditions and restrictions for its use in people, and must especially safeguard brain activity, as well as information derived from it. [6] The bill remains under legislative debate at this time.

In **Brazil**, in 2023, Senator Randolfe Rodrígues presented a constitutional reform bill to incorporate neurorights into the Brazilian constitution (Constitutional Amendment Proposal No. 29).[7] The bill seeks to ensure that scientific and technological development respects mental integrity and algorithmic transparency. Also in 2023, the state of Rio Grande do Sul incorporated neuro rights into its constitution. The reform states that every individual has the right to mental integrity as an inalienable protection against manipulation resulting from advances in neuroscience and neurotechnology. The reform was unanimously approved (PEC 298 of 2023) and amended the sole paragraph of Article 235 of the Constitution of the State of Rio Grande do Sul, which deals with science and technological research policy, to establish the protection of mental identity against any research affecting the brain and its activity without the consent of the individual. Also in Brazil, in 2022, a bill was presented in the Chamber of Deputies proposing the incorporation of neural data as sensitive data, with the aim of conceptualising neural data and regulating its protection.[8]

In **Mexico**, on 1 August 2023, a bill was presented to amend Article 4 of the Constitution, which seeks to protect the mental integrity of individuals. The initiative, led by Deputy María Eugenia Hernández Pérez with the support of Dr Arturo Sánchez, proposes guaranteeing the right to individual identity and physical and mental integrity, and establishing that no authority or individual may use technologies to modify, reduce or affect such integrity.[9] In September 2023, a comprehensive neuroprotection bill was presented. This bill proposed the creation of a General Law on Neuro-rights and Neurotechnologies.

6 Ley 21383, Promulgación: 14-OCT-2021. Publicación: 25-OCT-2021. Materias: Desarrollo Científico, Desarrollo Tecnológico, Actividad Cerebral, Respeto a la Vida, Respeto a la Integridad Física, Respeto a la Integridad Psíquica: https://www.bcn.cl/leychile/navegar?idNorma=1166983&idParte=10278855&idVersion=2021-10-25

7 Available at: https://www25.senado.leg.br/web/atividade/materias/-/materia/158095

8 https://www.camara.leg.br/proposicoesWeb/fichadetramitacao?idProposicao=2317524&fichaAmigavel=nao

9 Available at: https://drive.google.com/file/d/1BJ6qnyDWf_fGVMh0uzE3lsz65hEIMLvJ/view

The initiative sets out various provisions on neuro-rights and neurotechnologies and aims to propose general principles and foundations to guarantee the protection of human dignity and the recognition, respect, promotion and defence of human rights related to the central and peripheral nervous system.

These reforms embody the recognition of neuro-rights previously established in the 'Charter of Rights of the Person in the Digital Environment'[10], drawn up by the Personal Data Coordination Office of Mexico's National System for Transparency, Access to Public Information and Personal Data Protection. In Chapter VII, the Charter includes guidelines for the preservation of digital identity, the privacy of neural data, and the non-interference of neurotechnologies in personal autonomy and freedom of decision. It also incorporates the principle of equity in access to brain enhancement.

In **Peru**, on 25 August 2023, Ministerial Resolution No. 0270-2023-JUS[11], JUS41 published the draft regulations for Law No. 29733 on Personal Data Protection. These regulations incorporate mental and emotional health data as sensitive data, ensuring greater legal guarantees. Following an open public consultation process, in November 2024, the Supreme Decree 016-2024-JUS approved the new regulations. Similarly, in Uruguay, in June 2024, legislator and representative for Montevideo, Rodrigo Goñi Reyes, presented a reform bill to incorporate neuro data as sensitive data.[12]

**In Colombia,** in August 2024, a bill on personal data was presented.[13] It includes several references to the protection of neuro data. In addition, since mid-2024, a working group has been drafting a bill to be submitted to the Congress of the Republic. The bill is at an advanced stage and has received input from various experts. This bill regulates the principles governing neuroscience, neurotechnology and human rights.

In **Ecuador,** in March 2023, a draft Organic Law on Neuroprotection and the Ethical Application of Neurotechnologies was presented.[14]This law establishes that advances in neuroscience and neurotechnology pose ethical and regulatory challenges to fundamental human rights that could even redefine what it means to be human.

Meanwhile, in the **United States**, laws protecting consumer rights have been reformed to include neuro data as sensitive data. Reforms have been approved in the states of Colorado and California.[15] In Colorado, in April 2024, a legislative reform to the Colorado Privacy Act.[16] Furthermore, in September 2024, in the State of California, Bill SB 1223[17] added the term 'neural data,' which means 'information generated by measuring the activity of a consumer's central or peripheral nervous system, and not inferred from non-neural information,' as a category of sensitive personal information for the purposes of the California Consumer Privacy Act (CCPA), which grants this information additional protections under the law. In addition, legislative reforms are also progressing in the **States of Illinois, Massachusetts and Montana**.

Across the Atlantic, it is interesting to note the case of **Spain**, and the Charter of Digital

10 Available at: https://www.infocdmx.org.mx/doctos/2022/Carta_DDigitales.pdf

11 Available at: https://www.gob.pe/institucion/minjus/normas-legales/4574374-0270-2023-jus

12 https://parlamento.gub.uy/documentosyleyes/documentos/repartido/representantes/49/1152/0/PDF

13 https://www.camara.gov.co/regimen-proteccion-de-datos-personales

14 https://participa.asambleanacional.gob.ec/legislation/processes/64/draft_versions/81

15 https://neurorightsfoundation.org/united-states

16 https://leg.colorado.gov/sites/default/files/2024a_1058_signed.pdf

17 https://cppa.ca.gov/meetings/materials/20240716_item7_sb_1223.pdf

Rights presented in 2021, which includes a section dedicated to digital rights in the use of neurotechnologies.[18] This section highlights the need for specific regulation to guarantee the identity, self-determination, integrity, equality, non-discrimination and security of data related to these technologies.

Furthermore, in the **United Kingdom**, the 2022 report 'Tech Futures: Neurotechnology'[19] prepared by the Information Commissioner's Office, addresses security and protection standards in the development of neurotechnologies.

It should also be noted that **France** has a law on bioethics[20] that complements the public health law to prohibit any activities related to the modification of brain activity that present—or may be suspected of presenting—a serious danger to human health. France has also made progress on a letter of voluntary measures for the responsible development of neurotechnologies.[21]

### New Challenges

Looking ahead to the coming years, it is essential to create reliable scenarios for the development of new technologies, especially immersive technologies that use neurotechnology and AI. It is urgent that clear rules be established and that the rights of the most vulnerable people (children, adolescents, persons with disabilities, women and dissidents, and older persons) be duly protected. At the regional level, the IACJ has set an important precedent by addressing the specific challenges that neurotechnologies pose for human rights. In addition, it has recently been developing a work plan to address the development of neurotechnologies in convergence with artificial intelligence. This is why progress has been made within the IACJ in presenting the issue, with a focus on children and adolescents.[22] In addition, agenda items were established[23] to further develop a working document on artificial intelligence and children. Similarly, efforts are underway in a consultation with OAS member states to receive updated information on the status of specific regulations in each country, in order to analyse and design regulatory strategies at the regional level.

18 https://www.lamoncloa.gob.es/presidente/actividades/Documents/2021/140721-Carta_Derechos_Digitales_RedEs.pdf

19 https://ico.org.uk/about-the-ico/research-and-reports/ico-tech-futures-neurotechnology/

20 https://www.assemblee-nationale.fr/dyn/15/textes/l15t0343_texte-adopte-seance

21 https://www.enseignementsup-recherche.gouv.fr/sites/default/files/2023-04/french-charter-for-the-responsible-development-of-neurotechnologies-27500.pdf

22 https://www.oas.org/es/sla/cji/docs/CJI-doc_701-23_ESP.pdf

23 https://www.oas.org/es/sla/cji/docs/CJI-RES_293_CV-O-24_ESP.pdf

# Neurotecnologies and Human Rights: The Chilean Case in Light of International Instruments

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## The 'Insight' Case

In early August 2023, Chile made headlines when its Supreme Court admitted a constitutional protection action concerning the violation of constitutional guarantees to the detriment of a user. The action was brought against the company Emotiv Inc. for the sale and marketing in Chile of 'Insight,' a neural data capture device which designed like a "headband equipped with sensors that collect information about the electrical activity of the brain, obtaining data on gestures, movements, preferences, reaction times and cognitive activity of the user".[1]

The Court considered that constitutional guarantees relating to physical and mental integrity and the right to privacy had been violated to the detriment of the claimant, and that a product whose characteristics and impact on rights had not been previously studied had nonetheless been marketed. It therefore ordered a review by the health and customs authorities. It also ordered the company to delete all information related to the claimant's use of the product stored in its cloud or portals.[2]

This case shows how the debate around the collection and use of neural data is far from science fiction, given that 'Insight', like other devices, can be purchased online without restriction and used with no precautions other than its instructions. On another note, the case highlights the enormous contribution of national regulation in the field of neurotechnologies: in 2021, Chile became the first country to protect human rights in relation to neurotechnologies at a constitutional level, through a reform that establishes 'special protection for brain activity and the information derived from it.'[3] Given the massive impact of neurotechnologies – which are associated with AI – countries such as Brazil and Mexico are also currently considering changes to their constitutions, and Peru is promoting the protection of neural data by amending the Regulations of the Personal Data Protection Act.

## Some Regulatory Proposals from the International System

Where can we find guidance to justify constitutional reforms, regulatory adjustments and the design of public policies

1 Supreme Court of Chile, ruling of 9 August 2023, Third Chamber of the Supreme Court. Case N° 105.065-2023.

2 Supreme Court of Chile, ruling of 9 August 2023, Third Chamber of the Supreme Court. Case N° 105.065-2023.

3 In October 2021, Law No. 21.383 was passed, adding a final paragraph to Article 19(1), amending the Political Constitution of the Republic and providing that: 'Scientific and technological development will be at the service of the people and will be carried out with respect for life and physical and psychic integrity. The law will regulate the requirements, conditions and restrictions for its use in people, and must especially safeguard brain activity, as well as information derived from it'.

on such a complex issue? At the international level, we find at least three important instruments whose practical relevance we will illustrate in relation to the already mentioned case. These instruments are: a) the OECD Recommendations on Innovation in Neurotechnologies (2019)[4]; b) the Inter-American Declaration of Principles on Neuroscience, Neurotechnologies and Human Rights, of the OAS (2023)[5]; and c) the Report of the UNESCO International Bioethics Committee on the Ethics of Neurotechnologies (2021).[6] The OECD and OAS documents establish very general guidelines through a series of principles, while the UNESCO IBC report develops ethical, governance and human rights issues. From a legal perspective, it is also worth mentioning the 'International Human Rights Protection Gaps in the Age of Neurotechnology (2022)[7]" document compiled by the NeuroRights Foundation.

### *OECD Recommendations on Innovation in Neurotechnologies*

The OECD Recommendations aim to provide guidance at each stage of the innovation process following three pillars: values, institutions and processes. Among the values, trust, security and privacy stand out in the technological context. The aim is to develop the capacity of institutions to anticipate, supervise and advise and, finally, to promote processes for social deliberation, inclusive innovation and collaboration. What can we highlight from the OECD Recommendations in relation to the Chilean case? The recommendations set out 9 principles. In particular, principle 7 refers to the 'protection of personal data from the brain and other information obtained through neurotechnology' and develops guidelines on informed consent for the collection, storage, processing and potential use of personal brain data collected for health purposes; the prohibition of discrimination; the protection of confidentiality and privacy; and the traceability of collected and processed data, as well as of the medical acts in which neurotechnology is used.

It is important to remember that informed consent is a term that refers to two different but closely related realities: on the one hand, a communication process and, on the other, the name of the document that sets out the authorisation to act as a result of that communication process. Designed as a requirement for participation in research trials and the doctor-patient relationship, informed consent is a category that has been developed within bioethics and that assumes its vital importance in the context of neurotechnologies, even though, according to the Chilean court ruling, these are not necessarily used in a healthcare context or for therapeutic purposes.

### *The Inter-American Declaration of Principles on Neuroscience, Neurotechnology and Human Rights (OAS)*

For its part, the Inter-American Declaration of Principles on Neuroscience, Neurotechnology, and Human Rights (OAS), originating from the Inter-American Juridical Committee and approved in March 2023, lists 10 principles (Annex I), along with comments and annotations (Annex II). In the case under discussion, principles 3 and 4 are particularly applicable. Principle 3 establishes that neural data are highly sensitive personal data that require enhanced privacy and security measures, ensuring limits on decoding

4 OECD. *Recommendation of the Council on Responsible Innovatin on Neurotecnologies*. 2022. OECD/LEGAL/0457

5 OAS. Declaration of Principles on Neuroscience, Neurotechnology and Human Rights: New Juridical Challenges for the Americas OEA/Ser. Q 6 – 10 March, 2023 CJI/RES. 281 (CII-O/23) corr.1

6 IBC UNESCO. Report of the International Bioethics Committee on the Ethical issues of Neurotec- nologies (Dec. 2021) SHS/BIO/IBC-28/2021/3 Rev.Paris, 15 December 2021

7 Jared Genser, Stephanie Herrmann & Rafael Yuste. International Human Rights Protection Gaps in the Age of Neurotechnology. NeuroRights Fundation, mayo 6 2022.

techniques. Principle 4 requires express and revocable informed consent as an essential prerequisite for access to and collection of neural data. As mentioned elsewhere,[8] the cornerstone of the case was developed by the Chilean Court in its seventh recital, where it defined type of consent required to authorise the collection and use of neural data. The Court held that 'express consent is required for its use for scientific research purposes', as regulated under Article 11 of Law 20.120. The Court considers that informed consent is required under the terms of Law 20.120 and that this cannot be replaced by any other type of consent that is less protective, such as a contract with pre-determined clauses.

### *The Importance of the Business and Human Rights Framework*

Another very key element highlighted by the Court he obligation to regulate and supervise devices before granting marketing authorisation of any device that raises 'issues not previously studied' and that poses a risk to human rights. For this reason, the Court would order the customs authority and, above all, the health authority to analyse any such device. This action is consistent with principle 8 of the OAS on transparency and governance in neurotechnologies and principle 9 on oversight and monitoring. The latter principle, in particular, is in line with the United Nations Guiding Principles on Business and Human Rights (UN, 2011) and the recommendations of the REDESCA/IACHR Report on Business and Human Rights,[9] which is an important and innovative tool that can be used when considering flexible forms of protection in the area of technology and human rights in the region.

Under the current inter-American legal order, the legal content of human rights and the corresponding state obligations have an impact on companies, although to a different degree and with different levels of reach than the impacts on states, due to the nature of the system. According to the report, 'this relationship is crystallised when states formulate, monitor and assign explicit and binding legal responsibilities aimed at ensuring respect for human rights by companies at the domestic level and based on international human rights norms and specific standards that may be determined for this purpose by the competent bodies'.[10]

The importance of the Business and Human Rights in Technology Project (B-Tech Project) should also be noted. This project is led by the Office of the High Commissioner for Human Rights to provide a roadmap for applying the United Nations Guiding Principles on Business and Human Rights to the development and use of digital technologies. There is a consensus that we are no longer facing only moral dilemmas but rather, concrete risks to human rights. The B-Tech Project aims to 'find pragmatic and principled ways to prevent and address the socio-political harms that may arise, or do arise, during the application of digital technologies by the public and private sectors.'[11] Through the Guiding Principles framework, issues of governance, public policy, corporate responsibility and accountability can be addressed 'to prevent, address and remedy the wide range of actual or potential harms related to digital technologies'.[12]

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8 Siverino Bavio, Paula. "Neuroderechos: el consentimiento informado y la regulación sobre empresas y derechos humanos como herramientas de tutela". Fundación Kamanau, septiembre 2023

9 REDESCA/CIDH. Informe Empresas y Derechos Humanos: Estándares Interamericanos. OEA/Ser.L/V/II CIDH/REDESCA/INF.1/19 1 de Noviembre de 2019 parr. 195

10 Ibid.

11 Business and Human Rights in Technology Project ("B-Tech Project"): Applying the UN Guiding Principles on Business and Human Rights to Digital Technologies disponible en BTechprojectoverview.pdf consultado el 15 de noviembre de 2024.

12 Ibid.

### *The UNESCO IBC Report on Ethics of Neurotechnologies*

From late 2019 to September 2021, the UNESCO International Bioethics Committee (IBC UNESCO) worked on discussing and drafting a report on the ethics of neurotechnologies, which was approved in late December 2021. Considering the special mandate of the IBC to promote the Universal Declaration on Bioethics and Human Rights (UNESCO, 2005) and to address health issues from a bioethical and human rights perspective, the report develops some key issues.

The report begins with a brief description of existing neurotechnologies, before addressing issues of neurotechnology and ethics, including ethical principles, mental integrity and human dignity, personal identity and biological continuity, autonomy, mental privacy, accessibility and justice, and enhancement. It also reflects on the application of neurotechnologies and clinical ethics, as well as research ethics. The chapter on neurotechnologies and law focuses on the impact on human rights, consent for the use of brain data (a key issue in the Chilean Supreme Court ruling), freedom of thought, and the impact on democracy. The report includes a chapter on governance, which raises the issue of responsible innovation, public-private partnerships, and commitments from society and industry. The final chapter contains recommendations.

One of the main recommendations of the IBC report is that UNESCO should work on a Universal Declaration on Neurotechnologies and Human Rights and organise global expert meetings. The proposed Declaration was replaced by a draft Recommendation on the Ethics of Neurotechnologies. In July 2023, a first global meeting was held and throughout 2024 work was carried out on the first drafts of the Recommendation, which will be proposed for approval by states in 2025.

Among the IBC's main recommendations to states is the emphasis on the local protection of human rights impacted by neurotechnologies through appropriate regulatory mechanisms, as well as the promotion of education in this area. Several countries in the region are already taking steps in this direction, with Chile leading the way. Legal reform efforts are also underway, or under discussion, in Brazil, Mexico, and Peru, including proposals at the constitutional level. The report further recommends that researchers, among other things, adopt a code of conduct for responsible research and innovation in neurotechnologies, and work on finding ways to strengthen informed consent and preventing the dual use of neurotechnologies. The IBC also recommends that industry adopts a code of conduct, algorithmic transparency and data privacy, and states that industry should avoid giving false expectations to users and making unfair promises. The report also includes recommendations for the media and the public.

### Conclusions

Technological convergence is changing the way we understand the world. Neurotechnologies and AI have immense potential benefits, but they could also introduce high risks factors to human life and, many of which are difficult to even foresee in their true dimensions. Integrating perspectives from bioethics, science and technology ethics, and human rights is essential to guide scientific and technological development, protect survival, and to support human flourishing.

# Generative Artificial Intelligence, Natural Language Processing & the Complexity of Legal Reasoning. Views from Colombia

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This article analyses the legal and technical challenges of implementing generative artificial intelligence and natural language processing in the administration of justice. It explains the limitations of legal formalism and syllogistic reasoning for programming AI that emulates judicial discretion. It explores existing obstacles to natural language processing in capturing tacit legal reasoning. The piece concludes that, for now, AI can assist with simple tasks but cannot replace the work of interpretation and substantive decision-making performed by judges.

## Introduction

In his book *The Nature of the Judicial Process*, North American judge Benjamin Cardozo explained that, in the judicial process, judges operate between historical adherence and progressive adaptability of law as a discipline for governing a community normatively. At its core lies the complex interaction of precedent, legal reasoning, social norms, and the ever-evolving nature of justice.[1]

For Cardozo, although logic and reasoning are fundamental to legal decision-making, they do not operate in a vacuum. The lived experiences of judges and their understanding of social dynamics are essential in shaping their judgements.

This means that, although precedents ensure stability in the legal framework, overly strict adherence could consolidate outdated or unfair principles.

Cardozo's reflections on a myriad of subconscious and extra-legal forces that shape judicial decisions highlight the complexities and nuances inherent in the exercise of judicial discretion in the administration of justice. However, the era for which Cardozo wrote is radically different from ours. Profound social, cultural, political, technological, and economic changes have resulted in multiple constitutional revolutions, regulatory transformations, and continuous social demands, leading to an exponential increase in the work of judges.

This exponential increase in the workload of judicial offices is exacerbated by the growing increase in legal knowledge and its multiple sources. Drawing on the philosophy of information, I suggest that legal systems can experience legal 'entropy' through an increase in complexity due to the incorporation of new laws, regulations and judicial precedents. Lawyers and judges strive to interpret and apply the law to maintain order and consistency within this constantly evolving complexity.

The panorama for our judicial system is critical.[2] For example, the Constitutional

1 Cardozo, B.N (1921) *The Nature of the Judicial Process*. Yale University Press.

2 Castaño, D. (2018) *La Transformación Digital de la Justicia, ¿primer paso hacia la descongestión judicial?* https://www.ambitojuridico.com/noticias/columnista-impreso/educacion-y-cultura/la-transformacion-digital-de-la-tutela-primer-paso

Court reviewed around 600,000 legal actions in 2022.[3] According to the Excellence for Justice Corporation, the Colombia judiciary had more than 1.8 million cases pending in December 2020. [4] Needless to say, our judges could use some additional help, even if it comes from AI.

The implementation of artificial intelligence systems in law and the legal profession promises potential improvements in the efficiency and consistency of decision-making. However, the use of GenAI for judicial decision-making also raises profound technical, legal, social, and ethical risks, pressing the following questions: can AI truly replicate the balance between predictability and flexibility that is central to justice systems? Would integrating AI technologies increase the risk of amplifying, altering, or even obscuring the subconscious elements and biases that influence legal reasoning? This paper seeks to explore the implications of AI's integration into the judicial process, examining its potential benefits, challenges, and the fundamental questions it raises about the nature of justice in an increasingly interconnected and automated world.

The ideas I present here are an iteration of others that I have published in different contexts in the past.[5] For writing this paper, I have used various AI tools to analyse large-scale data, translate, summarise and edit some texts.

## Three Real-Life Use Cases of GenAI in Judicial Proceedings in Colombia

In the first case, the Unión Temporal de Servicios Integrados y Especializados de Tránsito y Transporte de Santa Marta (SIETT) sued the nation, the Ministry of National Defence, and the National Police, in a direct reparation proceeding before the administrative court. The initial hearing was scheduled to take place in person, but the claimant's representative requested that it be held in the metaverse.[6]

The Administrative Court of Magdalena analysed the regulations on data messages, electronic signatures and virtual proceedings, before concluding that the metaverse is the product of the synergy of data message exchange and that its use is legally endorsed in accordance with Law 527 of 1999. In order to fully understand the context and technical implications of this, the judge asked ChatGPT for a precise definition of the term 'avatar' and inquired about best practices for identity verification in the metaverse.

Given that the specific query focused on the authenticity of avatars in the metaverse, ChatGPT provided suggestions tailored to that environment, such as avatar identity verification, biometric authentication, and linking to previously verified social media accounts. Based on this information, the judge incorporated the relevant methodologies into her ruling to guarantee the identity and authenticity of the participants through their

3 https://www.corteconstitucional.gov.co/lacorte/estadisticas.php

4 https://cej.org.co/indicadores-de-justicia/efectividad/indice-de-congestion-de-la-rama-judicial-en-colombia-sector-jurisdiccional/

5 Castaño, D, (2024). The Language of Legality: The Role of Administrative Power in Building the Digital Republic. In B. Heller (Ed.), Existing Law and Extended Reality. Stanford Law School; Castaño, D. (2021), "La Gobernanza ética de los sistemas de inteligencia artificial", en "Derecho, Innovación y Tecnología: Fundamentos de una Lex Informática", T.3, Así Habla el Externado: Tecnología, Transformación Digital y Sociedad", Universidad Externado de Colombia, Bogotá. Se puede consultar en: https://www.uexternado.edu.co/wp-content/uploads/2021/04/Capitulo1Tomo3.pdf; Castaño, D. (2021), Marco Analítico Exploratorio – República de Colombia, CETyS – Universidad de San Andrés, Buenos Aires; Castaño, D. & Heller, B. (2023), Artificial Intelligence, Virtual Courts, Real Harms, publicado en Lawfare: https://www.lawfareblog.com/artificial-intelligence-virtual-courts-and-real-harms.

6 Tribunal Administrativo del Magdalena, auto de 9 February 2023, rad. 47-001-2333-000-2020-00014-00.

virtual representations in the hearing held in the metaverse. To this end, she identified the legal basis that would allow her to apply ChatGPT's recommendations in the process with full respect for the law and due process.

In the second case, in a ruling dated 3 April 2024, the Superior Court of the Judicial District of Pereira overturned a first instance acquittal in a manslaughter case, marking a milestone in the integration of artificial intelligence into the criminal process. The case, which originated from a tragic traffic accident in 2013, led the trial judge to acquit the driver on the grounds that the victim took action at his own risk, partly as a result of impaired cognitive faculties caused by alcohol consumption.

This case is of great significance for this report because the Court used ChatGPT to overturn the first instance ruling and substantiate its decision. To do so, the judges consulted AI on the equivalence between blood alcohol levels and the victim's consumption of alcoholic beverages: 'How many glasses of wine, brandy, rum or beer are equivalent to 20 milligrams of ethanol per 100 millilitres of blood?' ChatGPT provided a detailed analysis based on which it concluded that the level of 20 mg/100 ml detected in the victim was equivalent to approximately one standard drink. This information was crucial in dismissing the theory of self-endangerment or action at one's own risk accepted by the trial judge and strengthening the argument for the defendant's liability.

For this reason, the Court overturned the discharge handed down by the trial judge and, instead, convicted the driver based on the analysis provided by ChatGPT and the other evidence in the case, which allowed it to conclude that the level of alcohol found in the victim's blood was not significant enough to constitute reckless behaviour and that, on the contrary, the driver created an illegal risk by exceeding the speed limit in a school zone.

The third case concerns a child diagnosed with autism spectrum disorder (ASD), acting through his mother, who filed a constitutional appeal (action for protection) against a health service provider (EPS) requesting exemption from payments and authorisation for transport to his therapy sessions. The judge of first instance granted the claims. The EPS appealed. The judge of second instance upheld the decision, considering that the payments represented a barrier to accessing the health services required by the child. The decision was based on the jurisprudence of the Constitutional Court on similar cases.[7]

Additionally, the judge included excerpts constructed with artificial intelligence on legal questions, in an initiative to optimise the time spent drafting judgments. In effect, the second instance judge consulted ChatGPT-3 on several legal questions about the case. ChatGPT-3 not only answered the questions posed by the judge, but also provided a detailed analysis based on applicable regulations and case law, which was meticulously verified by the judge in terms of consistency and accuracy.

Analysed with GenAI, these excerpts were then integrated into the ruling as part of the legal grounds, with the aim of showing how the decision-making process in an action of 'tutela' can be accelerated with the use of this technology. It is crucial to emphasise that the judge's decision was not replaced by ChatGPT-3 at any time, but rather that this technology served as a complement to the argumentation and analysis of the case. The judge justified his innovative initiative on the basis of Law 2213 of 2022, which allows the use of technology by judges.

The Constitutional Court chose this case for review, the analysis of which I will present below.

7 Juzgado 1° Laboral del Circuito de Cartagena, sentencia de tutela de 30 de enero de 2023, rad. 13001410500420220045901.

## Judgement T – 323 of 2024 of the Colombian Constitutional Court

Upon reviewing the case, the Constitutional Court determined that there was no violation of due process, noting that the second instance judge had made his decision before using AI. However, the Court identified alleged deficiencies in the application of the principles of transparency and accountability when the second instance judge used an AI tool. With regard to the rights of the minor, the Court partially upheld the ruling, extending transport coverage and ordering the EPS to guarantee the effective exemption from co-payments and user fees.

Regardless of its content and scope, this ruling is a landmark case because it is the first ruling by a Constitutional Court in Latin America, and possibly the world, that specifically addresses the question: can judges use AI tools in judicial proceedings and decision-making?

The answer was yes, but under certain conditions. While this ruling does not definitively resolve all the concerns on the matter, it establishes a fundamental framework for the responsible and ethical use of AI in the administration of justice.

First, the Court articulated twelve essential guiding criteria for the use of AI tools such as ChatGPT in judicial offices. These criteria, which had been already developed by the legal doctrine, range from transparency and accountability to suitability and human control, and represent an important first step towards the implementation of AI. Transparency, for example, requires judges to clearly demonstrate the use, scope and location of AI-generated results in their proceedings or decisions.

The ruling is particularly emphatic in prohibiting the replacement of judicial reasoning by AI systems. This prohibition is crucial to maintaining the integrity of the judicial process and trust in the justice system, which is very much in line with the first report we published as part of this initiative in 2020. In fact, in line with what had already been developed by doctrine for several years, the Court pointed out that AI cannot replace judges in the interpretation of facts, the assessment of evidence, the reasoning behind decisions, or the adoption of rulings.

Based on the above, the Court warns of the potential risks of using AI systems such as ChatGPT, which are not owned by the Judicial Branch and whose internal workings may not be completely transparent. Specifically, the Court points out that the use of these tools could lead to a violation of the confidentiality of proceedings, the right to access information, and the right to privacy, especially if semi-private, private, or confidential data from the proceedings are entered into the AI systems.

The Court expressly authorised the use of AI in '(...) administrative and document management, as well as in support of judicial management and the correction and synthesis of texts',

provided that measures are implemented to protect fundamental rights such as privacy and judicial impartiality is maintained. Likewise, the ruling highlights that it will be imperative to identify a human responsible for each AI intervention, ensuring that the responses generated are understandable and verifiable.

To this end, it is noteworthy that the Constitutional Court pointed out that the 'guide, manual or guideline' must provide that '(...) the authorities and in the review of 'tutela' cases, controls must be applied to prevent the violation of the right to due process through the misuse of AI, including ethical self-regulation.' It is not very clear what the Court means by 'ethical self-regulation' because this expression is completely foreign to Article 230 of the 1991 Constitution, which clearly establishes the sources of law to which judges are subject in a peremptory manner.

In this vein, Judgement T-323 of 2024

established an initial framework for the use of AI in the judicial branch. While it does not resolve all doubts nor completely eliminate the associated risks, it provides guidance and fundamental principles that will serve as a basis for the development of more detailed policies and practices in the future by the Congress of the Republic and the Superior Council of the Judiciary.

In compliance with the aforementioned Judgement T-323, the Supreme Court of Justice (CSJ) issued Agreement PCSJA24-12243 on 16 December 2024, which although purporting to implement the ruling in question, ended up contradicting its spirit by creating barriers that effectively prevent the use of AI in the Colombian Judicial Branch. The disconnect between the Constitutional Court's intention and the CSJ's implementation is evident in three fundamental aspects.

This divergence between constitutional intent and administrative implementation is particularly evident in Article 8.2 of the Agreement, which requires knowledge of 'the origin of the data used for training, the manner in which it is processed, or its functioning' of AI tools intended for certain judicial uses described in Article 4. This requirement, which appears reasonable from an idealistic academic point of view, is disconnected from technical reality because it would be practically impossible to comply with using the currently available tools, which generally do not disclose this type of technical information.

With this, I believe that the CSJ restricted the use of the only commercial tools currently available to judicial officials without offering technically viable alternatives. In my opinion, this results in what we could call an 'implementation paradox': on the one hand, the CSJ recognises the need to modernise the administration of justice through AI; on the other, it establishes strict requirements that make modernisation impossible.

Thus, the aspiration for a generative AI tool specific to the Colombian judicial branch, as envisaged by the Constitutional Court and the CSJ, seems more like a judicial 'hallucination.' We will have to wait to see whether this holds true in the future.

## In Conclusion: How Can Judges Use AI Today?

In light of the above, I would consider that the optimisation of certain judicial actions specific to the administration of justice via the implementation of artificial intelligence that is able to perceive its environment and act with a certain degree of autonomy is not so close, for now.

Until that time comes, I recommend that judges should not use GenAI for tasks that involve exercising a high degree of discretion and contextual analysis, such as interpreting the law or evaluating evidence and making substantive decisions that affect a person's fundamental rights. I even consider it risky to leave in the hands of GenAI certain procedural decisions that could result in a denial of justice or violations of due process.[8]

I could also speculate, based on Alan Turing's postulates, that such a machine should have the ability to learn the criteria and parameters that allow it to distinguish *what is fair from what is not*, and the ability to do *something truly new* in accordance with the conceptual framework proposed by the aforementioned author.[9]

However, until computer science manages to design an AI system capable of overcoming these legal and technical obstacles, I believe that the judiciary could implement AI systems to optimise or improve the performance of easy tasks within the exercise of judicial

8 Interview with Professor Brittan Heller for Ámbito Jurídico on the use of AI in justice: https://www.ambitojuridico.com/noticias/invitado/sera-importante-tener-en-cuenta-las-capacidades-de-cualquier-ia-la-hora-de-crear

9 A. M. Turing (1950), Computing Machinery and Intelligence, 49 Mind 433, 1950.

functions involving the administration of justice.[10]

These tasks could be mechanical, repetitive and with a margin of error that does not pose a risk to the exercise of fundamental rights, such as scheduling and transcribing hearings, document drafting and management, precedent research and knowledge management, workflow improvement, electronic notifications and support in understanding technical issues in a case.

The dynamic between tradition and innovation in the judicial process, which Cardozo described as a necessary tension, has intensified exponentially in the digital era.

The continuous increase in legislation, case law and doctrine generate a form of 'legal entropy' that manifests itself in the growing complexity and unpredictability of judicial interpretations and outcomes. This entropy, far from being a flaw in the system, reflects the natural evolution of law in response to contemporary social demands.

As I have argued on several occasions, I am convinced that the responsible use of AI can become a tool to help mitigate this 'legal entropy' and guarantee the fundamental rights to due process, access to the administration of justice and an effective judicial remedy that concludes with a timely judgment to protect the fundamental rights of individuals. In my opinion, AI could contribute to managing entropy, without being limited to a simple calendar management tool or a means of summarising court hearings.

To this end, I believe that at this initial stage we can start by applying the so-called 'Presidio Recommendations on Responsible Generative AI' recently adopted by the World Economic Forum, which provide guidance for navigating this complex landscape.[11] Based on this, I consider it would be optimal to incorporate traceability mechanisms into generative AI systems used by the judicial system, to allow for audits of how decisions are made.

Users must have a clear understanding of the system's capabilities and limitations to prevent excessive reliance on AI. Continuous testing and human oversight can identify potential biases or errors. Similarly, multidisciplinary review teams should rigorously evaluate the system design, training data, and outputs before the integration of an AI system into legal procedures.

In addition, several recommendations in this document call for assessing and addressing the social impacts of AI integration. To this end, I believe it is necessary to ensure equitable access to generative AI capabilities across all judicial systems and to ensure that they do not perpetuate structural biases or inequalities. Public outreach can increase AI literacy in legal communities. Workforce training programmes should equip judicial employees to work effectively with AI.

However, the implementation paradox of Judgement T-323 of 2024, highlighted in CSJ Agreement PCSJA24-12243, reflects the practical challenges of this integration. The way forward requires a regulatory framework that is grounded in technology, not in the personal fantasies of a few, and that recognises both the current limitations of technology and its potential to help manage legal entropy.

The Colombian experience suggests that success in incorporating judicial AI will depend on our ability to develop protocols that allow us to take advantage of the available tools responsibly, while maintaining

10 RUSSEL, Stuart y NORVIG, Peter. Artificial Intelligence: A Modern Approach. 4 Ed. Hoboken, NJ: Pearson, 2021. p. 4 y 5.

11 https://www3.weforum.org/docs/WEF_Presidio_Recommendations_on_Responsible_Generative_AI_2023.pdf

judicial discretion as a fundamental pillar of the administration of justice.

I believe that the ethical and responsible use of AI has the potential to optimise or augment human intelligence, both individually and collectively, to promote creativity, diversify thinking and strengthen efficiency in the performance of many tasks that occupy our personal and professional lives in what Professor Ken Goldberg calls 'multiplicity'. I am convinced that justice is human and should remain so forever.

# Work in the Age of Machines

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## Introduction

Technological evolution is bringing us closer to a decisive moment, commonly described as a turning point. Various institutional entities refer to this phenomenon as the 'digital transition.' Other voices call it the 'digital age' or 'digital revolution.' There is no question that the integration of digital and related technologies will transform many areas of our existence, with work being one of the most affected.[1]

In this context, it is pertinent to reconsider some of the classical economic conceptions, such as those of Karl Marx, whose ideas have remained on the margins of mainstream economic thought. This marginalisation could be attributed to the widespread perception among economists, reinforced by empirical evidence, that his labour theory of value is an oversimplification. However, as Sen (2021, p. 213) points out, certain valuable concepts can be drawn from his work, such as the principle of non-exploitation, which advocates remuneration based on work, and the principle of needs, which advocates payment according to individual needs.

Extending this analysis, another relevant concept in Marx's thinking is the idea of the "industrial reserve army", also known as the 'surplus labour force.' Marx argues that mass and structural unemployment in capitalist societies play critical roles, including on regulating wages and ensuring an available workforce to respond to fluctuations and expansions in production.

This idea is reflected in today's vocabulary through terms such as 'precariousness,' 'precarious,' and 'precariat,' which have become part of our lexicon to describe working realities and lifestyles marked by vulnerability, fear, guilt, feelings of uselessness, instability, frustration, loneliness, and lack of recognition. These conditions and mental states resonate deeply with many workers and unemployed people today.

This situation can be better understood through the lens of Marx's 'industrial reserve army.' For example, if there is always someone willing to take your job for a lower wage, you will probably continue to work despite low pay, which hinders the possibility of building a dignified life project. This illustrates one of the key functions of this Marxist concept.

At the same time, emerging technologies

such as machine learning, computer vision and natural language processing are opening up unprecedented horizons for entrepreneurship and innovation. It does not take much effort to imagine this, as Artificial Intelligence (AI) has become a key economic pillar in our era. Its applications

1 See, for example, Susskind D. (2020). A World Without Work: Technology, Automation, and How We Should Respond. New York. Allen Lane; Danaher J. (2019). Automation and Utopia: Human Flourishing in a World Without Work. Cam. Mass. Harvard University Press.

span sectors as diverse as transport,[2] agriculture,[3] toursim,[4] health,[5] science,[6] finance and banking,[7] law,[8] among others, and they continue to expand their potential reach.

Some experts compare AI to general-purpose technologies such as fire, electricity or the internet, highlighting its cross-cutting nature and usefulness in multiple fields. AI undoubtedly promises to open up a range of opportunities, but at the same time, it could intensify existing inequalities. This raises a crucial question: who really benefits from these opportunities? Are workers the main beneficiaries?

The interaction between AI and the world of work is complex. While we seek to promote opportunities and employment for all, it is also essential not to slow down innovation. However, maintaining this balance is a huge challenge . If innovation leads to extensive automation, we could face the phenomenon of technological unemployment, which would challenge even the second function of the Marxist concept of the 'industrial reserve army'. In such a scenario, machines could completely replace human workers, rendering the need for a labour reserve obsolete.

Threfore, reflecting on work in the age of machines is not limited to technical or economic analyses[9], but rather invites us to rethink the role of work in the construction of identity and meaning. Automation, in this framework, does not necessarily imply an irreparable loss, but rather the possibility of reconfiguring the way we conceive of value and purpose in life.

## Job Opportunities

In his June 1930 lecture in Madrid, entitled 'Economic Possibilities for our Grandchildren', John M. Keynes addressed the notion of 'technological unemployment', defining it as unemployment resulting from our ability to save on labour faster than we can find new jobs for it. Keynes viewed this phenomenon in a positive light, arguing that automation would allow people to devote more time to contemplation, art, and self-actualisation.[10]

JJohn Danaher, a contemporary theorist on automation, argues in his 2019 book that we must accept and embrace the progressive and inevitable automation of work. Accordin g to Danaher, automation is

2 Townsend, A. (2013). *Smart Cities: Big Data, Civic Hackers, and the Quest for a New Utopia*. New York: W.W. Norton & Company

3 Alreshid E. (2019). Smart sustainable agriculture (SSA) solution underpinned by Internet of Things (IoT) and Artificial Intelligence (AI). *International Journal of Advanced Computer Science and Ap-plications*, 10 (5): 93-102.

4 Fang, H., Gao, S., y Zhang, F. (2023). Forecasting Inter-Destination Tourism Flow via a Hybrid Deep Learning Model. Recuperado de https://arxiv.org/abs/2305.03267

5 Goodman, Kenneth (2016). *Ethics, Med-icine and Information Technology: Intelligent Machines and the Trans-formation of Healthcare*. Cambridge: Cambridge University Press. Topol E. (2019). Deep Medicine: How Artificial Intelligence Can Make Healthcare Human Again. New York. Basic Books.

6 Acquaviva V. (2023). *Machine Learning for Physics and Astronomy*. New Jersey. Princeton University Press

7 Cao, L. (2022). AI in Finance: Challenges, Techniques, and Opportunities. *Comput. Surveys*, 55(3), 64:1–64:38

8 Susskind R. (2021). *Online Courts and the Future of Justice*. Oxford. Oxford University Press.

9 There is a relationship between work and meaning that is difficult to capture in econometric models, and right now we need to focus more on the non-economic aspects of work. Economists have traditionally focused on wages, consumption and utility, but the vital meaning that work can offer must be measured. In my humble opinion, none. If machines did our work and could create wealth, work would have no connection to meaning in life.

10 Keynes, J. M. (1930). Economic possibilities for our grandchildren. Published in Keynes (1972), also freely available online: http://www.econ.yale.edu/smith/econ116a/keynes1.pdf.

not only possible but also desirable, given that work is 'intrinsically bad.'[11] Although he personally enjoys his work as a philosophy professor, Danaher points out that, due to increasingly precarious working conditions and institutional and technical changes that are difficult to reverse, work has become harmful to many people.

This controversial perspective, which views work as fundamentally negative, challenges the belief that work is an integral part of a person's identity and a salvation from evils such as boredom, vice, and necessity, as Voltaire argued.[12] If automation makes human obsolescence imminent, this leads us to the question: How will we occupy our lives and our time? Danaher suggests that we should protect from automation only those facets of our lives that we consider important and meaningful.

In a scenario where machines take over the essential functions of human work and legal and economic systems are implemented to ensure a fair and equitable distribution of the wealth generated, I like anyone else, pace Voltaire, would be able to find alternative and meaningful ways to occupy my life and my time, thus challenging the notion that work is the only source of purpose and satisfaction. Personally, I would find more meaning in reading nineteenth century German poetry than in working. And this approach has a rich economic tradition dating back to Adam Smith.[13]

## Technological Unemployment

The phenomenon of technological unemployment will spread across all sectors and affect workers in all categories, from manual employees (for example, blue-collar workers) to office professionals (white-collar workers). In this context, it is crucial to pay special attention to a specific demographic group: women.

The automation of life affects gender equality in employment. When robots are not implemented, or tasks and jobs are not automated, companies are forced to hire women when there are no qualified men. However, when there is productivity and growth thanks to automation, companies can maintain discrimination.

The case of South Korea illustrates this: the high density of robots and the availability of highly skilled men willing to work long hours reduce the demand for women in the labour market, thereby perpetuating discrimination and patriarchal preferences.

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11 The etymology of the word 'work' comes from the Latin 'tripaliare,' which refers to a three-pronged yoke used to bind and whip slaves This symbolic origin is crucial to understanding the relationship between work and meaning, a vital topic of study in economics, philosophy, and other social and human sciences. As machines become integrated into the labour market and begin to replace us, thereby ensuring our material well-being, we are moving away from a myth that is deeply rooted in Judeo-Christian morality. Work, in reality, does not bring any more meaning to our lives than other activities. However, it provides us with a reward—money—which we often confuse with a deeper meaning or purpose. If each and every one of us had sufficient economic resources and machines did our work, who would work? As the Bard would say, that is the question.

12 Philosopher Raymund Geuss, in his book A Philosopher Looks at Work, 2021, proposes a three-part distinction—effort, necessity, and objectivity—that articulates the essential characteristics of work, forming a mirror image opposite to Voltaire's approach.

13 For Adam Smith, work is a source of 'fatigue and problems', Riqueza de las Naciones, Book 1, Chapter 5.

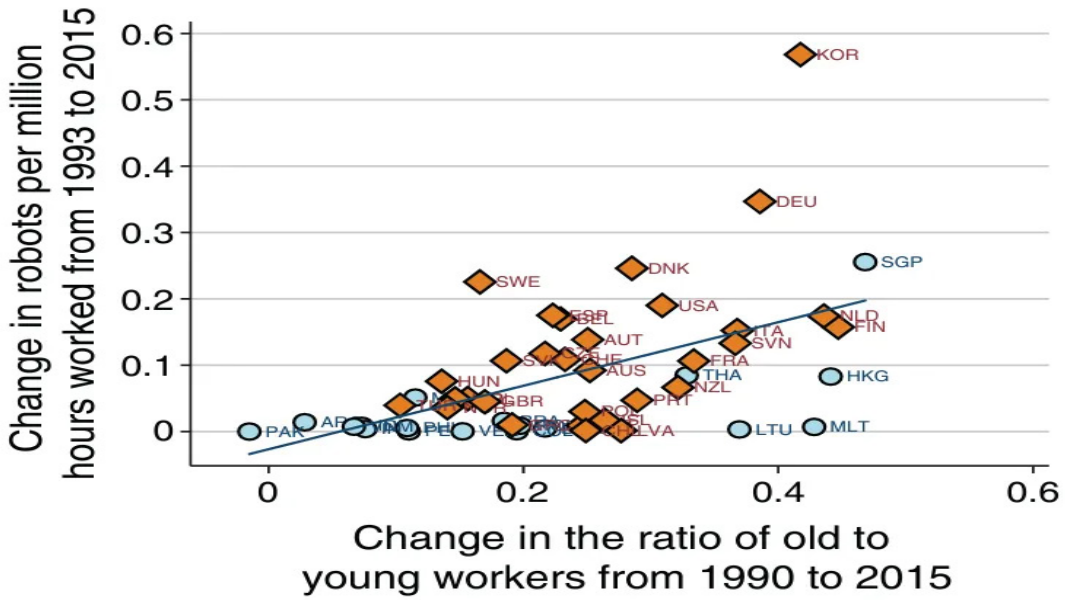



FIGURE 3. CORRELATION BETWEEN CHANGE IN THE RATIO OF OLD TO YOUNG WORKERS BETWEEN 1990 AND 2015 AND CHANGE IN ROBOTS PER MILLION HOURS WORKED BETWEEN 1993 AND 2014 (*From IFR*)

Fig. 1. The graph shows the correlation between the change in the ratio of older to younger workers from 1990 to 2015 and the change in the number of robots per million hours worked between 1993 and 2014. Countries represented with orange diamonds show changes in robot employment, while blue circles indicate the ratio of older workers to younger workers. For example, South Korea (KOR) shows a high increase in both robot density and the proportion of older workers, which could suggest a relationship between the adoption of automation and an ageing workforce. Source: Acemoglu and Restrepo (2017).

## Could unconditional basic income be key to adapting to changes in the labour market?

We live in a time when a post-work utopia remains science fiction, but the automation of a wide range of jobs, both physical and cognitive, is a tangible reality. This situation could trigger social unrest and mass unemployment.

This raises a crucial question: how will we deal with the situation of millions of people who could potentially find themselves out of work?

Unconditional basic income is proposed as a universal solution, covering single mothers, members of minority groups, ex-convicts, young people, the poor, the rich, and so on. Pilot schemes have debunked several myths about this measure, such as the supposed demotivating effectit has on seeking work.[14] On the contrary, it has been shown to encourage the creation of new job niches and provide greater economic freedom to start businesses.[15]

14 See Goodman-Bacon, A. y Palmer, V. (2024). How do low-income households respond to basic income? Experimental evidence from Minneapolis. Federal Reserve Bank of Minneapolis. https://researchdatabase.minneapolisfed.org/concern/publications/0p0967135

15 Van Parijs P. y Vanderborght Y. (2017). *Basic Income: A Radical Proposal for a Free Society and a Sane Economy*. Cam. Mass. Harvard University Press

The potential of unconditional basic income to generate new jobs is a strong argument in its favour. Added to this is the increasing automation of work, which reinforces the need for measures such as unconditional basic income. In a future where job scarcity is a reality and machines take on many roles in the workplace, unconditional basic income could be the only way to ensure that people have the means to pursue their life goals and maintain their well-being.

Regarding the possibility that automation and robotics will lead to technological unemployment, I am in favour of this transformation. I believe that work, in itself, is not the only source of meaning in our lives. Automation can offer us the opportunity to rediscover and redefine what we consider valuable and meaningful, beyond work obligations.[16]

16 Dedicated to Paul Lafargue, whose vision of leisure as a space for human fulfilment may become a reality thanks to automation and the rediscovery of meaning in life beyond work. I hope that technological progress will enable us to achieve this goal.

# How Can We Balance Instrumental Reason with Dialogical Reason in the Introduction of AI in Education?

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The first thing we must recognise is that AI is not only here to stay, but also poised to expand its use as widely as possible and across all areas of human endeavour. As Neil Postman wrote, technological changes are irreversible. Unlike other revolutions in the past, technological changes are increasingly viral (global and rapid) and are, therefore, more difficult for new generations to assimilate. It is clear that 'chip time' is not the same as 'human time,' and this is a principal handicap for education.

Secondly, we must also be aware that there is a clear imbalance between what in pedagogy we call the 'level of use' and 'the level of meaning' in the introduction of technology in education. Adela Cortina expressed this clearly in the title of her latest book, in which she referred to as 'the eclipse of communicative reason'. Other philosophers have also captured this same idea, describing it with other nuances such as the 'Promethean lag' (Anders), for example. What is clear is that there is a gap between practical reason and instrumental reason. This is why there is so much talk about the 'ethics' of AI. The problem with this imbalance is, paradoxically, that by talking so much about ethics, we end up completely diluting its meaning and importance. Or worse, we see how large companies use it as the technological version of 'green washing'.

In this sense, the rapid integration of AI into classrooms and educational processes, especially with its widespread use among students (ChatGPT, Gemini, and so on), poses numerous challenges to the already questioned issue of 'educational digitalisation.' This does not just refer to technical aspects, such as data use, data protection, mass surveillance, or the platformisation of teaching. There is also, quite simply, its misuse. In Spain there was a case of students at a secondary school in Almendralejo (Extremadura, Spain) that used an AI application on their female classmates to create fake nude images. This practice has been replicated in schools and colleges around the world.[1]

We must, therefore, carefully consider whether, given the obvious technological imbalance and the necessary integration of AI into the educational process, we should opt for facilitating and productive measures based on generalised use or, given the danger, opt for restrictive, prohibitive or even punitive measures, as some Western governments have done.[2] These measures can be viewed as 'containment dams', but more ambition and pedagogical depth are

1 Teen Girls Confront an Epidemic of Deepfake Nudes in Schools Vid. https://www.nytimes.com/2024/04/08/technology/deepfake-ai-nudes-westfield-high-school.html (Consultado el 22/12/2024).

2 From the Spanish government to the Australian government... reducing digitalisation in schools, not exposing children under 6 to screens, banning the use of social media for children under 16, etc. There is concern about the 'reach' of screens, multiplied by the use of AI, as a cause of the global mental health crisis among children, adolescents and young people.

needed to address the reality of emerging technologies in the education sector. Checks and balances that really help us focus on the essentials are that education is a universal human right and that the best interests of students must take precedence over other interests.

Indeed, one of the most significant challenges to understanding the phenomenon of education (as it has traditionally been understood) is the aforementioned 'platformisation of education'. Although they are convergent and can be interpreted as a continuum, digitalisation is not the same as platformisation. Platformisation involves the application of business models specific to the digital economy to the education sector. Furthermore, under no circumstances can we think that both phenomena are 'neutral', incidental, or superficial because they go to the very heart of the educational process: from who the educational agents are to how knowledge is generated and transmitted. In general, there are two types of arguments in favour of the type of digitalisation that leads to platformisation. On the one hand, there are those who advocate greater access, modernisation and efficiency in education and educational institutions and, on the other hand, those who advocate higher quality education, reflected in the personalised tailoring of education. Undoubtedly, the convergence of these arguments creates a 'perfect storm' for the education sector because they are both difficult arguments to refute. However, it is not entirely clear that the promises made in these arguments are true nor that they will achieve what they lay out. For this reason, it is necessary to address and correctly identify what we can classify as the 'collateral damage' of educational platformisation, which includes the following four points:

First, these processes simplify and redirect the complexity and variety of the educational phenomenon to that which happens within a platform that, in most cases, belongs to large private companies. In addition, these companies impose their own models, or techno-pedagogical logic, as well as a predetermined culture (including linguistic culture) for the production and creation of the materials and content they manage.

Secondly, we must be aware of the problem posed by the proliferation of these private actors (such as big tech companies) in the education sector and, above all, their involvement in public education, given that these are profit-driven companies operating in a sector that is as sensitive as education. In other words, the problem is not so much the existence of private actors as the fact that their main interest is their bottom line. These phenomena or processes are part of the global trends that make up what has been called the attention economy or 'surveillance capitalism,' which represents the triumph of an economic model in which all human experience can be converted into data—the fuel for this economic model.

Thirdly, the issue of data itself. Data is necessary to train the algorithms that power autonomous systems (AI). The more data there is on the entire educational process and its agents, the better the learning analytics for "improvement", that is, controlling, predicting, anticipating and even automating pedagogical decisions. That is why, in this context, big tech companies have a significant presence in the education sector—just think of Google or Microsoft, which have promoted their own platforms for the education sector—and not only as a means of facilitating teaching and learning, but because education platforms are unbeatable systems for the mass collection of personal data. A consequence of this dependence on platforms for teaching and learning, therefore, is that it can weaken the freedom of educational agents, from families to academic freedom and the pedagogical autonomy of teachers themselves. But what it really undermines is the educational experience of students, because the need for data forces an increase in screen time.

Finally, among these collateral effects, there is a risk of creating greater 'social inequality.'

While a proven advantage of technology is that it guarantees mass access to education (as demonstrated during the COVID-19 pandemic), it is also true that this creates and increases two types of inequality. Inequality of access, because not everyone has access to the internet and devices (something we also saw during the pandemic), but once the problem of access has been solved, there is also a less visible inequality that comes from the use of technology. Indeed, as technology reduces the cost of delivering education, it may lead to a divide beween two types of educational experiencess: a more "expensive" face-to-face education, enriched by human interactions and experiences, and a 'cheap' digital alternative education, driven by algorithms and platforms, primarily accesible to those who cannot access the first option.

On the other hand, from a pedagogical point of view, to incentivise the development of critical thinking and balance the level of use and the level of meaning, I believe we should be able to do three things: the first is to not let ourselves be persuaded by marketing and to thoroughly analyse the language used to refer to technology. We must expose the bubbles of technological language in education; secondly, we must help ask the right questions about emerging technology; and thirdly, perhaps the most pedagogical and most important of all, we must shape students' moral character through ethically valuable experiences, because it may well be that the best way to train students to deal with the digital world lies outside the digital world.

If we look at the various definitions of AI, we find that the most characteristic feature is the repetition of its instrumental traits, that is, what AI can 'do for us'. It is said that AI can, based on the input it receives (that is, data), make predictions, and generate content, recommendations or decisions that are also automatic or autonomous. The result is that it focuses more on these tasks and their results than on other 'less' productive but equally essential processes for human life, such as deliberation, reflection or even empathy. These elements are also essential for sustaining, among other things, human dignity and coexistence in a democratic culture.

There are many examples of how this technological language in education can be unmasked. In reality, I believe that we have adopted a results-based language where the supposed positive effects are more a marketing intention than the product of objective evaluation. The argument that legitimises this pro-AI discourse in educational processes is that it will improve education. When in reality, there is still no evidence for this, and international organisations such as UNESCO and the Special Rapporteur on the Right to Education are calling for impact assessments to be objective and not just be carried out by technology companies (which, obviously, only show the positive impact AI can have).

For example, in a World Bank report on the AI revolution in education in Latin America, terms such as "transformation", "unprecedented pace", "effectiveness" and "productivity" abound, along with statements about the efficiency of the technology (for example, "a study found that consultants using ChatGPT4 completed more tasks, worked faster and produced higher quality results"). There is also a lot of talk about "personalised education", but pedagogically speaking, in reality this "personalisation" is "individualised teaching", and is closer to Skinner's teaching machines of the 1950s than to Socratic dialogue, for example.

Thus, the first thing we must do is unmask such neoliberal language, the hype with which big tech companies sell us the supposed 'benefits' of technology for education while also hiding their concern for profits and access to greater amounts of data from educational agents.

In this sense, we must not only pay attention to the analysis of technological language in education, but also, secondly, we must

stimulate critical thinking by helping to ask the right questions. It is not only important to be aware of the type of language used by big tech companies, but also to question and ask the right questions. As a critical media theorist, Neil Postman raised questions about technology such as these that are still relevant today. Some of these questions could be: What problem does this technology [AI in education in this case] aim to solve, but also what new problems does its adoption create? Who could be harmed by a technological solution and what kind of harm could it cause? Who wins and who loses with technological changes?

Who gains more power through technological changes? And, thinking in educational terms, we can also ask ourselves, how does technology help us improve [if it really does] as people? Considering that by introducing AI, we ourselves as educators or learners 'stop doing something', we must think about what it is that we stop learning to do. What do we stop learning to be? What are we unlearning to help us live together? Some international organisations, such as the EU Commission and the OEI itself, have also opted for a question-based approach in their manuals or proposals on AI ethics for education. For example, in its document on 'ethical guidelines on AI and data usage in education', it asks questions at different levels and according to the educational agents that encourage reflection. The OEI's proposal, together with the Profuturo organisation, also offers questions and raises dilemmas rather than 'solutions'.

But knowledge or exposing the language alone is not enough to pedagogically address AI and other emerging technologies (although we cannot develop this point further), it is also necessary to instil a particular moral character in educational agents (from teachers to students, including families). In short, this means developing the virtue required to control and not be controlled by technology. Something that can only be acquired, as Aristotle tells us, from practice and through lived experience.

Given the importance of education not only for the full development of the human personality but also for maintaining and safeguarding democracy, human rights and the rule of law, we cannot think of education in terms other than as a universal human right, and we must continue to view the introduction of AI into educational processes through the same lens of democracy, human rights and the rule of law. Undoubtedly, in our "complex democracies", emerging technologies, AI and algorithms play an increasingly important role. However, we cannot hide the fact that the leading role in education must continue to belong to citizens and that, as Dewey reminded us, it is educational experiences that sustain democratic culture because democracy, like citizenship, cannot be learned in any other way than from one's own experience.

# Teachers, Artists and Entrepreneurs in Synthetic Times

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We already know that the eruption of artificial intelligence (AI) in education has brought opportunities and challenges that are reconfiguring teaching practices and their impact on learning. But how are educators in Latin America preparing for this? Is teaching in the region resistant to this technology? What do those in the classroom think about it? These questions guided the 'Teach Your Voice' campaign[1] which reached nearly 13,000 educators in 19 countries in the region. The initiative sought to gather the perceptions, challenges and opportunities faced by teachers in Latin America, with a special emphasis on the use of technologies such as artificial intelligence in education.

One of the main findings when addressing the relationship between teaching and artificial intelligence was that 97.4% of teachers believe that AI has an impact on their practice, but only 26.31% use AI tools for activities not related to their teaching work. In addition, 72.51% of educators believe that AI will have a positive impact on teaching practice (7 out of 10 of those surveyed teach in state-run schools), thus dispelling the myth of a negative bias or resistance to AI. Similarly, 53.51% are interested in incorporating AI into their teaching process but have not yet done so. Among the main reasons that hinder the incorporation of AI are the lack of time for training, lack of funds and/or resources to access paid licences (which allow for better use of AI, as most teachers use free versions that constantly change their usage and access policies), and not having the necessary knowledge about the risks and uses of information sources in applications, which refers to uncertainty about the accuracy of the information provided by AI tools and the lack of verified primary sources. This can be interpreted as a critical (and necessary) perspective, as well as the impact of educational technology policies that now have users who ask key and responsible questions.

We are, therefore, facing at a turning point: there is widespread awareness about the importance of AI in education, but structural difficulties remain, which hinder its effective integration. The first thing any technology development company does is survey the user experience: what do users already do with technology? How can we reduce the learning curve? With this mindset, we could read the data in relation to teaching practices and respond to the question: what do we already have? How can we enhance what we already know how to do? One relevant aspect that the campaign addressed was the use of technology in educational practice. The data is very

significant: nine out of ten teachers use some type of technological tool, with computers (86.15%) and mobile phones (85.56%) being the most common. Educators prioritise the use of tools with educational objectives,

1 Infobae. (2024, marzo 11). Almost all teachers in Latin America believe that AI affects education, but less than half have ever used it. Infobae. https://www.infobae.com/educacion/2024/03/11/casi-todos-los-docentes-de-latinoamerica-consideran-que-la-ia-afecta-a-la-educacion-pero-menos-de-la-mitad-la-ha-usado-alguna-vez/

such as online platforms with subject-specific content, specific software for the subject they teach, school management systems, virtual classrooms, digital libraries and digital content repositories. But isn't much of this software already powered by AI, even if it's not always recognised as such? In this regard, if the aim is to make effective use of AI, conducting a survey of what already exists, what is being done, and what works (in terms of practices and resources) is a necessary step before acquiring closed educational technology solutions that update old discourses, facilitated by the rise of AI. Public opinion, families, the world of work, and so on, all demand that the education system incorporate AI, but at what cost?

And who are the educators that use AI? Identifying and recognising user profiles is another strategy used by technology companies to offer personalised proposals and not underestimate their audience. The results of the 'Teach your voice' show that more experienced teachers have a higher level of academic qualifications, although this does not necessarily translate into more frequent use of hardware and software in the classroom. On the other hand, younger teachers tend to be self-taught in the adoption of technological tools, while more experienced teachers prefer certified training. However, despite these profiles and preferences, when asked about the current state of provision, a frequent reality in many classrooms in the region emerges: teachers are in the process of exploring how to integrate AI into their lessons, but they are doing so without a clear frame of reference and with few opportunities for sustained formal training.

Therefore, how are those who educate trained in technology? Most use their personal devices to access online courses and resources, prioritising self-directed learning that allows them to be flexible in terms of learning times and spaces, for example, training on their mobile phones while travelling to work (by bus, train or carpool). It should be noted that, although the survey met anonymity and identity preservation requirements, each teacher had to complete a profile that allowed them to be identified within a population group. Based on this the sample was formed. It showed that more than 60% of participants were women, and one in three were in the 31-40 age range. In addition, almost 50% had a university degree, and two out of three worked in secondary education. In terms of classroom experience, more than half had at least 10 years of teaching experience, and three out of four worked in state-run institutions. With these profiles, it can be inferred that self-directed learning is vital to balancing other responsibilities that this group has, such as caregiving tasks.

When asked about their motivations, 58.45% indicated that they seek training for professional reasons. So, what are we waiting for to leverage this motivational factor? What do we need to train with AI? What content can guarantee continuity in the face of digital turmoil?

During this time, we have seen different examples ranging from Khan Academy, with its AI-based assistant Khanmigo, which shows how technology can be used as a resource to enhance teaching rather than replace it. Khanmigo helps teachers to organise students according to their progress and to design personalised strategies for each student. Even influential teachers such as Edrei Robles and Adriana González (known as @lademateméatica) use platforms such as Twitch to broadcast their classes live, experiment with AI software and bring teaching closer to a wider community.Other teachers are experimenting with AI available on mobile phones, such as Meta AI on WhatsApp, where they ask their students to ask AI questions or request that it generate an image to help them research a topic, while also teaching students to challenge its accuracy and / or how to generate better questions and requests for AI. These experiences reflect that educational innovation does not depend exclusively on

technology, but on how teachers integrate it into their practice. We already know this, but AI confirms it. What and how can we best use AI?

In the media boom surrounding AI, we often hear the verb 'to prompt,' and if we take time to think about the art of prompting, it leads us to say that teachers are the architects of learning. Throughout history, teachers have always been the architects of learning, designing questions and guiding their students through processes of reflection and discovery. In this sense, we can say that teachers have been 'serial prompters' long before AI. The art of prompting (giving precise instructions to AI models to obtain useful responses) is a reflection of the pedagogical skill of designing questions that encourage critical thinking. But there is an essential difference: a teacher does not simply respond, but questions; does not merely structure knowledge, but challenges students to construct new ideas; does not just organise information, but creates learning experiences.

The advance of AI also poses challenges in terms of teacher leadership. As digital tools are incorporated into classrooms, teachers must take on new roles that go beyond the transmission of knowledge. Thus, pedagogical planning becomes an art of prompting, where designing strategic questions and guiding AI becomes an essential skill.

Entrepreneurial skills, which are so highly valued in the technological ecosystem that creates what we use on our screens, are now essential for navigating a constantly changing world. Leaders must innovate, manage uncertainty, lead transformation processes and adapt quickly. In this sense, teaching shares many characteristics with entrepreneurial leadership, as it requires adaptability, creativity and the ability to manage uncertainty.

Data shows that AI is already present in education, and we sometimes come across news stories in the media that have an impact on educational tech coexistence (such as cyberbullying) or raise awareness of new pathologies such as digital gaming disorder, addiction to scrolling, and changes in socio-emotional health with cases such as digital body dysmorphia, where the education system is where they are detected or catalysed, and/or a sector of society responds in one way or another to a broader problem with little jurisdiction.

Today, this is one more reason to generate new responses in educational settings, where those who educate are the leaders. During this time, proposals have been identified that not only range from developing specific training in AI applied to education, but also address the promotion of spaces for experimentation in the classroom, encouraging the exchange of experiences between teachers and other specialists in fields such as neuroscience applied to education, and the creation of communities of practice to accompany the technological transition and involve families. Those who integrate AI strategically go beyond the idea of personalising learning or optimising time; they focus on what really matters: developing critical and creative thinking at a cognitive and socio-emotional level. To do this, teachers become digital artists who focus not only on how to programme, but also on the experience they want to achieve in the educational spacethat will transform the opportunities of those who study.

In a context where more than half of the children in the region do not understand what they read, education is at a crossroads. AI is not the problem, but rather the lack of clear strategies for its integration into teaching and understanding that we are social beings and our hardware is designed to socialise, and education is the most sophisticated way we have to do this.

The future of education will depend on the ability of teachers to lead this transformation process and on whether they have the resources to carry out such a titanic task. AI will not replace educators, but it can be an

opportunity to redefine their role or return it to the centre of its social (and human) importance. The question is not whether we should adopt AI in education, but how we want to do so and for what purpose. Are we prepared to train teachers who not only teach with AI, but also question it and use it to generate new technological solutions? We know that educators not only promote skills for the future of work, but also for citizen participation and digital sovereignty. What happens when these definitions are unclear, lack consensus and are undergoing transformation? Today's educators can seize new, more ethical and responsible opportunities, go further and focus on generating a deep understanding of the role of AI in teaching and learning, not only in the future, but also in a present that requires urgent decisions.

# Artificial Intelligence for All: The Foucs of Human Rights in Times of Digital Transformation in Education

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The integration of artificial intelligence (AI) in education is currently a topic of both enormous interest great concern, as demonstrated by the wide range of learning resources that use AI, as well as the profusion of reflections, debates, alerts and published works on its projections and implications for individuals, communities and societies (UN, 2019).[1]

Although, strictly speaking, the presence of AI in education is not new, in recent years there has been a recurrence of discussions about opportunities and warnings regarding its impact on education and, beyond educational spaces, on work, daily life, the exercise of rights, citizenship, democracy and, in general, on the present and future of societies.

There are at least five crucial issues to consider when thinking about AI as an opportunity to improve learning, develop complex thinking, boost skills for global citizenship and, ultimately, improve people's quality of life, without losing sight of the challenges and risks involved. These issues are:

- digital transformation and the integration of AI as a global process with unimaginable magnitude, value and challenges;
- the importance of considering the integration of AI within the framework of understanding education as a universal human right;
- the opportunity offered by the incorporation of technologies and AI to rethink education: its approaches, public policies, modalities, strategies, pedagogical practices, interpersonal relationships and coexistence with other human beings and with nature in general;
- the role of managers and teachers in turning AI into a strategy for improving learning and inclusion from a human rights perspective; and
- the role of states in promoting, regulating and using technologies to ensure their contribution to sustainable development.

Digital transformation and the integration of AI in education have grown exponentially in recent years and are expected to bring about profound changes in the development of humanityto the point that some specialists compare these phenomena to the great industrial revolutions, even expecting AI to have a greater impact than these given its speed and scale. The rapid integration of ICTs and, more specifically, AI in educational spaces during the current century increased

1 UN (2019). The Age of Digital Interdependence: report of the UN Secretary-General's High-Level Panel on Digital Cooperation. Executive Summary, June 2019. https://www.un.org/sites/www.un.org/files/uploads/files/es/HLP%20on%20Digital%20Cooperation%20Report%20Executive%20Summary%20-%20ES%20.pdf

after the pandemic, when countries had to implement online and hybrid education modalities to maintain educational services (Inter-American Dialogue, 2020)[2] despite the fact that, in many cases, this was an exercise of trial-and-error due to insufficient prior knowledge and experience, and limitations in connectivity, equipment, and the preparedness of education systems.

Three aspects stand out in this area: 1) the benefits of AI in education for learning; 2) the risks and challenges posed by the integration of AI in classrooms without an explicit conceptual approach and adequate teacher preparation; and 3) the persistence of barriers to access to ICT.

According to UNESCO (2024), '[D]igital innovation has demonstrated powers to complement, enrich and transform education, and has the potential to speed up progress towards Sustainable Development Goal 4 (SDG 4) for education and transform modes of provision of universal access to learning. It can enhance the quality and relevance of learning, strengthen inclusion, and improve education administration and governance."[3]

However, technological development is not without risks and challenges that, despite warnings, persist in the education systems of various countries in Latin America and the Caribbean, such as widening learning gaps, the instrumental use of AI resources, the homogenisation of education without considering contextual differences when creating or promoting the use of AI strategies, the invisibility of gender diversities, cultural, linguistic and ethnic diversity, ethnicities, etc., and ethical questions about the use of personal data, among others.

Similarly, the effective integration of AI in schools and classrooms faces persistent barriers that were starkly revealed during the pandemic, namely: infrastructure and connectivity deficits, lack of policies for the integration of ICT in education, scarce resources for teacher training, and so on, as shown by studies by ECLAC/UNESCO,[4] UNICEF/UIT,[5] and the OEI.[6] EThese digital and AI access gaps in the countries of the region are expressions of economic, social, political, cultural, educational, gender and other inequalities; they therefore affect, above all, the poorest populations, women, persons with disabilities, indigenous peoples and Afro-descendants, groups in situations of human mobility, and so on.

In addition, concerns and questions have been raised about two important aspects: first, the absence of regulations on the expansion and intervention of big tech corporations,whose goal is economic profitability, in educational policy decisions related to curriculum priorities, teaching resources, learning assessment, teacher training, etc.; and second, ethics in the access and management of personal and institutional databases that are easily accessible to specialised personnel.

To summarise, while AI is bringing about profound change in the educational ecosystem and presents immense opportunities for comprehensive learning beyond what can be imagined even today, it carries with it a number of risks and challenges that must be identified and addressed so that

2 Inter-American Dialogue (2020). Education Technology and Innovation in the Context of the Pandemic: Lessons Learned. Available at: https://thedialogue.org/wp-content/uploads/2020/08/La-tecnologia-e-innovacion-educativa-en-el-marco-de-la-pandemia-Lecciones-aprendidas.pdf

3 https://www.unesco.org/es/digital-education

4 CEPAL/UNESCO (2020). La educación en tiempos de la pandemia delCovid-19. https://repositorio.cepal.org/server/api/core/bitstreams/c29b3843-bd8f-4796-8c6d-5fcb9c139449/content

5 UNICEF/UIT (2020). How many children and young people have internet access at home? Estimating digital connectivity during the COVID-19 pandemic.

6 OEI (2021). Educación inclusiva hoy: Iberoamérica en tiempos de pandemia. https://oei.int/wp-content/uploads/2021/12/educacion-inclusiva-hoy-ib-en-tiempos-de-pandemia.pdf

technological development can effectively contribute to the progress of humanity and the sustainability of all forms of life on the planet.

The integration of AI with an approach to education as a universal human right implies, as UNESCO argues, talking about 'AI for all' as a commitment by states to guarantee lifelong learning opportunities and outcomes through policies and strategies for inclusion and equality, and through education that uses technological resources to advance the well-being and happiness of human beings, respect for nature, the consolidation of democratic societies with equality and social justice, and the sustainability of the planet.

In this sense, the discussion on AI is not only focused on aspects of the technology but is also a discussion on universal access to human rights and the role of states in ensuring they are fulfilled.

The integration of AI is an opportunity to rethink the meaning of education. AI has the potential to address some of the greatest challenges facing education today, to develop innovative teaching and learning practices, and ultimately to accelerate progress towards Sustainable Development Goal (SDG) 4. However, these rapid technological advances inevitably involve numerous risks and challenges, which policy debates and regulatory frameworks are still struggling to overcome.[7]

'The challenge of the digital transformation of education lies not only in the lack of infrastructure, but also in the way in which technological solutions have been implemented in the education sector. For years, digital initiatives have been developed in a fragmented manner and have been guided by technological innovation rather than human and educational needs. Furthermore, in most cases, these initiatives have ignored key issues of equity, sustainability and cultural relevance.'[8]

AI offers us the opportunity to open up and/or deepen our vision of the meaning and purpose of education, of the need for comprehensive training in complex and critical thinking, problem solving, life planning and participation in democratic life at all levels. Learning understood solely as the acquisition of skills restricts our exercise of full rights and relegates the majority of the school population to an education that will barely help them find jobs to survive.

The use of AI-based learning resources, adopted mechanically and without reflection on the relevance andappropriateness of their adaptation and integration into cultures, languages and diversity, can deepen the instrumentalisation of learning and, in turn, move education in the opposite direction of complex and independent critical thinking skills in students and teachers.

The OEI refers to the need to move forward with a 'process of change aimed at improving quality and equity in education, supported by technology and connectivity, but which goes beyond the digitisation of the system by requiring that it be accompanied by profound pedagogical review'. (...) 'a new model of teaching and learning that is more flexible and better able to serve everyone (inclusively), through collaborative, more experiential methodologies that seek to develop students' skills. Only an inclusive and resilient type of education will be able to successfully reach everyone equally and achieve more just and egalitarian

7 Miao, F.; Holmes, W.; Huang, R.; Zhang, H. (2021). AI and education: guidance for policy-makers. UNESCO. Available at: https://unesdoc.unesco.org/ark:/48223/pf0000376709

8 https://profuturo.education/observatorio/enfoques/seis-pilares-para-la-transformacion-digital-de-la-educacion/#:~:text=Esta%20cifra%20ejemplifica%20la%20magnitud,calidad%20y%20el%20impacto%20social.

societies.'[9]

TAccording to UNESCO, all of this compels states to adopt a human-centred approach to AI that can help to reverse the paradoxical persistence of current inequalities in access to knowledge, research and the diversity of cultural expressions, and to ensure that AI does not widen the technological divide within and between countries, and within and between schools.[10]

*In this endeavour, school leaders and teachers are key to making AI a strategy for improving learning with a human rights approach.* Their work is crucial to achieving the objectives of educational policy agendas, even more so in the current context, where information and technological resources circulate through various channels and not only in school settings. Teachers are also trying to navigate a process of understanding, appropriating and integrating AI, with the same risks of instrumentalising their learning and reducing AI to the use of 'novel' teaching resources.

It is, therefore, essential to promote comprehensive teacher policies that include high-quality initial and in-service training to harness the potential of AI and work in highly diverse contexts; systematic technical support strategies; decent working conditions; and recognition, social value and strengthening of the teaching profession. These are all key elements for overcoming teachers' fears and resistance to the integration of AI and strengthening their preparedeness and ethical commitment to student learning.

The role of states is crucial in promoting, regulating and integrating technologies in education to ensure their contribution to sustainable development. Currently, states have a set of guidelines, frameworks and consensuses participatively built to advance the implementation of global and regional agreements[11] that promote the formulation of multisectoral public policies, programmes and strategies with a rights-based approach and with the necessary public funding for the integration of technologies and AI in education systems.

Equally important is government action to establish regulations, supervisory mechanisms and information systems, to monitor and evaluate programme policies, and to apply personal data protection laws etc., with the objective of tackling the identified risks of AI.

Lastly, discussions about the future of education involve many competing actors within and outside the education system, including governments to varying degrees, ministries, leading scholars, teachers, students, families, entrepreneurs, civil society organisations, and many others. All of them share a collective responsibility to seize the opportunities that AI offers, to respond to the challenges and risks it poses, and uphold a human-rights based approach in decisions about how AI can be promoted, advanced, and integrated into education.

9 OEI (2021). Ibero-American Programme for Digital Transformation in Education https://oei.int/oficinas/secretaria-general/programas/programa-regional-de-transformacion-digital-en-educacion-2/

10 https://www.unesco.org/en/artificial-intelligence.

11 Beijing Consensus (https://unesdoc.unesco.org/ark:/48223/pf0000368303), Recomendación sobre la ética de la inteligencia artificial (https://www.unesco.org/es/artificial-intelligence/recommendation-ethics?hub=99488)

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